



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

I/IV B.TECH

(With effect from 2017-2018 Admitted Batch onwards)

Under Choice Based Credit System

GROUP-B (CSE, ECE & IT)

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 BS 1101	English – I*	3	3	1	--	4	30	70	100
B17 BS 1102	Mathematics – I*	3	3	1	--	4	30	70	100
B17 BS 1103	Mathematics-II	3	3	1	--	4	30	70	100
B17 BS 1104	Engineering Physics	3	3	1	--	4	30	70	100
B17 CS 1101	Computer Programming Using C	3	3	1	--	4	30	70	100
B17 CE 1101	Environmental Studies *	2	2	1	--	3	30	70	100
B17 BS 1106	Engineering Physics Lab	2	--	--	3	3	50	50	100
B17 BS 1108	English Communication Skills Lab – I *	2	--	--	3	3	50	50	100
# DL1	Department Lab	2	--	--	3	3	50	50	100
B17 BS 1110	Engineering Physics Virtual Labs-Assignments	--	--	--	2	2	--	--	--
B17 BS 1112	NCC	--	--	--	2	2	--	--	--
Total		23	17	6	13	36	330	570	900

* Common to both Group - A and Group - B

#DL 1	CSE & IT	B17 CS 1102	C Programming Lab & Hardware Fundamentals
	ECE	B17 CS 1103	C Programming Lab

Code: B17 BS 1101

ENGLISH - I
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To improve the language proficiency of the students in English with emphasis on LSRW skills.
2. To enable the students to study and comprehend the prescribed lessons and subjects more effectively relating to their theoretical and practical components.
3. To develop the communication skills of the students in both the formal and informal situations.
4. To expose the students to various forms of writing in formal settings in order to build confidence in standard grammar
5. To introduce the students various types of reading comprehension practices.

Course Outcomes: At the end of the Course, Student will be able to:

1. Understand the rudiments of LSRW Skills, comprehension and fluency of speech.
2. Gain confidence and competency in vocabulary and grammar.
3. Listen, speak, read and write effectively in both the academic and non- academic environment.
4. Extend his/her reading skills towards literature.
5. Strengthen his/her analytical and compositional skills.

SYLLABUS

Life through Language: An Effective Learning Experience

Life through Language has a systematic structure that builds up communicative ability progressively through the chapters. It will enable the learner to manage confusion; frame question for themselves and others; develop new ideas; support ideas with evidence; express themselves with poise and clarity; and think critically. Acquisition of skill leads to confidence.

UNIT-I

People and Places:- Word search - Ask yourself-Self-assessment-I -Self-assessment-II - Sentence and its types- Describing people, places and events-Writing sentences-Self-awareness-Self-motivation, Dialogue writing.

UNIT-II

Personality and Lifestyle:- Word quiz – Verbs-Adverbs-Negotiations-Proving yourself-Meeting Carl Jung- Describing yourself- Living in the 21st century- Using your dictionary-Communication-Adaptability.

UNIT-III

Media and Environment: - A list of 100 basic words – Nouns- Pronouns- Adjectives-News report- Magazine article- User's Manual for new iPod- A documentary on the big cat- Why we need to save our tigers: A dialogue- Global warming- Paragraph Writing-Arguing a case-Motivation- Problem solving.

UNIT-IV

Entertainment and Employment:- One word substitutes- Parts of speech- Gerunds and infinitives- An excerpt from a short story an excerpt from a biography- A consultant interviewing employees- Your first interview- Reality TV- Writing an essay-Correcting sentences- Integrity Sense of humor.

UNIT-V

Work and Business:- A list of 100 difficult words- Articles, Quantifiers- Punctuation - Open letter to the Prime Minister Business dilemmas: An email exchange- A review of *IPL: The Inside Story*, Mark Zuckerberg: World's Youngest Billionaire- A conversation about a business idea- Pair work: Setting up a new business- Recession- Formal letters-Emails- Reports- Professionalism-Ethics, Fill in the blanks.

Text Book:

1. *Life through Language: A Holistic Approach to Language Learning*. Board of Editors, Pearson Publishers, India. 2013.

Reference Books:

1. Basic Vocabulary. Edgar Thorpe, Showick Thorpe. Pearson P. 2008.
2. Quick Solutions to Common Errors in English, Angela Bunt. MacMillan P. 2008.
3. Know Your English (Volume 1&2), by Dr. S. Upendra, Universities Press, India 2012
4. Business Communication Strategies. Mathukutty Monippally. Tata Mc Grahill P. 2009.

MATHEMATICS - I
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that is essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Learn about linear first order and higher order ordinary differential equations and their applications.
4. Acquire knowledge of Laplace transform, partial differentiation and their applications.
5. Learn certain first order and higher order partial differential equations.

Course Outcomes: At the end of the Course, Student will be able to:

1. Solve linear ordinary differential equations of first order and first degree. Also will be able to apply the knowledge in simple applications such as Newton's law of cooling, orthogonal trajectories and simple electrical circuits.
2. Solve linear ordinary differential equations of second order and higher order. Also will be able to apply the knowledge in simple applications such as LCR circuits and Simple harmonic motion.
3. Determine Laplace transform and inverse Laplace transform of various functions.
4. Use Laplace transforms to solve a linear ODE.
5. Calculate total derivative, Jacobian and maxima/minima of functions of two variables.
6. Form partial differential equations and solve some standard types of first order PDEs. Find complimentary function and particular integral of linear higher order homogeneous and non-homogeneous PDEs.

SYLLABUS

UNIT I: Differential equations of first order and first degree:

Linear, Bernoulli, Exact, Reducible to exact types.

Applications: Newton's Law of cooling, Law of natural growth and decay, Orthogonal trajectories, Simple electrical circuits, Chemical reactions.

UNIT II: Linear differential equations of higher order:

Non-homogeneous equations of higher order with constant coefficients with RHS term of the type e^{ax} , $\sin ax$, $\cos ax$, polynomials in x , $e^{ax}V(x)$, $xV(x)$, Method of Variation of parameters.

Applications: LCR circuit, Simple Harmonic motion.

UNIT III: Laplace transforms:

Laplace transforms of standard functions, transforms of $tf(t)$, $f(t)/t$, properties, transforms of derivatives and integrals, transforms of unit step function, Dirac delta function, Inverse Laplace transforms, convolution theorem (without proof).

Applications: Solving ordinary differential equations (initial value problems) using Laplace transforms.

UNIT IV: Partial differentiation:

Introduction, Homogeneous functions, Euler's theorem, Total derivative, Chain rule, which variable is to be treated as constant, Functional dependence, Jacobians, Taylor series for a function of two variables, Leibnitz rules for differentiation under the integral sign.

Applications: Errors and Approximations, Maxima and Minima of functions of two variables without constraints, Lagrange's method (with constraints)

UNIT V: First order and higher order partial differential equations:

Formation of partial differential equations by elimination of arbitrary constants and arbitrary functions, solutions of first order Lagrange linear equation and nonlinear equations of standard types (excluding Charpit's method).

Solutions of Linear homogeneous and non-homogeneous Partial differential equations with constant coefficients - RHS terms of the type e^{ax+by} , $\sin(ax+by)$, $\cos(ax+by)$, $x^m y^n$.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, A Text book of Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
2. Michael Greenberg, Advanced Engineering Mathematics, 9th edition, Pearson.
3. Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press.
4. Peter O'Neil, Advanced Engineering Mathematics, Cengage Learning.
5. Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

MATHEMATICS – II
(Common to CSE, ECE& IT)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that are essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Understand some basic numerical methods to solve algebraic and transcendental equations.
4. Learn interpolation methods for equally spaced and unequally spaced data points. Learn methods for numerical evaluation of integrals and for solving first order ODEs.
5. Learn concepts of a Fourier series and Fourier transform.

Course Outcomes: At the end of the Course, Student will be able to:

1. Find a real root of algebraic and transcendental equations using different methods.
2. Know the relation between the finite difference operators. Determine interpolation polynomial for a given data.
3. Evaluate numerically certain definite integrals applying Trapezoidal and Simpson's rules.
4. Solve a first order ordinary differential equation by Euler and RK methods.
5. Find Fourier series of a given function satisfying Dirichlet conditions. Find half range cosine and sine series for appropriate functions.
6. Find Fourier transforms, Fourier cosine and sine transforms of appropriate functions and evaluate certain integrals using inverse transforms and Fourier integral.

SYLLABUS

UNIT I: Solution of Algebraic and Transcendental Equations:

Introduction, Bisection method, Method of false position, Iteration method, Newton- Raphson method (One variable and simultaneous Equations).

UNIT II: Interpolation:

Introduction, Errors in polynomial interpolation, Finite differences, Forward differences, Backward differences, Central differences and Symbolic relations between the operators, Differences of a polynomial, Newton's formulae for interpolation, Interpolation with unequal intervals, Lagrange's interpolation formula.

UNIT III: Numerical Integration and solution of Ordinary Differential equations:

Trapezoidal rule, Simpson's $1/3^{\text{rd}}$ and $3/8^{\text{th}}$ rules, Solution of ordinary differential equations by Taylor series method, Picard's method of successive approximations, Euler's method, Runge-Kutta methods (second order and fourth order).

UNIT IV: Fourier Series:

Introduction, Periodic functions, Fourier series of a periodic function, Dirichlet's conditions, Even and odd functions, Change of interval, Half-range sine and cosine series, Parseval's formula.

UNIT V: Fourier Transforms:

Fourier integral theorem (without proof), Complex form of Fourier integral, Fourier sine and cosine integrals, Fourier transform, Fourier sine and cosine transforms, properties, inverse transforms, Parseval's identities, Finite Fourier transforms.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, A Text book of Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press.
2. V.Ravindranath and P. Vijayalakshmi, Mathematical Methods, Himalaya Publishing House.
3. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
4. David Kincaid, Ward Cheney, Numerical Analysis-Mathematics of Scientific Computing, 3rd Edition, Universities Press.
5. Srimanta Pal, Subodh C. Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

ENGINEERING PHYSICS
(Common to CSE, ECE& IT)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

To re-orient Physics curriculum to the needs of the circuit and non circuit branches of Engineering / Technology courses offered by SRKREC(A) Bhimavaram that serves to understand the general and branch specific basic topics and introduce the related advanced technologies to the stake holders.

1. To impart the knowledge of physical optics phenomena like Interference and Diffraction required to understand the design and working of optical instruments with high resolution.
2. To understand the concept of coherence and generation of high intensity light sources and propagation of light waves in fibers for technological applications.
3. To study and analyze the behavior of electric and magnetic fields and their mutual interaction, and to understand the science of ultrasonics.
4. To understand the basic concepts of Quantum mechanics to know the behavior of fundamental particles in materials and to understand the basic nature of materials in general and classify them.
5. To study the structure property relationship of crystalline solids and understand basics of nanotechnology.

Course Outcomes: At the end of the course the students-

1. Learn the basic concepts of interference and diffraction of light and its applications.
2. Understand the science of producing high intensity light beams for technological applications and also understand the propagation of light waves in optical fibers in various applications.
3. Understand the inter relationship of electric and magnetic fields and learn ultrasonic's as a tool for technological applications
4. Learn the behavior of particles at the very microscopic level by using wave nature of particles and understand the behavior of materials and be able to classify them using the band theory of solids.
5. Learn the basics of structures of solid materials and nano material preparation Techniques/methods.

SYLLABUS

UNIT I: Interference and Diffraction

Principle of superposition-coherence-interference in thin films (reflected system) – Wedge shaped film-Newton's rings-Michelson's interferometer. Fraunhofer's diffraction at single slit, Diffraction grating-Resolving power of a grating.

UNIT- II: Lasers and Optical Fibers

Introduction, Spontaneous emission and Stimulated emission – Einstein's relation – Requirements of Laser device- Ruby laser- He-Ne gas laser- Characteristics of laser- Applications.

Description of optical fiber, Principle of light propagation- Optical fiber –Acceptance angle- Numerical aperture of optical fiber- Modes of propagation- Classification of fibers- Applications of fiber.

UNIT- III: Electro Magnetic Fields and Ultrasonics

Concept of Electromagnetic induction , Faraday's law, Lenz's law, Electric fields due to time varying magnetic fields, Magnetic fields due to time varying electric fields, Displacement current, Modified Ampere's law, Maxwell's equations and their significance (without derivation).

Definition of Ultrasonics-Methods of Producing Ultrasonics- Detection of Ultrasonics- Applications of Ultrasonics.

UNIT- IV: Quantum Mechanics and Band Theory of Solids

Introduction, de Broglie matter waves- properties-Experimental confirmation, wave function-significance- Schrodinger's time dependent and time independent wave equations- Eigen values and functions, Particle in a box.

Band theory of Solids- Introduction- Kronig Penney model (Qualitative)- Energy bands of crystalline solids- Distinction between Conductors, Semi conductors and insulators.

UNIT-V: Crystallography and Nano Materials

Basis and Lattice, Crystal systems, Bravais lattice, Unit cell Coordination number – Packing fraction for SC ,FCC, and BCC lattices, Miller indices- Diffraction of X rays from crystals- Bragg's law.

Introduction to Nanomaterials – Synthesis methods : Condensation, ball milling, sol-gel, chemical vapour deposition methods, properties and applications.

Text Books:

1. Physics by Resnick & Halliday. Wiley - Eastern (India) Ltd.
2. Engineering Physics By M.N. Avadhanulu & PG Kshirasagar, S.Chand & Co.
3. Engineering Physics by V. Rajendran, Mc Graw Hill Education (India) Pvt.Ltd.

Reference Books:

1. Engineering Physics by MR Srinivasan, New International Publishers.
2. Solid State Physics – A.J. Dekkar, MacMillan (India) Ltd.
3. Engineering Physics by Gaur Gupta, Dhanpat Rai Publications, Meerut, India.
4. Engineering Physics by P k Palanisamy, Scitech Publications (India) Pvt.Ltd

(Note: Assignment Marks of **Engineering Physics** are to be considered from the Internal marks of **Engineering Physics-- Virtual Labs – Assignments B17 BS 1110**)

COMPUTER PROGRAMMING USING C
(Common to CSE, ECE & IT)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Formulating algorithmic solutions to problems and implementing algorithms in C

1. Notion of Operation of a CPU, Notion of an algorithm and computational procedure, editing and executing programs in Linux
2. understanding branching, iteration and data representation using arrays
3. Modular programming and recursive solution formulation
4. Understanding pointers and dynamic memory allocation
5. Understanding miscellaneous aspects of C
6. Comprehension of file operations

Course Outcomes:

1. Understand the basic terminology used in computer programming
2. Write, compile and debug programs in C language.
3. Use different data types in a computer program.
4. Design programs involving decision structures, loops and functions.
5. Explain the difference between call by value and call by reference
6. Understand the dynamics of memory by the use of pointers
7. Use different data structures and create/update basic data files.

SYLLABUS

UNIT I:

Unit objective: Notion of Operation of a CPU, Notion of an algorithm and computational procedure, editing and executing programs in Linux

Introduction: Computer systems, Hardware and Software Concepts.

Problem Solving: Algorithm / Pseudo code, flowchart, program development steps, computer languages: machine, symbolic and highlevel languages, Creating and Running Programs: Writing, Editing(vi/emacs editor), Compiling(gcc), Linking and Executing in under Linux.

BASICS OF C: Structure of a c program, identifiers, basic data types and sizes. Constants, Variables, Arithmetic , relational and logical operators, increment and decrement operators, conditional operator, assignment operator, expressions, type conversions, Conditional Expressions, precedence and order of evaluation, Sample Programs.

UNIT II:

Unit objective: understanding branching, iteration and data representation using arrays

SELECTION – MAKING DECISION: TWO WAY SELECTION: if-else, null else, nested if, examples, Multi-way selection: switch, else-if, examples.

ITERATIVE: loops- while, do-while and for statements , break, continue, initialization and updating, event and counter controlled loops, Looping applications: Summation, powers, smallest and largest.

ARRAYS: Arrays- concepts, declaration, definition, accessing elements, storing elements, Strings and String Manipulations, 1-D arrays, 2-D arrays and character arrays, string manipulations, Multidimensional arrays, array applications: Matrix operations, checking the symmetricity of a Matrix. **STRINGS: concepts, c strings.**

UNIT III:

Objective: Modular programming and recursive solution formulation

FUNCTIONS- MODULAR PROGRAMMING: functions, basics, parameter passing, storage classes extern, auto, register, static, scope rules, block structure, user defined functions, standard library functions, recursive functions, Recursive solutions for fibonacci series, towers of Hanoi, header files, C Preprocessor, example c programs, Passing 1-D arrays, 2-D arrays to functions.

UNIT IV:

Objective: Understanding pointers and dynamic memory allocation

POINTERS: pointers- concepts, initialization of pointer variables, pointers and function arguments, passing by address- dangling memory, address arithmetic, character pointers and functions, pointers to pointers, pointers and multi-dimensional arrays, dynamic memory management functions, command line arguments

UNIT V:

Objective: Understanding miscellaneous aspects of C

ENUMERATED, STRUCTURE AND UNION TYPES: Derived types- structures declaration, definition and initialization of structures, accessing structures, nested structures, arrays of structures, structures and functions, pointers to structures, self referential structures, unions, typedef, bit-fields, program applications

BIT-WISE OPERATORS: logical, shift, rotation, masks.

Objective: Comprehension of file operations

FILEHANDLING: Input and output- concept of a file, text files and binary files, Formatted I/O, File I/O operations, example programs

Text Books:

1. Problem Solving and Program Design in C, Hanly, Koffman, 7thed, PERSON
2. Programming in C, Second Edition PradipDey and Manas Ghosh, OXFORD Higher Education
3. Programming in C, A practical approach Ajay Mittal PEARSON
4. The C programming Language by Dennis Richie and Brian Kernighan
5. Programming in C, B. L. Juneja, Anith Seth, Cengage Learning.

Reference Books:

1. C Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE
2. Programming with C, Bichkar, Universities Press
3. Programming in C, ReemaThareja, OXFORD
4. C by Example, Noel Kalicharan, Cambridge

ENVIRONMENTAL STUDIES
(Common to all Branches)

Lecture	: 2 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 2

Course Objectives:

1. Developing an awareness and sensitivity to the total environment and its related problems.
2. Motivating students for active participation in environmental protection and improvement.
3. Developing skills for active identification and development of solutions to environmental problems
4. Evaluation of environment programmes in terms of social, economic, ecological and aesthetic factors.
5. Creating a “CONCERN AND RESPECT FOR THE ENVIRONMENT”.

Course Outcomes:

1. To bring awareness among the students about the nature and natural ecosystems
2. Sustainable utilization of natural resources like water, land, energy and air
3. Resource pollution and over exploitation of land, water, air and catastrophic (events) impacts of climate change, global warming, ozone layer depletion, marine, radioactive pollution etc to inculcate the students about environmental awareness and safe transfer of our mother earth and its natural resources to the next generation
4. Safe guard against industrial accidents particularly nuclear accidents
5. Constitutional provisions for the protection of natural resources

SYLLABUS

UNIT – I**Global Environmental Crisis:**

Environmental Studies - Definition, Scope and importance, Need for public awareness. Global Environmental Crisis

Ecosystems:

Basic Concepts - Structure and Functions of Ecosystems: Producers, Consumers and Decomposers. Types of Ecosystems: Forest Ecosystems, Grassland Ecosystems Desert Ecosystems and Aquatic Ecosystems

UNIT-II**Biodiversity:**

Introduction to Biodiversity, Values of Bio-diversity, Bio-geographical classification of India, India as a Mega-diversity habitat, Threats to biodiversity, Hotspots of Biodiversity, Conservation of Biodiversity: In-situ and Ex-situ conservation of Biodiversity.

UNIT-III

Environmental and Natural Resources Management:

Land Resources: Land degradation, soil erosion and desertification, Effects of modern agriculture. **Forest Resources:** Use and over exploitation-Mining and Dams-their effects on forest and tribal people. **Water resources:** Use and over utilization of surface and ground water, Floods, droughts, conflict over water, water logging and salinity, dams – benefits and problems. **Energy Resources:** Renewable and non-renewable energy sources, use of alternate energy sources-impact of energy use on environment.

UNIT-IV

Environmental Pollution:

Causes, Effects and Control measures of - Air pollution, Water pollution, Soil pollution, Marine Pollution, Thermal pollution, Noise pollution, Nuclear Hazards; Climate change and Global warming, Acid rain and Ozone layer depletion. Solid Waste Management: Composting, Vermiculture, Urban and Industrial Wastes, Recycling and Reuse.

Environmental Problems in India:

Drinking water, Sanitation and Public health, Population growth and Environment; Water Scarcity and Ground Water Depletion; Rain water harvesting, Cloud seeding and Watershed management.

UNIT-V

Institutions and Governance:

Regulations by Government- Environmental Protection Act, Air (Prevention & Control of Pollution) Act, Water (Prevention and control of Pollution) Act, Wildlife Protection Act, Forest Conservation Act. Environmental Impact Assessment (EIA)

Case Studies:

Chipko Movement, Narmada Bachao Andolan, Silent Valley Project, Mathura Refinery and Taj Mahal, Industrialization of Patancheru, Nuclear reactor at Nagarjuna Sagar, Tehri Dam, Ralegaon Siddhi (Anna Hazare), Kolleru lake – Aquaculture, Fluorosis in Andhra Pradesh & Telangana.

Field Work:

Visit to a local area to document and mapping environmental assets. Visits to Industries, Water Treatment Plants, Effluent Treatment Plants.

Text Books:

1. Environmental Studies, K. V. S. G. Murali Krishna, VGS Publishers, Vijayawada
2. Environmental Studies, R. Rajagopalan, 2nd Edition, 2011, Oxford University Press.
3. Environmental Studies, Dr. M. Sita Rama Reddy, Dr. K. Anji Reddy, Silicon Publications, ASR Nagar, Bhimavaram
4. Environmental Studies, P. N. Palanisamy, P. Manikandan, A. Geetha, and K. Manjula Rani; Pearson Education, Chennai.

Reference Books:

1. Text Book of Environmental Studies, Deeshita Dave & P. Udaya Bhaskar, Cengage Learning.
2. A Textbook of Environmental Studies, Shaashi Chawla, TMH, New Delhi
3. Environmental Studies, Benny Joseph, Tata McGraw Hill Co, New Delhi
4. Perspectives in Environment Studies, Anubha Kaushik, C P Kaushik, New Age International Publishers, 2014

**ENGINEERING PHYSICS LAB
(Common to CSE, ECE& IT)****Lab : 3 Periods**
Exam : 3 Hrs.**Int. Marks : 50**
Ext. Marks : 50
Credits : 2

Course Objectives:

Training the field oriented Engineering graduates to handle instruments and their design methods to improve the accuracy of measurements.

1. To impart hands-on experience to the students entering engineering/Technology education about handling sophisticated equipment/ instruments.
2. To make the students understand the theoretical aspects of various phenomena experimentally.

Course Outcomes:

Physics lab curriculum gives fundamental understanding of design of an instrument with targeted accuracy for physical measurements.

1. Students get hands on experience in setting up experiments and using the instruments/equipment individually.
2. Get introduced to using new/ advanced technologies and understand their significance.

**LIST OF EXPERIMENTS
(Any 10 of the following listed experiments)**

1. Determination of wavelength of a source-Diffraction Grating-Normal incidence.
2. Newton's rings – Radius of Curvature of Plano - Convex Lens.
3. Determination of thickness of a spacer using wedge film and parallel interference fringes.
4. Determination of Rigidity modulus of a material- Torsional Pendulum.
5. Determination of Acceleration due to Gravity and Radius of Gyration-Compound Pendulum.
6. Melde's experiment – Transverse and Longitudinal modes.
7. Verification of laws of vibrations in stretched strings – Sonometer.
8. Determination of velocity of sound – Volume Resonator.
9. L- C- R Series Resonance Circuit.
10. Study of I/V Characteristics of Semiconductor diode.
11. I/V characteristics of Zener diode.
12. Characteristics of Thermistor – Temperature Coefficients.
13. Magnetic field along the axis of a current carrying coil – Stewart and Gee's apparatus.
14. Energy Band gap of a Semiconductor p - n junction.
15. Hall Effect in semiconductors.
16. Time constant of CR circuit.
17. Determination of wavelength of laser source using diffraction grating.
18. Determination of Young's modulus by method of single cantilever oscillations.
19. Determination of lattice constant – lattice dimensions kit.
20. Determination of Planck's constant using photocell.
21. Determination of surface tension of liquid by capillary rise method.

Reference Book:

1. Advanced Practical Physics –Vol 1 & 2 By S.Venkata Raman, S.Chand & Co.

ENGLISH COMMUNICATIONSKILLS LAB- I
(Common to All Branches)

Lab	: 3 Periods	Int.Marks	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

Course Objectives:

1. To enable the students to learn through practice the communication skills of listening, speaking, reading and writing.
2. To make students recognize the sounds of English through Audio- Visual aids.
3. To familiarize the students with stress and intonation.
4. To help the students build their confidence in speaking skills.

Course Outcomes:

1. A study of the communicative items in the laboratory will help the students become successful in the competitive world.
2. Students improve their speaking skills in real contexts.
3. Students learn standard pronunciation and practice it daily discourse.
4. Students give up their communicative barriers.

SYLLABUS

- ❖ WHY study Spoken English?
- ❖ Making Inquiries on the phone, thanking and responding to Thanks - Practice work.
- ❖ Responding to Requests and asking for Directions - Practice work.
- ❖ Asking for Clarifications, Inviting, Expressing Sympathy, Congratulating
- ❖ Apologising, Advising, Suggesting, Agreeing and Disagreeing - Practice work.
- ❖ Letters and Sounds-Practice work.
- ❖ The Sounds of English-Practice Work
- ❖ Pronunciation
- ❖ Stress and Intonation-Practice work.

Lab Manual:

1. '**INTERACT:** English Lab Manual for Undergraduate Students' Published by Orient Blackswan Pvt Ltd.

Reference Books:

1. Strengthen your communication skills by Dr M Hari Prasad, Dr Salivendra Raju and Dr G Suvarna Lakshmi, Maruti Publications.
2. English for Professionals by Prof Eliah, B.S Publications, Hyderabad.
3. Unlock, Listening and speaking skills 2, Cambridge University Press
4. Spring Board to Success, Orient BlackSwan
5. A Practical Course in effective english speaking skills, PHI
6. Word power made handy, Dr shalini verma, Schand Company
7. Let us hear them speak, Jayashree Mohanraj, Sage texts
8. Professional Communication, Aruna Koneru, Mc Grawhill Education
9. Cornerstone, Developing soft skills, Pearson Education.

**C PROGRAMMING LAB& HARDWARE FUNDAMENTALS
(Common to CSE & IT)**

Lab	: 3 Periods	Int.Marks	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

Course Objectives:

1. Understand the basic concept of C Programming, and its different modules that includes conditional and looping expressions, Arrays, Strings, Functions, Pointers, Structures and File programming.
2. Acquire knowledge about the basic concept of writing a program.
3. Role of constants, variables, identifiers, operators, type conversion and other building blocks of C Language.
4. Use of conditional expressions and looping statements to solve problems associated with conditions and repetitions.
5. Role of Functions involving the idea of modularity.

Course Outcomes:

1. Apply and practice logical ability to solve the problems.
2. Understand C programming development environment, compiling, debugging, and linking and executing a program using the development environment.
3. Analyzing the complexity of problems, Modularize the problems into small modules and then convert them into programs.
4. Understand and apply the in-built functions and customized functions for solving the problems.
5. Understand and apply the pointers, memory allocation techniques and use of files for dealing with variety of problems.
6. Document and present the algorithms, flowcharts and programs in form of user manuals.
7. Identification of various computer components, Installation of software

List of Programs**Exercise - 1 Basics**

- a) What is an OS Command, Familiarization of Editors - vi, Emacs
- b) Using commands like mkdir, ls, cp, mv, cat, pwd, and man
- c) C Program to Perform Adding, Subtraction, Multiplication and Division of two numbers
From Command line

Exercise - 2 Basic Math

- a) Write a C Program to Simulate 3 Laws at Motion
- b) Write a C Program to convert Celsius to Fahrenheit and vice versa

Exercise - 3 Control Flow - I

- a) Write a C Program to Find Whether the Given Year is a Leap Year or not.
- b) Write a C Program to Add Digits & Multiplication of a number

Exercise – 4 Control Flow - II

- a) Write a C Program to Find Whether the Given Number is

- i) Prime Number
 - ii) Armstrong Number
- b) Write a C program to print Floyd Triangle
- c) Write a C Program to print Pascal Triangle.

Exercise – 5 Functions

- a) Write a C Program demonstrating of parameter passing in Functions and returning values.
- b) Write a C Program illustrating Fibonacci, Factorial with Recursion without Recursion.

Exercise – 6 Control Flow – III

- a) Write a C Program to make a simple Calculator to Add, Subtract, Multiply or Divide Using switch...case
- b) Write a C Program to convert decimal to binary and hex (using switch call function the function)

Exercise – 7 Functions - Continued

- a) Write a C Program to compute the values of $\sin x$ and $\cos x$ and e^x values using Series expansion. (use factorial function)

Exercise – 8 Arrays

Demonstration of arrays

- a) Search-Linear.
- b) Sorting-Bubble, Selection.
- c) Operations on Matrix.

Exercises - 9 Structures

- a) Write a C Program to Store Information of a Movie Using Structure
- b) Write a C Program to Store Information Using Structures with Dynamically Memory Allocation
- c) Write a C Program to Add Two Complex Numbers by Passing Structure to a Function

Exercise - 10 Arrays and Pointers

- a) Write a C Program to Access Elements of an Array Using Pointer
- b) Write a C Program to find the sum of numbers with arrays and pointers.

Exercise – 11 Dynamic Memory Allocations

- a) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using malloc () function.
- b) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using calloc () function.

Understand the difference between the above two programs

Exercise – 12 Strings

- a) Implementation of string manipulation operations **with** library function.
 - i) copy
 - ii) concatenate
 - iii) length
 - iv) compare
- b) Implementation of string manipulation operations **without** library function.
 - i) copy
 - ii) concatenate
 - iii) length
 - iv) compare

Exercise -13 Files

- a) Write a C programming code to open a file and to print its contents onscreen.
- b) Write a C program to copy files

Exercise - 14 Files Continued

- a) Write a C program merges two files and stores their contents in another file.
- b) Write a C program to delete a file.

Exercise - 15

- a) System Assembling, Disassembling and identification of Parts / Peripherals.
- b) Operating System Installation-Install Operating Systems like Windows, Linux along with necessary Device Drivers.

Exercise - 16

- a) MS-Office / Open Office
 - i. Word - Formatting, Page Borders, Reviewing, Equations, symbols
 - ii. Spread Sheet-Organize data, usage of formula, graphs, charts.
 - iii. Powerpoint - features of power point, guidelines for preparing an effective presentation.
- b) Network Configuration & Software Installation-Configuring TCP/IP, Proxy, and firewall settings. Installing application software, system software & tools.

Note:

- a) All the Programs must be executed in the Linux Environment. (Mandatory)
- b) The Lab record must be a print of the LATEX (.tex) Format.

**C PROGRAMMING LAB
(For ECE)**

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. Understand the basic concept of C Programming, and its different modules that includes conditional and looping expressions, Arrays, Strings, Functions, Pointers, Structures and File programming.
2. Acquire knowledge about the basic concept of writing a program.
3. Role of constants, variables, identifiers, operators, type conversion and other building blocks of C Language.
4. Use of conditional expressions and looping statements to solve problems associated with conditions and repetitions.
5. Role of Functions involving the idea of modularity.

Course Outcomes:

1. Apply and practice logical ability to solve the problems.
2. Understand C programming development environment, compiling, debugging, and linking and executing a program using the development environment.
3. Analyzing the complexity of problems, Modularize the problems into small modules and then convert them into programs.
4. Understand and apply the in-built functions and customized functions for solving the problems.
5. Understand and apply the pointers, memory allocation techniques and use of files for dealing with variety of problems.
6. Document and present the algorithms, flowcharts and programs in form of user manuals.
7. Identification of various computer components, Installation of software

Programming

Exercise - 1 Basics

- a) What is an OS Command, Familiarization of Editors - vi, Emacs
- b) Using commands like mkdir, ls, cp, mv, cat, pwd, and man
- c) C Program to Perform Adding, Subtraction, Multiplication and Division of two numbers
From Command line

Exercise - 2 Basic Math

- a) Write a C Program to Simulate 3 Laws at Motion
- b) Write a C Program to convert Celsius to Fahrenheit and vice versa

Exercise - 3 Control Flow - I

- a) Write a C Program to Find Whether the Given Year is a Leap Year or not.
- b) Write a C Program to Add Digits & Multiplication of a number

Exercise – 4 Control Flow - II

- a) Write a C Program to Find Whether the Given Number is
 - i. Prime Number
 - ii. Armstrong Number
- b) Write a C program to print Floyd Triangle
- c) Write a C Program to print Pascal Triangle.

Exercise – 5 Functions

- a) Write a C Program demonstrating of parameter passing in Functions and returning values.
- b) Write a C Program illustrating Fibonacci, Factorial with Recursion without Recursion.

Exercise – 6 Control Flow – III)

- a) Write a C Program to make a simple Calculator to Add, Subtract, Multiply or Divide Using switch...case
- b) Write a C Program to convert decimal to binary and hex (using switch call function the function)

Exercise – 7 Functions - Continued

- a) Write a C Program to compute the values of $\sin x$ and $\cos x$ and e^x values using Series expansion. (use factorial function)

Exercise – 8 Arrays

Demonstration of arrays

- a) Search-Linear.
- b) Sorting-Bubble, Selection.
- c) Operations on Matrix.

Exercises - 9 Structures

- a) Write a C Program to Store Information of a Movie Using Structure
- b) Write a C Program to Store Information Using Structures with Dynamically Memory Allocation
- c) Write a C Program to Add Two Complex Numbers by Passing Structure to a Function

Exercise - 10 Arrays and Pointers

- a) Write a C Program to Access Elements of an Array Using Pointer
- b) Write a C Program to find the sum of numbers with arrays and pointers.

Exercise – 11 Dynamic Memory Allocations

- a) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using `malloc ()` function.
 - b) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using `calloc ()` function.
- Understand the difference between the above two programs.

Exercise – 12 Strings

- a) Implementation of string manipulation operations **with** library function.
 - i) copy
 - ii) concatenate
 - iii) length
 - iv) compare
- b) Implementation of string manipulation operations **without** library function.
 - i) copy
 - ii) concatenate
 - iii) length
 - iv) compare

Exercise -13 Files

- a) Write a C programming code to open a file and to print its contents onscreen.
- b) Write a C program to copy files

Exercise - 14 Files Continued

- a) Write a C program merges two files and stores their contents in another file.
- b) Write a C program to delete a file.

Note:

- a) All the Programs must be executed in the Linux Environment. (Mandatory)
- b) The Lab record must be a print of the LATEX (.tex) Format.

ENGINEERING PHYSICS - VIRTUAL LABS – ASSIGNMENTS
(Common to CSE, ECE & IT)

Lab : 2 Periods

Int.Marks: 5

Course Objective: Training Engineering students to prepare a technical document and improving their writing skills.

Course Outcome: Physics Virtual laboratory curriculum in the form of assignment ensures an engineering graduate to prepare a /technical/mini-project/ experimental report with scientific temper.

LIST OF EXPERIMENTS

1. Hall Effect
2. Crystal Structure
3. Hysteresis
4. Brewster's angle
5. Magnetic Levitation / SQUID
6. Numerical Aperture of Optical fiber
7. Photoelectric Effect
8. Simple Harmonic Motion
9. Damped Harmonic Motion
10. LASER – Beam Divergence and Spot size
11. B-H curve
12. Michelson's interferometer
13. Black body radiation

URL: www.vlab.co.in

(Note: Internal Marks of **Engineering Physics - Virtual Labs – Assignments** are to be considered as Assignment marks in the Internal Marks of **Engineering Physics- B17 BS 1104**)

NCC
(Common to CSE, ECE & IT)

Practice : 2 Periods

The NCC- National Integration and Awareness- Drill- Personality Development Life Skills- Leadership- Disaster Management-Social Awareness and Community Development- Health and Hygiene- Environment Awareness and Conservation.

(**Note:** It is an uncredited course. It will not be included in the Grade Memo / Certificate. The Certificate will be issued based on the performance and attendance. This course attendance will be counted in the semester overall attendance.)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R17)

I/IV B.TECH
(With effect from **2017-2018** Admitted Batch onwards)
Under Choice Based Credit System

GROUP-B (CSE, ECE & IT)
II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 BS 1201	English – II*	3	3	1	--	4	30	70	100
B17 BS 1203	Mathematics – III*	3	3	1	--	4	30	70	100
B17 BS 1205	Engineering Chemistry	3	3	1	--	4	30	70	100
B17 ME 1201	Engineering Drawing	3	1	--	3	4	30	70	100
# DS 2	Department Subject	3	3	1	--	4	30	70	100
# DS 3	Department Subject	3	3	1	--	4	30	70	100
B17 BS 1207	Engineering Chemistry Lab	2	--	--	3	3	50	50	100
B17 BS 1208	English Communication Skills Lab – II*	2	--	--	3	3	50	50	100
# DL2	Department Lab	2	--	--	3	3	50	50	100
B17 BS 1212	Inner Engineering	--	--	--	2	2	--	--	--
Total		24	16	5	14	35	330	570	900

*** Common to both Group - A and Group - B**

# DS 2	CSE & IT	B17 CS 1202	Object Oriented Programming Through C++
	ECE	B17 CS 1203	Data Structures
#DS 3	CSE & IT	B17 EC 1201	Elements of Electronics Engineering
	ECE	B17 EE 1203	Elements of Electrical Engineering
#DL2	CSE & IT	B17 CS 1205	Object Oriented Programming Lab
	ECE	B17 BS 1209	Engineering Workshop & IT Workshop

ENGLISH - II
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To improve the language proficiency of the students in English with emphasis on LSRW skills.
2. To enable the students to study and comprehend the prescribed lessons and subjects more effectively relating to their theoretical and practical components.
3. To develop spoken and written forms of the students in both formal and informal situations.
4. To enrich the vocabulary of the students through the usage of the lexis in appropriate contexts.
5. To expose them to the correct structural patterns of English.

Course Outcomes: At the end of the Course, Student will be able to:

1. To comprehend the speech of people belonging to different backgrounds and regions.
2. Understand the importance of speaking and writing for personal and professional communication and practice it in real contexts.
3. To express fluently and accurately in social discourse.
4. Participate in group activities like role-plays, discussions and debates.
5. Identify the discourse features, and improve intensive and extensive reading skills.

SYLLABUS

UNIT I:

- A. Detailed-Text: Unit 1: ' The Greatest Resource- Education'
- B. Non-Detailed Text: Lesson 1: ' A P J Abdul Kalam' from The Great Indian Scientists.

UNIT II:

- A. Detailed-Text: Unit 2: ' A Dilemma'
- B. Non-Detailed Text: Lesson 2:'C V Raman' from The Great Indian Scientists.

UNIT III:

- A. Detailed-Text: Unit 3: 'Cultural Shock': Adjustments to new Cultural Environments
- B. Non-Detailed Text: Lesson 3:'Homi Jehangir Bhabha' from The Great Indian Scientists.

UNIT IV:

- A. Detailed-Text: Unit 4: 'The Lottery'
- B. Non-Detailed Text: Lesson 4: 'Jagadish Chandra Bose' from The Great Indian Scientists.

UNIT V:

- A. Detailed-Text: Unit 5: ' The Chief Software Architect'
- B. Non-Detailed Text: Lesson 5: ' Prafulla Chandra Ray' from The Great Indian Scientists

Detailed Textbook:

1. English Encounters Published By Maruthi Publishers.

Non-Detailed Text Book:

1. The Great Indian Scientists Published by Cengage learning.

MATHEMATICS - III
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that are essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Understand important concepts of Linear algebra including solution of linear simultaneous equations, eigen values and eigen vectors.
4. Learn Beta, Gamma functions and how to evaluate double and triple integrals.
5. Learn fundamental concepts of vector calculus that help understand other Engineering courses.

Course Outcomes: At the end of the Course, student will be able to:

1. Determine rank, and solve a system of linear simultaneous equations numerically using various matrix methods.
2. Determine Eigen values and Eigen vectors of a given matrix, Reduce a Quadratic form to its canonical form and classify.
3. Evaluate double integrals over a region and triple integral over a volume.
4. Use the knowledge of Beta and Gamma functions in evaluation of different integrals.
5. Find gradient of a scalar function, divergence and curl of a vector function. Use vector identities for solving problems.
6. Evaluate line, surface and volume integrals by the use of Green's, Stokes' and Gauss divergence theorems.

SYLLABUS

UNIT I: Linear systems of equations:

Rank, Echelon form, Normal form, Solution of linear systems, Gauss elimination, Gauss-Jordan, Jacobi and Gauss-Seidel methods.

Applications: Finding the current in electrical circuits.

UNIT II: Eigen values - Eigen vectors and Quadratic forms:

Eigen values, Eigen vectors, Properties, Cayley-Hamilton theorem, Inverse and powers of a matrix by using Cayley-Hamilton theorem, Diagonalization, Quadratic forms, Reduction of a Quadratic form to Canonical form, Rank, Positive, Negative, Semi-Definite and indefinite forms of a Quadratic form, Index and Signature of a Quadratic form.

Applications: Free vibration of a two-mass system.

UNIT III: Multiple integrals:

Double and triple integrals, Change of variables, Change of order of integration. Application to finding Areas, Moment of Inertia and Volumes.

Beta and Gamma functions, Properties, Relation between Beta and Gamma functions, Application to evaluation of improper integrals. The error function and the complimentary error function.

UNIT IV: Vector Differentiation:

Gradient, directional derivative, Divergence, Curl, Incompressible flow, solenoidal and irrotational vector fields, second order operators, vector identities.

UNIT V: Vector Integration:

Line integral, Work done, Potential function; Area, Surface and volume integrals, Flux, Vector integral theorems: Greens, Stokes and Gauss Divergence theorems (without proof) and related Problems.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Michael Greenberg, Advanced Engineering Mathematics, 9th edition, Pearson edn.
2. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
3. Peter O'Neil, Advanced Engineering Mathematics, 7th edition, Cengage Learning.
4. D.W. Jordan and T. Smith, Mathematical Techniques, Oxford University Press.
5. Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

ENGINEERING CHEMISTRY
(Common to CSE, ECE & IT)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Knowledge of basic concepts of chemistry for engineering students will help them as professional engineers later in design and material selection as well as utilizing the available resources.

Course Objectives:

1. To understand the physical and mechanical properties of Polymers/Plastics/elastomers helps in selecting suitable materials for different purpose.
2. To create awareness on fuels as a source of energy for industries like thermal power stations, steel industry, fertilizer industry etc.
3. To understand the concepts of galvanic cells and corrosion with theories like electro chemical theory.
4. To understand the importance of water.
5. To understand about the materials which are used in major industries like steel and metallurgical manufacturing industries, construction and electrical equipment manufacturing industries.

Course Outcomes:

1. At the end of the course the students learn the advantages and limitations of plastic materials and their use in design.
2. Fuels which are used commonly and their economics, advantages and limitations are discussed.
3. Students gained Knowledge reasons for corrosion and some methods of corrosion control.
4. Students understands the impurities present in raw water, problems associated with them and how to avoid them.
5. Similarly students understand liquid crystals and semi conductors. Students can gain the building materials , solar materials, lubricants and energy storage devices.

SYLLABUS

UNIT-I: High Polymers and Plastics; Rubbers & Elastomers:

Polymerization Definition, Types of Polymerization, Mechanism of addition polymerization, Plastics as engineering materials, Thermoplastics and Thermosetting plastics, Compounding of plastics, Fabrication of plastics (4 techniques); Preparation, Properties and applications of Polyethylene, PVC, Bakelite, Nylon - 6,6, Bullet Proof plastics -polycarbonate and Kelvar; Fiber reinforced plastics, conducting polymers, Biodegradable Polymers - PHBV, Nylon 2, Nylon 6. Natural rubber – Vulcanization – Compounding of Rubber; Preparation, properties and applications of Buna – S; Buna – N;

UNIT-II :Fuel Technology& Lubricants:

Fuels: - Introduction – Classification of fuels, Calorific value – HCV and LCV, Determination of Calorific value by bomb calorimeter; Proximate and ultimate analysis of coal, coke: manufacture of coke by Otto – Hoffmann’s by-product coke oven process; Refining of Petroleum, Knocking-octane number of gasoline, cetane number of diesel oil. Synthetic Petrol; LPG , CNG.

Lubricants:- Definition, Mechanism of Lubrication, Properties of Lubricants (Definition and significance)

UNIT-III: Electrochemical cells and Corrosion:

Galvanic cell, single electrode potential, Calomel electrode; Modern batteries: - Lead – Acid battery; Fuel cells- Hydrogen – Oxygen cell, Lithium battery Theories of corrosion (i) dry Corrosion (ii) wet corrosion. Types of corrosion - differential aeration corrosion, pitting corrosion, galvanic corrosion, stress corrosion, Factors influencing corrosion, Protection from corrosion-material selection & design, cathodic protection, Protective coatings- metallic coatings – Galvanizing, Tinning, Electroplating; Electroless plating ; Paints.

UNIT-IV: Water technology:

Sources of water – Hardness of water – Estimation of hardness of water by EDTA method; Boiler troubles – sludge and scale formation, Boiler corrosion, caustic embrittlement, Priming and foaming; Softening of water by Lime – Soda Process, Zeolite Process, Ion – Exchange Process; Municipal water treatment; Desalination of sea water by Electrodialysis and Reverse osmosis methods.

UNIT-V :Chemistry of Engineering Materials& Advanced Engineering materials

Cement:- Manufacture of Portland cement, setting and hardening of cement, Deterioration of cement concrete.

Refractories: - Definition, Characteristics, classification, Properties and failure of refractories.

Solar Energy: - Construction and working of Photovoltaic cell, applications.

Solid State Materials: Crystal imperfections, Semi Conductors, Classification and chemistry of semi conductors: Intrinsic semiconductors; Extrinsic semiconductors; Defect semiconductors, Compound Semiconductors and Organic Semiconductors.

Liquid Crystals: - Definition – Classification with examples – Applications

Text Books:

1. Engineering Chemistry by Jain and Jain, Dhanpat Rai publishing co.
2. Engineering Chemistry by Willy India Pvt Ltd.
3. Engineering Chemistry by Dr.K.Anji Reddy and Dr.M.S.R.Reddy; Silicon Publications.

Reference Books:

1. Engineering Chemistry by Shikha Agarwal; Cambridge University Press,2015 edition.
2. A text of Engineering Chemistry by S.S.Dara; S.Chand & Co Ltd.

ENGINEERING DRAWING
(Common to CSE, ECE & IT)

Lecture	: 1 Period.	Int.Marks	: 30
Practice	: 3 Periods.	Ext. Marks	: 70
Exam	: 3Hrs.	Credits	: 3

Course Objectives:

1. To highlight the significance of universal language of engineers.
2. To impart basic knowledge and skills required to prepare engineering drawings.
3. To impart knowledge and skills required to draw projections of solids in different contexts.
4. To visualize and represent the pictorial views with proper dimensioning and scaling.

Course Outcomes:

Students will be able to

1. Apply principles of drawing to represent dimensions of an object.
2. Construct polygons and engineering curves.
3. Draw projections of points, lines, planes and solids.
4. Represent the object in 3D view through isometric views.
5. Convert the isometric view to orthographic view and vice versa.

SYLLABUS

UNIT I

Polygons: Constructing regular polygons by general methods, inscribing and describing polygons on circles.

Curves: Parabola, Ellipse and Hyperbola by general methods, cycloids, involutes, tangents & normals for the curves.

UNIT II

Orthographic Projections: Horizontal plane, vertical plane, profile plane, importance of reference lines, projections of points in various quadrants, projections of lines, lines parallel either to one of the reference planes (HP, VP or PP)

Projections of straight lines inclined to both the planes, determination of true lengths, angle of inclination and traces- HT, VT

UNIT III

Projections of planes: regular planes perpendicular/parallel to one plane and inclined to the other reference plane; inclined to both the reference planes.

UNIT IV

Projections of Solids – Prisms, Pyramids, Cones and Cylinders with the axis inclined to one of the planes.

UNIT V

Conversion of isometric views to orthographic views; Conversion of orthographic views to isometric views.

Text Books:

1. Engineering Drawing by N.D. Butt, Chariot Publications.
2. Engineering Drawing by Agarwal & Agarwal, Tata McGraw Hill Publishers

Reference Books:

1. Engineering Drawing by K.L.Narayana & P. Kannaiah, Scitech Publishers.
2. Engineering Graphics for Degree by K.C. John, PHI Publishers.
3. Engineering Graphics by P. Varghese, McGrawHill Publishers.
4. Engineering Drawing + AutoCad – K Venugopal, V. Prabhu Raja, New Age

OBJECT-ORIENTED PROGRAMMING THROUGH C++
(Common to CSE & IT)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. This course is designed to provide a comprehensive study of the C++ programming language. students are able to write efficient, maintainable and portable code to design algorithmic solutions to problems.
2. Expertise in object oriented principles and their implementation in C++
3. To acquire an understanding of basic object-oriented concepts and the issues involved in effective class design.
4. To Write C++ programs that use arrays, structures, pointers, object oriented concepts such as information hiding, constructors, destructors, inheritance.

Course Outcomes:

1. Write, compile and debug programs in C++ language. Use different data types in a computer program.
2. Design programs involving decision structures, loops and functions.
3. Explain classes and abstract classes and objects, abstraction and encapsulation, inheritance, polymorphism, constructors, access control and overloading.
4. Solve a given application problem by going through the basic steps of program specifications, analysis, design, implementation and testing within the context of the object-oriented paradigm.

SYLLABUS

UNIT-I: Introduction to C++, Classes and Objects.

Difference between C and C++, Disadvantage of Conventional Programming, Basic Concepts of Object Oriented Programming, Advantage of OOP, Object Oriented Languages, Functions in C++, Operators in C++. **Classes and Objects:** Declaring Objects, Access Specifiers and their Scope, Static data members, static member functions, arrays of objects, local classes, Nested classes.

UNIT-II: Constructors, Destructors and Operator Overloading.

Constructors and Destructors: Introduction- Constructors and Destructor- types of constructors, Constructors with default Arguments, Dynamic initialization of objects, Dynamic constructors. **Operator Overloading** Introduction, Overloading Unary Operators and Binary Operators, Overloading Unary Operators and Binary Operators using friend function, Overloading Assignment Operator (=), Overloading insertion(<<) and extraction(>>) operators, Manipulation of Strings using Operators, Rules for Overloading Operators, Type Conversions.

UNIT-III: Inheritance , Pointers, Virtual Functions and Polymorphism.

Inheritance: Reusability, Types of Inheritance, Virtual Base Classes, Abstract Classes, Advantages of Inheritance, Disadvantages of Inheritance, and constructors in derived classes. **Pointers Introduction:** Pointers to Objects, “this” Pointer, Pointers to Derived Classes, including Polymorphisms and Virtual Functions, Rules for Virtual Functions, pure virtual functions.

UNIT-IV: Manipulating Strings, Managing console I/O operations and Exception Handling.

Strings: Creating String Objects, Manipulating String Objects, Relational operations, String Characteristics, Accessing Characters in Strings. **C++ Stream** Classes, Unformatted I/O operations, Formatted I/O operations, managing output with Manipulators, **Exception Handling:** Principles of Exception Handling, Exception Handling Mechanism, throwing and catching Mechanism.

UNIT-V: Generic Programming with Templates, Standard Template Library and Files.

Generic Programming with Templates, Need for Templates, Definition of class Templates, Normal Function Templates, Over Loading of Template Function-Bubble Sort Using Function Templates, Difference between Templates and Macros, Overview of **Standard Template Library**, STL Programming Model, Containers, Algorithms, Iterators, Vectors, Lists, Maps. **FILES:** Introduction, File Stream Classes, File Operations, File Pointers and Manipulators, Sequential Access Files, Random File Access Operation, Detecting End-of File, Command-Line Arguments.

Text Books:

1. A complete Guide to programming in C++, Ulla Kirch-Prinz, Peter Prinz, Jones and Bartlett Publishers (2002).
2. Programming in C++, 2nd Edition, Ashok N Kamthane, and Pearson.
3. Object Oriented Programming C++, 4th Edition, Joyce Farrell, Cengage Learning.

Reference Books:

1. C++ Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE
2. The Complete Reference C++, 4th Edition, Herbert Schildt, TMH.
3. Programming solving with C++, 9th Edition Walter Savitch.
4. Object Oriented Programming in C++, 4th Edition, Robert Lafore

**DATA STRUCTURES
(For ECE)**

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To be familiar with basic techniques handling problems with Data structures
2. Solve problems using data structures such as linear lists, stacks, queues, hash tables
3. Create and traverse different types of trees and graphs.
4. To practice different searching algorithms.

Course Outcomes:

Students will be able to

1. Apply advanced data structure strategies for exploring complex data structures.
2. Compare and contrast various data structures and design techniques in the area of Performance.
3. Implement all data structures like stacks, queues, trees, lists and graphs and compare their performance and tradeoffs.
4. Implement different operations on trees.
5. Apply graphs to real time applications.
6. Perform sorting using different algorithms.

SYLLABUS

UNIT-I

Arrays and Structures

Array as an Abstract Data Type, Polynomial Abstract Data Type, Introduction to Sparse Matrix, Sparse Matrix Abstract Data Type, Representation of Multidimensional Arrays, Structures and Unions, Internal Implementation of Structures, Self-Referential Structures.

Recursion, Simple Searching and Sorting Techniques

Recursive functions, Introduction to Searching, Sequential Search, Binary Search, Interpolation Search, Selection Sort, Bubble Sort, Insertion Sort, Quick Sort, Introduction to Merge Sort, Iterative Merge Sort, Recursive Merge Sort, Heap sort.

UNIT-II

Stacks and Queues

Stack Abstract Data Type, Queue Abstract Data Type, Stacks and Queues using arrays, , Introduction to Evaluation of Expressions, Evaluating Postfix Expressions, Infix to Postfix and Prefix conversion, Multiple Stacks and Queues, Circular Queues using arrays.

UNIT-III

Linked Lists

Pointers, Dynamically Allocated Storage using pointers, Singly Linked Lists, Dynamically Linked Stacks and Queues, Polynomials, Representing Polynomials as Singly Linked Lists, Adding Polynomials, Erasing Polynomials, Polynomials as Circularly Linked Lists, Additional List Operations, Operations for Singly Linked Lists, Operations for Doubly Linked Lists, Radix Sort.

UNIT-IV

Trees

Representation of Trees, Binary Trees Abstract Data Type, Properties of Binary Trees, Binary Tree Representations, Binary Tree Traversals, Additional Binary Tree Operations, Threaded Binary Trees, Heap Abstract Data Type, Priority Queues, Insertion into a max heap, Deletion from a max heap, Heap Sort, Introduction to Binary Search Trees, Searching a Binary Search Tree, Inserting an Element into a Binary Search Tree, Deleting an Element from a Binary Search Tree, Height of a Binary Search Tree, Counting Binary Trees.

UNIT-V

Graphs

Graph Abstract Data Type, Definitions, Graph Representations, Elementary Graph Operations, Depth First Search, Breadth First Search, Connected Components, Spanning Trees, Minimum Cost Spanning Trees, Prim's and Kruskal's Algorithms, Shortest Paths and Transitive Closure, Single Source All Destination - Dijkstra's Algorithm, All Pairs Shortest Paths - Floyd's Algorithm, Transitive Closure using Warshall's Algorithm.

Text Books:

1. Fundamentals of Data Structures in C, 2nd edition, Horowitz, Sahni and Anderson-Freed, Universities Press, 2008.

Reference Books:

1. Data Structures using C by Aaron M. Tenenbaum, Y. Langsam and M.J. Augenstein, Pearson Education, 2009.
2. Data Structures with C by Seymour Lipschutz, Schaum Outline series, 2010.
3. Data Structures using C by R. Krishna Moorthy G. Indirani Kumaravel, TMH, New Delhi, 2008.

ELEMENTS OF ELECTRONICS ENGINEERING
(Common to CSE & IT)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To give the exposure to the students on semiconductor physics of the intrinsic and extrinsic semiconductors and basics of P-N junction diode.
2. To give the exposure to the students on the basics of special purpose diodes like Zener diode, photo diode, LED, and tunnel diode and rectifier circuits using diodes.
3. To give the exposure to the students on basics of BJT, and transistor circuit configurations.
4. To give exposure to the students on transistor biasing and thermal stabilization.
5. To give exposure to the students on the basics of JFET, MOSFET and FET biasing.

Course Outcomes:

After completion of the course the students will be able to

1. Understand the basic concepts of transport of charge carriers in semiconductors, drift and diffusion currents, physical structure, operation, V-I characteristics of semiconductor diode.
2. Understand the basic concepts of special types of diodes like Zener Diode, LED, Photo Diode and tunnel diode, rectifier circuits with and without filters.
3. Understand the physical structure, operation, input and output characteristics of BJT in CE, CB, CC circuit configurations.
4. Understand the basic concepts of transistor biasing and thermal stabilization.
5. Understand the physical structure, operation, characteristics and circuit models of JFET's and MOSFET's.

SYLLABUS

UNIT I: Semiconductors and P-N junction diode:

Intrinsic and extrinsic semiconductors, charge densities in semiconductors, Drift and Diffusion currents, Hall Effect, Mass action law. Basic operation and V-I Characteristics of semiconductor diode, Diode current equation, Avalanche breakdown and Zener breakdown phenomenon.

UNIT II: Special Diodes and Diode Rectifiers:

Zener Diode, LED, Photo Diode and tunnel diode, Half wave and Full wave Rectifiers- with and without filters, Bridge Rectifier, Expressions - Ripple factor, Efficiency, Capacitor filters

UNIT III: Bipolar Junction Transistor:

Introduction, construction, basic operation of npn and pnp transistors, Transistor circuit configurations- CE, CB, CC- Input and output Characteristics in various configurations. h-parameter model for transistor amplifier. (*Introductory Treatment only*).

UNIT IV: Transistor Biasing and Thermal Stabilization:

Transistor Biasing, Thermal runaway, stabilization, Different methods of Biasing-Fixed Bias, collector feedback bias, self-bias, Bias compensation.

UNIT V: Field Effect Transistors: Junction field Effect Transistors (JFET)- JFET characteristics, JFET Parameters, Small Signal model of FET, Depletion and Enhancement type MOSFET's.

Text Books:

1. Electronic Device and Circuits by SanjeevGuptha, Dhanpatrai&Co.Pvt.Ltd.
2. Electronic Device and Circuits by K.Satya Prasad, VGS.

Reference Books:

1. Integrated Electronics- Millman&Halkias, TMH.

ELEMENTS OF ELECTRICAL ENGINEERING (For ECE)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To learn the fundamentals of electrical and magnetic circuits
2. To understand the principle of operation and construction details of DC machines.
3. To understand the principle of operation of Transformers.
4. To understand the principle of operation and construction details of three phase Induction motor
5. To understand the principle of operation of synchronous generator and different measuring instruments

Course Outcomes:

1. Able to understand the basics of Magnetic Circuits and Kirchhoff's laws.
2. Able to understand the operation of DC Machines and to conduct different Tests
3. Able to analyze the Performance of Transformers.
4. Able to explain the operation of three phase induction motor.
5. Able to analyze the operation of alternator and different measuring instruments.

SYLLABUS

UNIT I: Electrical and Magnetic Circuits:

Basic definitions, Types of network elements, Ohm's Law, Kirchhoff's Laws, Series and parallel Circuits and star-delta and delta-star transformations-simple problems. Magnetic flux, MMF, Reluctance, Faraday's laws, Lenz's law, statically induced EMF, dynamically induced EMF.

UNIT – II: DC Machines:

Principle of operation of DC Generator - EMF equation – Construction-Types of DC generator- OCC of DC Generator-DC motor types - Torque equation –Losses-Efficiency-speed control methods- applications

UNIT – III: Transformers:

Principle of operation of single phase transformer - EMF equation - equivalent circuit –losses - efficiency and regulation- Open circuit and Short circuit tests.

UNIT – IV: Induction Motors:

Construction-Principle of operation of induction motor-slip- rotor frequency, slip - torque characteristics - Power flow diagram-Efficiency-Applications

UNIT – V: Synchronous Generator and Measuring Instruments:

Construction-Principle of operation of alternator-EMF equation of alternator- Regulation by Synchronous impedance method.

Classification –Deflecting, controlling, damping Torque, ammeter, voltmeter, wattmeter, MI, MC instruments-Energy meter

Text Books:

1. Electrical Machinery By Dr.P.S Bhimbra,Khanna Publications, January 2011
2. Electrical Technology by Surinder Pal Bali, Pearson Publications. January 1, 2013
3. Principles of Electrical Engineering, V.K mehta, Rohit Mehta, S. Chand Publications. Revised Edition 2012
4. A Course in Electronic Measurements and Instrumentation by A.K. Sawhney Dhanpat Rai & Co. (P) Limited; 2014 edition (2015)

Reference Books:

1. Basic Electrical Engineering, by J.B Gupta, K Kataria and Sons; Reprint 2013 edition.
2. Fundamentals of Electrical Engineering and Electronics, by Theraja B.L, S Chand Multicolor edition (1 December 2006)
3. A Course in Electronics and Electrical Measurements and Instrumentation by J.B. Gupta S K Kataria and Sons; Reprint 2013 edition (2013)

ENGINEERING CHEMISTRY LAB
(Common to CSE,ECE& IT)

Lab : 3 Periods
Exam : 3Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To investigate and understand Physical behavior in the laboratory using scientific reasoning and logic and interpret the result of simple experiments and demonstration of chemical Principle and also evaluate the impact of chemical discoveries on how we view the world.
2. Effectively communicate experimental results and solutions to application problems through oral and written reports.
3. Understand the basic concepts, definitions, characteristics and phenomena.
4. Recognize the classical ideas and chemical phenomena and also define and analyse the concepts.

Course Outcomes:

1. An understanding of Professional and develop confidence on recent trends.
2. Able to gain technical knowledge of measuring, operating and testing of chemical instruments and equipments.
3. Acquire ability to apply knowledge of chemistry.
4. Exposed to the real time working environment.
5. Demonstrate the ability to learn Principles, design and conduct experiments.
6. Ability to work on laboratory and multidisciplinary tasks.

List of Experiments

Introduction to chemistry Laboratory

1. Estimation of HCl using standard Sodium Hydroxide.
2. Determination of total hardness of water by EDTA method.
3. Estimation of Ferrous Iron by KMnO_4 .
4. Estimation of oxalic acid by KMnO_4
5. Estimation of Mohr's salt by $\text{K}_2\text{Cr}_2\text{O}_7$
6. Estimation of Dissolved oxygen by Winkler's method.
7. Determination of pH by pH meter and universal indicator method.
8. Conductometric titration of strong acid Vs strong base
9. Conductometric titration of strong acid Vs weak base.
10. Potentiometric titration of strong acid Vs strong base
11. Potentiometric titration of strong acid Vs weak base
12. Preparation of Phenol formaldehyde resin.
13. Determination of saponification value of oils
14. Determination of pour and cloud points of lubricating oil.
15. Determination Acid value of oil.

Text Books:

1. Engineering Chemistry Lab Manual Prepared by Chemistry Faculty of S.R.K.R.Engineering College.

Reference Books:

1. Laboratory manual on Engineering Chemistry by Dr.Sudha Rani ;Dhanpat Rai Publishing Company.
2. Engineering Chemistry Laboratory manual – I & II by Dr. K.Anji Reddy; Tulip Publications.

ENGLISH COMMUNICATION SKILLS LAB- II
(Common to All Branches)

Lab : 3 Periods
Exam : 3Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To enable the students to learn demonstratively the communication skills of listening, speaking, reading and writing.
2. To enable the students participate in group interactions.
3. To improve the presentation skills of the students.
4. To help the students gain their confidence in attending the interviews.

Course Outcomes:

1. A study of the communicative items in the laboratory will help the students become successful in the competitive world.
2. Students enhance their presentation skills.
3. Students participate in group discussions and improve their team skills.
4. Students confidently face the interviews.

SYLLABUS

- ❖ Debating & Practice.
- ❖ Group Discussions & Practice.
- ❖ Presentation Skills & Practice
- ❖ Interview Skills & Practice
- ❖ Email
- ❖ Curriculum Vitae & Practice
- ❖ Idiomatic Expressions
- ❖ Common Errors in English & Practice

LAB MANUAL:

1. **‘INTERACT: English Lab Manual for Undergraduate Students’** Published by Orient Blackswan Pvt Ltd.

Reference Books:

1. Strengthen your communication skills by Dr M Hari Prasad, Dr Salivendra Raju and Dr G Suvarna Lakshmi, Maruti Publications.
2. English for Professionals by Prof Eliah, B.S Publications, Hyderabad.
3. Unlock, Listening and speaking skills 2, Cambridge University Press
4. Spring Board to Success, Orient BlackSwan
5. A Practical Course in effective english speaking skills, PHI
6. Word power made handy, Dr shalini verma, Schand Company
7. Let us hear them speak, Jayashree Mohanraj, Sage texts
8. Professional Communication, Aruna Koneru, Mc Grawhill Education
9. Cornerstone, Developing soft skills, Pearson Education

OBJECT ORIENTED PROGRAMMING LAB
(Common to CSE & IT)

Lab : 3 Periods

Exam : 3Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

1. To strengthen their problem solving ability by applying the characteristics of an object-oriented approach.
2. To introduce object oriented concepts in C++ and Java.

Course Outcomes:

1. Explain what constitutes an object-oriented approach to programming and identify potential benefits of object-oriented programming over other approaches.
2. Apply an object-oriented approach to developing applications of varying complexities.

LIST OF PROGRAMS

1. Write a Programme that computes the simple interest and compound interest payable on principal amount (in Rs.) of loan borrowed by the customer from a bank for a given period of time (in years) at specific rate of interest. Further determine whether the bank will benefit by charging simple interest or compound interest
2. Write a Programme to calculate the fare for the passengers traveling in a bus. When a Passenger enters the bus, the conductor asks "What distance will you travel?" On knowing distance from passenger (as an approximate integer), the conductor mentions the fare to the passenger according to following criteria.
3. Write a C++ Program to illustrate Enumeration and Function Overloading
4. Write a C++ Program to illustrate Scope and Storage class
5. Implementation of ADT such as Stack and Queues
6. Write a C++ Program to illustrate the use of Constructors and Destructors and Constructor Overloading
7. Write a Program to illustrate Static member and methods
8. Write a Program to illustrate Bit fields
9. Write a Program to overload as binary operator, friend and member function
10. Write a Program to overload unary operator in Postfix and Prefix form as member and friend function
11. Write a C++ Program to illustrate Iterators and Containers
12. Write a C++ Program to illustrate function templates
13. Write a C++ Program to illustrate template class
14. Write C++ Programs and incorporating various forms of Inheritance
15. Write a C++ Program to illustrate Virtual functions
16. To write a C++ program to find the sum for the given variables using function with default arguments.
17. To write a C++ program to find the value of a number raised to its power that demonstrates a function using call by value.
18. To write a C++ program and to implement the concept of Call by Address
19. To write a program in C++ to prepare a student Record using class and object

20. To implement the concept of unary operator overloading by creating a C++ program.
21. Write a C++ program for swapping two values using function templates
22. Write a C++ program to implement a file handling concept using sequential access.

Reference Books:

1. C++ Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE
2. Programming in C++, 2nd Edition, Ashok N Kamthane, and Pearson.
3. Programming solving with C++, 9th Edition Walter Savitch.

**ENGINEERING WORKSHOP & IT WORKSHOP
(For ECE)**

Lab : 3 Periods

Exam : 3Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

PART-A ENGINEERING WORKSHOP

Course Objective:

To impart hands-on practice on basic engineering trades and skills.

Course Outcomes:

Student will be able to

1. Use various tools to prepare basic carpentry and fitting joints.
2. Prepare jobs of various shapes using black smithy.
3. Make basic house wire connections.
4. Fabricate simple components using tin smithy.

SYLLABUS

Carpentry	Fitting
1. T-Lap Joint 2. Cross Lap Joint 3. Dovetail Joint 4. Mortise and Tenon Joint	1. Vee Fit 2. Square Fit 3. Half Round Fit 4. Dovetail Fit
Black Smithy	Tin Smithy
1. Round rod to Square 2. S-Hook 3. Round Rod to Flat Ring 4. Round Rod to Square headed bolt	1. Taper Tray 2. Square Box without lid 3. Open Scoop 4. Funnel
House Wiring	
1. Parallel / Series Connection of three bulbs 2. Stair Case wiring 3. Florescent Lamp Fitting 4. Measurement of Earth Resistance	

Note: At least two exercises to be done from each trade.

Reference:

1. Elements of workshop technology, Vol.1 by S. K. and H. K. Choudary.

PART B: IT WORKSHOP:

Course Objectives:

1. Understand the basic components and peripherals of a computer.
2. To become familiar in configuring a system.
3. Learn the usage of productivity tools.
4. Acquire knowledge about the netiquette and cyber hygiene.
5. Get hands on experience in trouble shooting a system

Course Outcomes:

At the end of the course the students can able to

1. Assemble and disassemble the systems
2. Use the Microsoft office tools
3. Install various software
4. Know about various search engines
5. Trouble shoot various Hardware and Software problems
6. Use MATLAB and LATEX software

LIST OF EXERCISES

1. System Assembling, Disassembling and identification of Parts / Peripherals
2. Operating System Installation-Install Operating Systems like Windows, Linux along with necessary Device Drivers.
3. MS-Office / Open Office
 - a) Word - Formatting, Page Borders, Reviewing, Equations, symbols.
 - b) Spread Sheet - organize data, usage of formula, graphs, charts.
 - c) Power point - features of power point, guidelines for preparing an effective presentation.
 - d) Access- creation of database, validate data.
4. Network Configuration & Software Installation-Configuring TCP/IP, proxy and firewall settings. Installing application software, system software & tools.
5. Internet and World Wide Web-Search Engines, Types of search engines, netiquette, cyber hygiene.
6. Trouble Shooting-Hardware trouble shooting, Software trouble shooting.
7. MATLAB- basic commands, subroutines, graph plotting.
8. LATEX-basic formatting, handling equations and images.

Reference Books:

1. Computer Hardware, Installation, Interfacing, Troubleshooting and Maintenance, K.L. James, Eastern Economy Edition.
2. Microsoft Office 2007: Introductory Concepts and Techniques, Windows XP Edition By Gary B. Shelly, Misty E. Vermaat and Thomas J. Cashman (2007, Paperback).
3. LATEX- User's Guide and Reference manual, Leslie Lamport, Pearson, LPE, 2/e.
4. Getting Started with MATLAB: A Quick Introduction for Scientists and Engineers, Rudraprathap, Oxford University Press, 2002.
5. Scott Mueller's Upgrading and Repairing PCs, 18/e, Scott. Mueller, QUE, Pearson, 2008
6. The Complete Computer upgrade and repair book, 3/e, Cheryl A Schmidt, Dreamtech.
7. Comdex Information Technology course tool kit Vikas Gupta, WILEY Dreamtech.
8. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education.

INNER ENGINEERING
(Common to CSE, ECE & IT)

Practice : 2 Periods

Course Objectives:

Student should learn:

1. Human values and tools to lead a happy, stress-free life.
2. Yoga asanas, Pranayama, Sudarshan Kriya & Meditation
3. At-least two creative arts out of photography, sketching, craft-making, singing, clay molding, upcycling etc.
4. Concentration pranayama, Ego bursting process.
5. To take up responsibility for society and teach classes of their choice to school children.
6. About good food habits for good health.

Course Outcomes:

Student should be capable of

1. To improve his concentration levels and improve his public speaking abilities.
2. To balance his academic and non-academic activities (Work Life Balance).
3. To widen his vision and increase his breadth of perspective in his journey of 4 years.
4. To improve his communications skills, leadership, teamwork and decision-making abilities.
5. To inculcate creativity & innovation, planning & organizing as part of their life.
6. Taking responsibility for themselves and people around them.
7. To make their journey more fun and enjoyable.

SYLLABUS

Unit-I

YES!+ Workshop:

Yoga Postures – Seven Layers To our Existence – Puzzles – Sources Of Energy – Live in the present Moment – Importance of Breath – Ujjai Breath – Pranayama – Sudarshana Kriya.

Unit-II

YES!+ Workshop:

Yoga Postures (Suryanamaskars) – Giving 100% in everything – Time management – Happiness point – Opposite Values – Pranayama – Sudarshan kriya

Unit-III

YES!+ Workshop:

Yoga Postures – Knowledge points (Acceptance, opinions discretion and handling mistakes) – Eye Gazing Process – Dance – Life Story process – Sudarshana Kriya (short) – Eternal life – Ego Bursting – Relationships – Parents – Studies – Compliments/Praising process.

Unit-IV

Creative Arts:

Photography – Sketching – Handy-crafts – Clay molding – Singing – Upcycling – Communing with nature – Creative writing.

Unit -V

Service:

Leadership in action – Contributing to society – Take up Responsibility –Empowerment – Public Speaking – Art of Teaching.

REFERENCE BOOKS:

1. Discourse on Patanjali Yoga Sutras by H.H Sri SriRavishankar
2. Human values and professional ethics byRRGaur,RSangal,GPBagaria
3. The Art of Photography by AL Judge
4. Hand made in India : Crafts of India by Ranjan Aditi

(Note: It is an uncredited course. It will not be included in the Grade Memo / Certificate. The Certificate will be issued based on the performance and attendance. This course attendance will be counted in the semester overall attendance.)



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

II/IV B.TECH

(With effect from **2017-2018** Admitted Batch onwards)

COMPUTER SCIENCE AND ENGINEERING

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B17 CS 2101	Data Structures	3	3	1	--	4	30	70	100
B17 BS 2102	Probability, Statistics and Queuing Theory	3	3	1	--	4	30	70	100
B17 BS 2103	Discrete Mathematical Structures	3	3	1	--	4	30	70	100
B17 CS 2102	Computer Graphics	3	3	1	--	4	30	70	100
B17 CS 2103	Digital Logic Design	3	3	1	--	4	30	70	100
B17 CS 2104	Data Analysis and Visualization using Python	3	3	1	--	4	30	70	100
B17 CS 2105	Data Structures Lab.	2	--	--	3	3	50	50	100
B17 CS 2106	Data Analysis and Visualization using Python Lab	2	--	--	3	3	50	50	100
B17CS 2107	Industry oriented Training : R Programming Lab	1	--	--	2	2	50	---	50
B17 BS 2107	English Proficiency-I	--	1	1	--	2	--	--	--
Total		23	19	7	8	34	330	520	850

DATA STRUCTURES

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. Be familiar with basic techniques of algorithm analysis
2. Master the implementation of data structures like stacks, queues, linked lists, binary trees, graphs.
3. Be familiar with basic techniques for algorithm development like recursion.
4. Be familiar with several sub-quadratic sorting algorithms including quick sort, merge sort and heap sort
5. Master analyzing problems and writing program solutions to problems using the above techniques.

Course Outcomes:

1. Describe how arrays, records, linked structures, stacks, queues, trees, and graphs are represented in memory and used by algorithms.
2. Describe common applications for arrays, records, linked structures, stacks, queues, trees, and graphs. Write programs that use arrays, records, linked structures, stacks, queues, trees, and graphs. Demonstrate different methods for traversing trees [ABET (a)].
3. Compare alternative implementations of data structures with respect to performance [ABET (a, b, c)].
4. Compare and contrast the benefits of dynamic and static data structures implementations [ABET (a, b, c)].
5. Describe the concept of recursion, give examples of its use, describe how it can be implemented using a stack [ABET (a, c)].
6. Discuss the computational efficiency of the principal algorithms for sorting, searching.

SYLLABUS

UNIT -I Basic Concepts, Arrays, Structures and Recursion:

System Life Cycle, Algorithm Specification, Data Abstraction, Performance Analysis, Space Complexity, Time Complexity, Asymptotic Notation, Comparing Time Complexities. Array as an Abstract Data Type, Polynomial Abstract Data Type, Representation of Multidimensional Arrays, Structures and Unions, Internal Implementation of Structures, Self-Referential Structures, Recursive functions.

UNIT –II Simple Searching and Sorting Techniques:

Introduction to Searching, Sequential Search, Binary Search, Interpolation Search, Selection Sort, Bubble Sort, Insertion Sort, Quick Sort, Introduction to Merge Sort, Iterative Merge Sort, Recursive Merge Sort, Shell Sort, Radix Sort.

UNIT -III Stacks, Queues and Linked Lists:

Stack Abstract Data Type, Queue Abstract Data Type, Stacks and Queues using arrays, Introduction to Evaluation of Expressions, Evaluating Postfix Expressions, Infix to Postfix and Prefix conversion, Circular Queues using arrays. Pointers, Dynamically Allocated Storage using pointers, Singly Linked Lists, Dynamically Linked Stacks and Queues, Polynomials, Representing Polynomials as Singly Linked Lists, Adding Polynomials, Erasing Polynomials.

UNIT -IV Trees: Representation of Trees, Binary Trees Abstract Data Type, Properties of Binary Trees, Binary Tree Representations, Binary Tree Traversals, Additional Binary Tree Operations, Threaded Binary Trees, Heap Abstract Data Type, Insertion into a max heap, Deletion from a max heap, Heap Sort, Introduction to Binary Search Trees, Searching a Binary Search Tree, Inserting an Element into a Binary Search Tree, Deleting an Element from a Binary Search Tree, Height of a Binary Search Tree, Counting Binary Trees.

UNIT -V Graphs: Graph Abstract Data Type, Definitions, Graph Representations, Elementary Graph Operations, Depth First Search, Breadth First Search, Connected Components, Spanning Trees, Minimum Cost Spanning Trees, Prim's and Kruskal's Algorithms, Shortest Paths and Transitive Closure, Single Source All Destination - Dijkstra's Algorithm, All Pairs Shortest Paths - Floyd's Algorithm, Transitive Closure using Warshall's Algorithm.

Text Books:

1. Fundamentals of Data Structures in C, 2nd edition, Horowitz, Sahani and Anderson-Freed, Universities Press, 2008.

Reference Books:

1. Data Structures using C by Aaron M. Tenenbaum, Y. Langsam and M.J. Augenstein, Pearson Education, 2009.
2. Data Structures with C by Seymour Lipschutz, Schaum Outline series, 2010.
3. Data Structures using C by R. Krishna Moorthy, G. Indirani Kumaravel, TMH, New Delhi, 2008.

PROBABILITY, STATISTICS & QUEUING THEORY

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To illustrate the concept of a random variable, generating functions and their properties.
2. To learn different probability functions and analyse various statistical measures of a few discrete/continuous distributions.
3. To understand and compute the correlation coefficient, and estimation techniques from regression lines.
4. To fit a linear or nonlinear curves using method of least squares.
5. To develop a framework for testing of hypothesis in giving inferences about Population Parameters.
6. To study Queuing models and their Characteristics.

Course Outcomes:

At the end of the course a student will be able to

1. Identify the random variable as discrete/continuous and analyse it.
2. Predict the distribution suitable for the given data from its moments.
3. Measure the intensity of association between the variables.
4. Fit a best suitable Curve for the given data.
5. Decide the test applicable for giving inference about Population Parameter based on Sample statistic.
6. Make business decisions about the resources needed to provide a service in day-to-day life applications including telecommunication, traffic engineering, computing and the design of factories, shops, offices and hospitals.

SYLLABUS**UNIT -I**

Random Variables and Probability functions: Review on basic concepts of Probability (no questions will be set on review), Definition of a random variable, Distribution function, Properties of Distribution Function, Discrete Random Variable, Probability Mass Function, Discrete Distribution Function, Continuous Random Variable, Probability Density Function, Continuous Distribution Function.

Mathematical Expectation: Mathematical Expectation of a Random Variable, Expected Value of function of a Random Variable, Addition Theorem and Multiplication Theorem of Expectation(**without proofs**), Statistical Measures like Mean, Variance, Moments and Covariance in terms of Expectations.

Generating functions: Moment generating Function, Characteristic Function and Probability generating Function of a Random Variable.

UNIT II

Discrete Distributions: Binomial distribution and Poisson distribution - Definition, Mean, Variance, moments, m.g.f., Characteristic function, p.g.f., Fitting of distributions.

Continuous Distributions: Normal Distribution - Definition, Standard Normal Variate, Mean, Variance, m.g.f., Characteristic function Applications of Normal Distribution, Importance of Normal distribution. Exponential Distribution, Definition, Mean, Variance and Memory less property of Exponential distribution.

UNIT III

Curve fitting: Method of least Squares, fitting of a Straight line, Fitting of a Parabola.

Correlation: Definition, Karl Pearson's Coefficient of Correlation, Limits for correlation coefficient, Rank Correlation, Spearman's formula for rank correlation coefficient.

Regression Analysis: Regression Lines, Regression Coefficients and their properties (without proofs)

UNIT IV

Sampling Theory: Sample, population, statistic, parameter, Sampling distribution, standard error, point and interval estimation. Testing of Hypothesis: Formulation of Null hypothesis, Alternative hypothesis, Critical region, level of significance, Errors in sampling- Type-I-error, Type-II-error, One-tailed and Two-tailed tests.

Large Sample Theory: Test of significance of single sample proportion, Test of significance for difference of proportions.

Small Sample Theory: Degrees of freedom, Student's-t-distribution: definition, t-test for single mean, t-test for difference of means, Paired t-test for difference of means.

F-distribution: definition, F-test for equality of two population variances. Chi-square distribution: definition, Chi-square test for goodness of fit, Chi-square test for Population Variance.

UNIT V

Queueing Theory: Queue description, Birth and Death Process, Distribution of Inter-arrival times, Distribution of service times, Kendall's representation of a queueing model, Operating characteristics of a queueing model, steady-state solutions of $\{M/M/1: \infty/FCFS\}$ Model and $\{M/M/1; N/FCFS\}$ Model.

Text Books:

1. Fundamentals of Mathematical Statistics by S.C.Gupta and V.K.Kapoor, Sultan Chand & Sons Publishers.
2. Probability, Statistics and Random Processes by T.Veerarajan, Tata Mc Graw Hill Pub.

Reference Books:

1. Probability & Statistics with Reliability, Queueing and Computer Applications by Kishore.S.Trivedi, Prentice Hall of India, 1999.
2. Probability and statistics for Engineers, Miller and Freund, 7th edition, Prentice-Hall India.
3. Probability and statistics for Engineers and Scientists by Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying Ye, Eighth edition, Pearson Education.

DISCRETE MATHEMATICAL STRUCTURES

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To understand the mathematical arguments using logical connectives and quantifiers and verify the validity of arguments using propositional, predicate logic and truth tables.
2. To understand various types of relations and discuss various properties of the relations.
3. To know about the concepts of counting techniques
4. To know how to solve the recurrence relations
5. To understand the concepts in graphs & also in Number theory

Course Outcomes:

At the end of the course students will be able to :

1. Rewrite the mathematical arguments using logical connectives and quantifiers and verify the validity of the arguments using propositional and predicate logic.
2. Identify and give examples of various types of relations and describe various properties of relations.
3. Solve different counting problems.
4. Solve the recurrence relations which occur in many fields.
5. Utilize the concepts in graphs and Number theory in their fields.

SYLLABUS**UNIT -I: Mathematical Logic:**

Propositional Calculus: Statements and Notations, Connectives, Well Formed Formulas, Truth Tables, Tautologies, Equivalence of Formulas, Duality Law, Normal Forms, Theory of Inference for Statement Calculus, Consistency of Premises. Predicate Calculus: Predicative Logic, Statement Functions, Variables and Quantifiers, Free and Bound Variables, Inference Theory for Predicate Calculus.

UNIT -II:

Relations: Definition of Relation, Properties of Binary Relations, Relation Matrix and Digraph, Operations on Relations, Partition and Covering, Transitive Closure, Equivalence, Compatibility and Partial Ordering Relations, Hasse Diagrams.

Algebraic Structures: Algebraic Systems, Examples, General Properties, Semi Groups and Monoids, Homomorphism of Semi Groups and Monoids, Group, Subgroup, Abelian Group, Homomorphism, Isomorphism.

UNIT -III Combinatorics:

Basics of Counting, Permutations, Permutations with Repetitions, Circular Permutations, Restricted Permutations, Combinations, Restricted Combinations, Generating Functions of Permutations and Combinations, Binomial and Multinomial Coefficients, Binomial and Multinomial Theorems, The Principles of Inclusion–Exclusion, Pigeonhole Principle and its Application.

UNIT -IV: Recurrence Relations:

Generating Functions, Function of Sequences, Partial Fractions, Calculating Coefficient of Generating Functions, Recurrence Relations, Formulation as Recurrence Relations, Solving Recurrence Relations by Substitution and Generating Functions, Method of Characteristic Roots, Solving Inhomogeneous Recurrence Relations

UNIT -V:

Graph Theory: Basic Concepts of Graphs, Sub graphs, Matrix Representation of Graphs: Adjacency Matrices, Incidence Matrices, Isomorphic Graphs, Paths and Circuits, Eulerian and Hamiltonian Graphs, Multigraphs, Bipartite graphs, Planar Graphs, Euler's Formula.

Number Theory : Properties of Integers, Division theorem, Greatest Common Divisor, Euclidean Algorithm, Least Common Multiple, Testing for Prime Numbers, The Fundamental Theorem of Arithmetic, Modular Arithmetic (Fermat's Theorem and Euler's Theorem).

Text Books:

1. Discrete Mathematical Structures with Applications to Computer Science, J. P. Tremblay and P. Manohar, Tata McGraw Hill.
2. Discrete Mathematics for Computer Scientists and Mathematicians, J. L. Mott, A. Kandel, T.P. Baker, 2nd Edition, Prentice Hall of India.

Reference Books:

1. Elements of Discrete Mathematics-A Computer Oriented Approach, C. L. Liu and D.P. Mahopatra, 3rd Edition, Tata McGraw Hill.
2. Discrete Mathematics and its Applications with Combinatorics and Graph Theory, K. H. Rosen, 7th Edition, Tata McGraw Hill.
3. Discrete Mathematical Structures, Bernard Kolman, Robert C. Busby, Sharon Cutler Ross, PHI.
4. Discrete Mathematics, S. K. Chakraborty and B.K. Sarkar, Oxford, 2011.

COMPUTER GRAPHICS

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. Provide a comprehensive introduction to computer graphics leading to the ability to understand contemporary terminology, technology, progress and trends.
2. Design of algorithms for digitization of graphic primitives.
3. Apply computer graphics techniques for two-dimensional and three-dimensional transformations.
4. Visualize viewing transformations.
5. Demonstrate working of I/O devices.

Course Outcomes: At the end of the course the student will be able to

1. Summarize the application areas of computer graphics
2. Implement algorithms for scan converting graphic primitives in a graphic package.
3. Apply direct and indirect methods for two-dimensional transformations using matrices.
4. Construct three-dimensional geometric transformations using matrices.
5. Visualize two-dimensional viewing transformations
6. Produce views of three-dimensional scenes.
7. Visualize the working of I/O devices

SYLLABUS

UNIT-I Overview of Graphics Systems:

Applications of Computer Graphics-Graphical User Interfaces-Video Display Devices-Raster Scan Systems-Random Scan Systems-Graphics Monitors and Workstations -Input Devices-Logical Classification of Input Devices-Hard Copy Devices- Graphics Software-Overview of C-Graphics, Open GL and PHIGS.

UNIT-II Output Primitives and it attributes:

Points and Lines-Line Drawing Algorithms- Circle Generating Algorithms- Parallel Line Algorithms-Functions in C-Graphics for Output Primitives-Color and Gray Scale Levels - Boundary Fill Algorithm- Flood Fill Algorithm -Anti-aliasing Techniques.

UNIT-III Two-Dimensional Geometric Transformations:

Basic Transformations- Matrix Representations- Homogeneous Coordinates- Composite Transformations- Reflection- Shear- Transformations between Coordinate Systems- Affine Transformations- Raster Methods for Transformations.

UNIT-IV Two-Dimensional Viewing:

The Viewing Pipeline-Viewing Coordinate Reference Frame-Window-to-Viewport Coordinate Transformation-Clipping Operations-Point Clipping-Line Clipping-Polygon Clipping-Curve Clipping- Text and Exterior Clipping

UNIT-V Three-Dimensional Geometric Transformations and Viewing:

Translation- Rotation- Scaling- Reflection -Shear-Composite Transformations-Modeling and Coordinate Transformations-3D Display Methods- Spline Representations-Natural Cubic Spline - Bézier Curves and Surfaces-3D Viewing Pipeline- Viewing Coordinates- Projections- View Volumes- General Projection Transformations.

Case studies- Implementation of algorithms in C-graphics**Text Book:**

1. Computer Graphics C Version, Donald Hearn& M. Pauline Baker, Pearson Education, New Delhi, 2004

Reference Books:

1. Procedural Elements for Computer Graphics, David F. Rogers, Tata McGraw Hill Book Company, New Delhi, 2003
2. Computer Graphics: Principles & Practice in C, J.D. Foley, S.K. Feiner, A. Van Dam. F.H John, Pearson Education, 2004
3. Computer Graphics with Open-GL, Donald Hearn, M. Pauline Baker& Warren Carithers,4th Edition, 2011
4. Computer Graphics, Zhigang Xiang and Roy A. Plastock, McGraw-Hill Education, 2nd Edition, ,2015
5. Mathematical and computer programming techniques for computer graphics, Peter Comninos, Springer -Verlag, 2006.

DIGITAL LOGIC DESIGN

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To introduce the basic principles for design of combinational circuit and sequential circuits.
2. To learn simple digital circuits in preparation for computer engineering.
3. To introduce the Hardware Description Language

Course Outcomes:

At the end of course student can know about

1. An Ability to define different number systems, binary addition and subtraction, 2's complement representation and operations with his representation.
2. An Ability to understand different Boolean Algebra theorems and apply them for logic functions.
3. An Ability to design the Karnaugh map for a few variables and perform an algorithmic reduction of logic functions.
4. An Ability to define the following combinational circuits: multiplexer, de-multiplexers, encoders/decoders, comparators, arithmetic-logic units and to be able to build simple circuits.
5. An ability to understand asynchronous and synchronous sequential circuits like counters and registers.
6. An ability to understand memories like RAM and ROM, Programmable Logic Devices.

SYLLABUS**UNIT- I Binary Systems, Boolean Algebra and Logic Gates:**

Digital Systems -- Binary Numbers -- Number Base Conversions -- Octal and Hexadecimal Numbers -- Complements -- Signed Binary Numbers -- Binary Codes -- Binary Storage and Registers -- Binary Logic -- Basic Definitions -- Axiomatic Definition of Boolean Algebra -- Basic Theorems and Properties of Boolean Algebra -- Boolean Functions -- Canonical and Standard Forms -- Other Logic Operations -- Digital Logic Gates -- Integrated Circuits.

UNIT- II Gate-Level Minimization, Combinational Logic Design:

The Map Method -- Four-Variable Map -- Five-Variable Map -- Product of Sums Simplification -- Don't-Care Conditions -- NAND and NOR Implementation -- Other Two- Level Implementations -- Exclusive-OR Function -- Hardware Description Language(HDL) -- Combinational Circuits -- Analysis Procedure -- Design Procedure -- Binary Adder- Subtractor -- Decimal Adder -- Binary Multiplier -- Magnitude Comparator -- Decoders -- Encoders -- Multiplexers -- HDL For Combinational Circuits.

UNIT- III Sequential Logic Design, Synchronous Sequential Logic:

Sequential Circuits -- Latches -- Flip-Flops -- Analysis of Clocked Sequential Circuits -- State Reduction and Assignment -- Design Procedure -- HDL For Sequential Circuits.

UNIT- IV :Registers ad Counters, Fundamentals of Asynchronous Sequential Logic:

Registers -- Shift Registers -- Ripple Counters -- Synchronous Counters -- Other Counters -- HDL for Registers and Counters -- Introduction to Asynchronous Sequential Logic -- Analysis Procedure -- Circuits With Latches -- Design Procedure. Hazards

UNIT-V Memory and Programmable Logic:

Introduction -- Random-Access Memory -- Memory Decoding -Error Detection and Correction - Read-Only Memory -- Programmable Logic Array -- Programmable Array Logic -- Sequential Programmable Devices.

Text Book :

1. Digital Design, 3rd Edition, M. Morris Mano, Pearson Education, Inc., 2002

Reference Books:

1. Digital Logic Design Principles, Norman Balabanian and Bradley Carlson, John Wiley & Sons(Asia) Pte. Ltd., 2002.
2. Fundamentals of Digital Circuits, A. Ananda Kumar, PHI, 2002
3. Digital Circuits and Design, 2nd Edition, S Salivahanan and S Arivazhagan, Vikas Publishing House Pvt. Ltd., 2003.
4. Fundamentals of Digital Logic with VHDL Design, Stephen Brown and Zvonko Vranesic, Tata McGraw-Hill Edition, 2002

DATA ANALYSIS AND VISUALIZATION USING PYTHON

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To introduce the basics of Python Programming Language.
2. To learn about Numpy, Pandas, Matplotlib libraries.
3. To acquire knowledge about data analysis using various statistical approaches

Course Outcomes:

At the end of the course, a student should be able to:

1. Acquire knowledge on Basics of Python
2. Acquire knowledge on OOP of Python
3. Acquire knowledge on NumPy and Basics of Statistics
4. Use library such as Pandas
5. Acquire knowledge on Graph Visualizations in Python
6. Acquire knowledge on Data analysis

SYLLABUS**UNIT-I**

Introduction to Python: About Python, History, Features of Python, Who uses Python, What can we do with Python, Variables, Data Types, Operations, Operators.

Python OOP: Python Classes, Methods, constructors, Inheritance, Data Hiding, Exceptions, Modules, Packages, Files

UNIT-II

NumPy Arrays and Vectorized Computation: NumPy arrays, Array creation, Indexing and slicing, Fancy indexing, Numerical operations on arrays, Array functions, Data processing using arrays, Loading and saving data, Saving an array, Loading an array, Lists, Tuples, Dictionary, Sets, Control Loops, Linear algebra with NumPy, NumPy random numbers

UNIT-III

Data Analysis with Pandas: An overview of the Pandas package, The Pandas data structure-Series, The DataFrame.

The essential basic functionality: Reindexing and altering labels, Head and tail, Binary operations, Functional statistics, Function application, Sorting,

Indexing and selecting data, Computational tools, Working with missing data, Advanced uses of Pandas for data analysis - Hierarchical indexing, The Panel data

UNIT-IV

Statistics for Data Analysis :: Fitting aggregated counts to the Poisson distribution, Determining confidence intervals for mean, variance, and standard deviation, Correlating variables with Pearson's correlation, Correlating variables with the Spearman rank correlation, Evaluating relations between variables with ANOVA, Learning About Models- Models and experiments, The cumulative distribution function, Working with distributions, The probability density function, Multivariate distributions, **Regression-Introducing linear regression**-Getting the dataset, Testing with linear regression, **Multivariate regression**-Adding economic indicators, **Logistic regression**

UNIT-V

Data Visualization: The matplotlib API primer-Line properties, Figures and subplots, Exploring plot types-Scatter plots, Bar plots, Histogram plots, Legends and annotations, Plotting functions with Pandas, Additional Python data visualization tools-Bokeh, MayaVi

Interacting with Databases and Data Analysis Application:: Interacting with data in text format: Reading data from text format, Writing data to text format, Interacting with data in binary format:HDF5, Interacting with data in MongoDB, Interacting with data in Redis:The simple value, List, Set, Ordered set, Data munging: Cleaning data, Filtering , Merging data, Reshaping data, Data aggregation , Grouping data

Text Books

1. Vamsi Kurama, Python Programming-a modern approach, Pearson Publisher
2. Phuong Vo.T.H , Martin Czygan, Getting Started with Python Data Analysis, Packt Publishing Ltd
3. Ivan Idris, Python Data Analysis Cookbook, Packt Publishing Ltd
4. Magnus VilhelmPersson and Luiz Felipe Martins, Mastering Python Data Analysis, Packt Publishing Ltd

Reference Books

1. S.C. Gupta and V.K. Kapoor, Fundamentals of Mathematical Statistics, sultan chand & sons Educational Publishers New Delhi
2. Jake VanderPlas, Python Data Science, Handbook, Essential tools for working with data, Oreilly Publisher
3. Mark Summerfield, Programming in Python 3--A Complete Introduction to the Python Language, Second Edition, Additson Wesley

DATA STRUCTURES LAB

Lab : 3 Periods

Exam : 3 Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

1. To implement stacks and queues using arrays and linked lists.
2. To develop programs for searching and sorting algorithms.
3. To write programs using concepts of various trees.
4. To implement programs using graphs.

Course Outcomes:

1. Student will be able to write programs to implement stacks and queues.
2. Ability to implement various searching and sorting techniques.
3. Ability to implement programs using trees and graphs.

SYLLABUS

Implement the following programs using C-Language.

1. Write a program for sorting a list using Bubble sort and then apply binary search.
2. Write a program to implement the operations on stacks.
3. Write a program to implement the operations on circular queues.
4. Write a program for evaluating a given postfix expression using stack.
5. Write a program for converting a given infix expression to postfix form using stack.
6. Write a program for implementing the mazing problem.
7. Write a program for the representation of polynomials using linked list and for the addition of two such polynomials.
8. Write a program for quick sort .
9. Write a program for Merge sort.
10. Write a program for Heap sort .
11. Write a program to create a binary search tree and for implementing the in order, preorder, post order traversal using recursion.
12. Write a program for finding the transitive closure of a digraph.
13. Write a program for finding the shortest path from a given source to any vertex in a digraph using Dijkstra's algorithm.
14. a)Write a program for finding the Depth First Search of a graph.
b)Write a program for finding the Breadth First Search of a graph

Reference Book:

1. Fundamentals of Data Structures in C, 2nd edition, Horowitz, Sahani and Anderson-Freed, Universities Press, 2008.

DATA ANALYSIS AND VISUALIZATION USING R AND PYTHON LAB

Lab : 3 Periods

Exam : 3 Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

1. To implement Lists, dictionaries, Sorting, Searching, text handling and file handling
2. To develop Python programs for analyzing data using Statistical methods.
3. To develop Python programs by importing Numpy, Pandas, Matplotlib libraries.

Course Outcomes:

At the end of the course, a student should be able to:

1. Acquire Programming knowledge on Basics of Python
2. Acquire Programming knowledge on Searching and sorting using Python
3. Acquire Programming knowledge on Text and File Handling
4. Develop Python Programs to Mean, Median, Mode, Correlation, Regression and Probability distributions
5. Acquire Programming knowledge on NumPy, Pandas Library
6. Acquire Programming knowledge on Graph Visualizations in Python and Data for Analysis

SYLLABUS

1. Python Programs on lists & Dictionaries
2. Python Programs on Searching and sorting
3. Python Programs on Text Handling
4. Python Programs on Files Handling
5. Python Programs for Mean, Mode, Median, Variance, Standard Deviation
6. Python Programs for Karl Pearson Coefficient of Correlation, Rank Correlation, Regression, Distribution,
7. Python Programs for **NumPy Arrays**, Linear algebra with NumPy
8. Python Programs for Data Frame using Pandas Library
9. **Write a Python program for the following**
 - Importing matplotlib,
 - Simple Line Plots,
 - Adjusting the Plot: Line Colors and Styles, Axes Limits,
 - Labeling Plots,
 - Simple Scatter Plots,
 - Histograms,
 - Customizing Plot Legends,
 - Choosing Elements for the Legend,
 - Multiple Legends,
 - Customizing Colorbars,
 - Multiple Subplots,
 - Text and Annotation,
 - Customizing Tick.

10. Python Programs for Interacting with data in text format, Interacting with data in binary format, Interacting with data in MongoDB, Interacting with data in Redis, Cleaning, Filtering and Merging Data

Reference Books:

1. Wesley J Chen, Core Python Programming,
2. Jake VanderPlas, Python Data Science, Handbook, Essential tools for working with data, Oreilly Publisher
3. Mark Summerfield, Programming in Python 3--A Complete Introduction to the Python Language, Second Edition, Addison Wesley
4. Phuong Vo.T.H , Martin Czygan, Getting Started with Python Data Analysis, Packt Publishing Ltd
5. Ivan Idris, Python Data Analysis, Packt Publishing Ltd
6. Magnus Vilhelm Persson and Luiz Felipe Martins, Mastering Python Data Analysis, Packt Publishing Ltd

**INDUSTRY ORIENTED TRAINING
(R Programming Lab)**

Lab : 2 Periods

Exam : 3 Hrs.

Sessionals : 50

Credits : 1

Course Objectives:

1. Understand the basics of R programming Language.
2. Able to use R programming for the analysis of various data sets
3. A hands-on experience in the use of R Libraries for visualization of data sets.

Course Outcomes:

At the end of the course the students can able to

1. Install and find documentation for R functions and libraries. Search for and find domain-specific R packages.
2. Use and understand the R data types (vectors, matrices, dataframes, strings)
3. Reshape data and use visual exploratory graphics. Practice good data management.
4. Write their own functions in R and break a problem into a set of functions.
5. Be fluent in programming concepts such as functional programming, code reuse, object-oriented programming, recursion, regular expressions, and split-transform-recombine data manipulation.
6. Engage in good code and data organization practices and use a consistent programming style.

SYLLABUS

INSTALLING R

Downloading and Installing R from CRAN

Installing R on Your Windows Computer

Running the R Program

The Help Command in R

Help for Windows Users

Command Packages

Standard Command Packages

How to Get Extra Packages of R Commands

How to Install Extra Packages for Windows Users

Running and Manipulating Packages

Loading Packages

Windows-Specific Package Commands

Removing or Unloading Packages

Reading and Getting Data into R

Using the combine Command for Making Data

Entering Numerical Items as Data

Entering Text Items as Data

Using the scan Command for Making Data

Entering Text as Data
Using the Clipboard to Make Data
Reading a File of Data from a Disk
Reading Bigger Data Files
The read.csv() Command
Alternative Commands for Reading Data in R
Missing Values in Data Files
Viewing Named Objects
Viewing Previously Loaded Named-Objects
Viewing All Objects
Viewing Only Matching Names
Removing Objects from R
Types of Data Items
Number Data
Text Items
Converting Between Number and Text Data
The Structure of Data Items
Vector Items
Data Frames
Matrix Objects
List Objects

Saving Your Work in R
Saving the Workspace on Exit
Saving Data Files to Disk
Save Named Objects
Save Everything
Reading Data Files from Disk
Saving Data to Disk as Text Files
Writing Vector Objects to Disk
Writing Matrix and Data Frame Objects to Disk
Writing List Objects to Disk
Converting List Objects to Data Frames

Manipulating Objects
Manipulating Vectors
Selecting and Displaying Parts of a Vector
Sorting and Rearranging a Vector
Returning Logical Values from a Vector
Manipulating Matrix and Data Frames
Selecting and Displaying Parts of a Matrix or Data Frame
Sorting and Rearranging a Matrix or Data Frame
Manipulating Lists
Viewing Objects within Objects
Looking Inside Complicated Data Objects
Opening Complicated Data Objects
Quick Looks at Complicated Data Objects
Viewing and Setting Names
Rotating Data Tables

Constructing Data Objects

Making Lists

Making Data Frames

Making Matrix Objects

Re-ordering Data Frames and Matrix Objects

Forms of Data Objects: Testing and Converting

Testing to See What Type of Object You Have

Converting from One Object Form to Another

Convert a Matrix to a Data Frame

Convert a Data Frame into a Matrix

Convert a Data Frame into a List

Convert a Matrix into a List

Convert a List to Something Else

Box-whisker Plots

Basic Boxplots

Customizing Boxplots

Horizontal Boxplots

Scatter Plots

Basic Scatter Plots

Adding Axis Labels

Plotting Symbols

Setting Axis Limits

Using Formula Syntax

Adding Lines of Best-Fit to Scatter Plots

Pairs Plots (Multiple Correlation Plots)

Line Charts

Line Charts Using Numeric Data

Line Charts Using Categorical Data

Pie Charts

Bar Charts

Single-Category Bar Charts

Multiple Category Bar Charts

Stacked Bar Charts

Grouped Bar Charts

Horizontal Bars

Bar Charts from Summary Data

Creating Data for Complex Analysis

Data Frames

Matrix Objects

Creating and Setting Factor Data

Making Replicate Treatment Factors

Adding Rows or Columns

Summarizing Data

Simple Column and Row Summaries

Complex Summary Functions

The rowsum() Command

The apply() Command
Using tapply() to Summarize Using a Grouping Variable
The aggregate() Command

Adding Elements to Existing Plots
Adding Legends to Graphs
Color Palettes
Placing a Legend on an Existing Plot
Adding Text to Graphs
Making Superscript and Subscript Axis Titles
Orienting the Axis Labels
Making Extra Space in the Margin for Labels
Setting Text and Label Sizes
Adding Text to the Plot Area
Adding Text in the Plot Margins
Creating Mathematical Expressions
Adding Points to an Existing Graph
Adding Various Sorts of Lines to Graphs
Adding Straight Lines as Gridlines or Best-Fit Lines
Making Curved Lines to Add to Graphs
Plotting Mathematical Expressions
Adding Short Segments of Lines to an Existing Plot
Adding Arrows to an Existing Graph
Matrix Plots (Multiple Series on One Graph)
Multiple Plots in One Window
Splitting the Plot Window into Equal Sections
Splitting the Plot Window into Unequal Sections
Exporting Graphs
Using Copy and Paste to Move a Graph
Saving a Graph to a File

Reference Book :

1. Beginning R: The Statistical Programming Language by Dr. Mark Gardener, Wiley publishers

ENGLISH PROFICIENCY-I
(Common to All Branches)

Lecture	: 1Period		
Tutorial	: 1 Periods	Int.Marks	: --
Exam	: ---	Credits	: --

AIM:

Enriching the communicative competency of the students by adopting the activity-based as well as the class-oriented instruction with a view to facilitate and enable them to enhance their language proficiency skills.

Course Objectives:

The Students will be able to

1. Communicate their ideas and views effectively.
2. Practice language skills and improve their language competency.
3. Know and perform well in real life contexts.
4. Identify and examine their self attributes which require improvement and motivation.
5. Build confidence and overcome their inhibitions, stage freight, nervousness, etc.
6. Improve their innovative practices in speaking.

Course Outcomes:

The Students will

1. Improve speaking skills.
2. Enhance their listening capabilities.
3. Learn and practice the skills of composition writing.
4. Enhance their reading and understanding of different texts.
5. Improve their inter-personal communication skills.
6. Be confident in presentation skills.

SYLLABUS

UNIT-1:LISTENING

Selected Motivational Speeches

Selected Moral Stories

UNIT-2:SPEAKING

Book Review

Skit Presentation

PowerPoint Presentations

Describing event/place/thing

Extempore

Group Discussion

Picture Perception and Describing Test

UNIT-3:READING

Speeded Reading

Reading Comprehension

UNIT-4:WRITING

Paragraph Writing

Literary Appreciation – Understanding the Language of Literature

UNIT-5:PROJECT

Ad Making

Reference Books:

1. Classic Short Stories-A Reader's Digest Selection
2. English for Colleges by Brendan J. Carroll, Macmillan Publications
3. The World's Great Speeches- edited by B.S. Sekhar, Jeet Publications
4. Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
5. English and Communication Skills for Students of Science and Engineering, by S.P. Dhanavel, Orient Blackswan Ltd. 2009
6. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
7. The Oxford Guide to Writing and Speaking by John Seely OUP

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R17)

II/IV B.TECH
(With effect from **2017-2018** Admitted Batch onwards)

COMPUTER SCIENCE AND ENGINEERING

II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/Week	Internal Marks	External Marks	Total Marks
B17 CS 2201	Computer Organization	3	3	1	--	4	30	70	100
B17 CS 2202	Operating Systems	3	3	1	--	4	30	70	100
B17 CS 2203	Microprocessors	3	3	1	--	4	30	70	100
B17 CS 2204	Data Communications	3	3	1	--	4	30	70	100
B17 CS 2205	Advanced Data Structures	3	3	1	--	4	30	70	100
B17 ME 2207	Operations Research	3	3	1	--	4	30	70	100
B17 CS 2206	Operating Systems & Unix programming Lab	2	--	--	3	3	50	50	100
B17 CS 2207	Digital Electronics & Microprocessors Lab	2	--	--	3	3	50	50	100
B17 CS 2208	Competitive Programming Lab	1	--	--	2	2	50	--	50
B17 BS 2204	Professional Ethics & Human Values	--	2	--	--	2	--	--	--
B17 BS 2206	English Proficiency-II	--	1	1	--	2	--	--	--
Total		23	21	7	8	36	330	520	850

COMPUTER ORGANIZATION

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To study about structure and functional components of a computer.
2. Understanding the hierarchical organization of a computer system which consists of instruction set of commands.
3. Learn about the architecture of a computer from a programming view.
4. To design a balance system that minimizes performance and utilization of all elements.

Course Outcomes:

1. Knowledge about major components of a computer such as processor, memory and I/O modules along with their interconnections internally with outside world.
2. Detailed idea about architecture of central processing unit, functions of control unit, memory, I/O devices and their issues.
3. Simple and multiple processor organization and their issues.

SYLLABUS**UNIT-I**

Register Transfer and Micro operations: Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro operations, Logic Micro operations, Shift Micro operations, Arithmetic Logic Shift Unit.

UNIT-II

Basic Computer Organization and Design: Instruction Codes, Computer Registers, Computer Instructions, Timing and Control, Instruction Cycle, Memory-Reference Instructions, Input-Output and Interrupt, Complete Computer Description, Design of Basic Computer.

Micro programmed Control: Control Memory, Address Sequencing, Micro program Example, Design of Control Unit.

UNIT-III

Central Processing Unit: Introduction, General Register Organization, Stack Organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Program Control, Reduced Instruction Set Computer(RISC)

Pipeline and Vector Processing: Parallel Processing, Pipelining, Arithmetic Pipeline, Instruction Pipeline, RISK Pipeline, Vector Processing, Array Processors.

UNIT-IV

Input/output Organization: Peripheral Devices, I/O interface, Asynchronous data transfer, Modes of transfer, priority Interrupt, Direct memory access, Input-Output Processor (IOP), Serial Communication.

UNIT-V

Memory Organization: Memory Hierarchy, Main memory, Auxiliary memory, Associate Memory, Cache Memory, and Virtual memory.

Text Book:

1. Computer System Architecture, M. Morris Mano, Prentice Hall of India Pvt. Ltd., Third Edition, Sept.2008.

Reference Books:

1. Computer Architecture and Organization, William Stallings, PHI Pvt. Ltd., Eastern Economy Edition, Sixth Edition, 2003.
2. Computer Organization and Architecture, Linda Null, Julia Lobur, Narosa Publications ISBN 81- 7319-609-5.
3. Computer System Architecture”, John. P. Hayes.

OPERATING SYSTEMS

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To understand operating system as a layer of abstraction above physical hardware that facilitates usage convenience and efficient resource management.
2. To learn design and implementation of policies and mechanisms of OS.
3. To investigate case studies to understand the design paradigms for popular multiuser or single user operating system.

Course Outcomes:

1. The student understands OS evolution, its structure and services provided by it.
2. Learn process life cycle, process scheduling objectives, policies and mechanisms, process synchronization, inter process communication, deadlocks and other process subsystem related concepts.
3. Learn memory hierarchy, allocation, de-allocation policies and mechanism for main and auxiliary memory, file system design and implementation issues.
4. Investigate UNIX/ LINUX and Windows OS platforms w.r.t similarities and differences in design paradigms.

SYLLABUS**UNIT-I****Introduction to Operating Systems**

Over View of Operating Systems, Types of Operating Systems, Operating System Structures, Operating System Services, System Calls, Virtual Machines, Operating System Design and Implementation.

UNIT-II**Process Management and Process Synchronization**

Process Concepts, Operations on Processes, Co-operating Processes, Threads, Inter Process Communication, Process Scheduling, Scheduling Algorithms, Multiple - Processor Scheduling, Thread Scheduling. The Critical Section Problem, Peterson's Solution, Synchronization Hardware, Semaphores, Classical Problems of Synchronization, Critical Regions, Monitors.

UNIT-III**Deadlocks**

System Model, Deadlock Characterization, Methods For Handling Deadlocks, Deadlock Prevention, Avoidance, Deadlock Detection, Recovery from Deadlocks

UNIT-IV

Memory Management

Logical versus Physical Address, Swapping, contiguous memory allocation, paging, structure of the page table , segmentation, Virtual Memory, Demand Paging, Page Replacement, Allocation of Frames, Thrashing, Memory-Mapped files

UNIT-V

File Systems, Implementation, and Secondary-storage Structure and Case study

Concept of a file, Access Methods, Directory Structure, Protection, File System Structure, Allocation Methods, Free Space Management, Directory Management, Device Drivers, overview of Mass-storage structure, Disk structure, disk attachment, disk scheduling, swap-space management, Case Study of UNIX, MS-DOS and Windows

Text Book:

1. Operating Systems, Abraham Silberschatz, Peter Baer Galvin, and Greg Gagne, John Wiley Publ., Seventh Edition.

Reference Books:

1. Modern Operating Systems, Andrew S. Tanenbaum, , 2nd edition, 1995, PHI.
2. Operating Systems, William Stallings 5th Edition - PHI.
3. Operating Systems: A Design-Oriented Approach', Charles Crowley, _Tata Hill Co.,1998 edition.

MICROPROCESSORS

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives :

The objectives of this course are as follows:

1. To introduce the microprocessors (8085 & 8086).
2. To Introduce memory interfacing and memory mapping.
3. To introduce peripheral interfacing devices 8255 and keyboard/display interface device 8279.
4. To provide knowledge of address decoding and bus transactions.
5. To develop programming skills in 8085 and 8086 microprocessor

Course Outcomes:

1. Students can able to understand **The 8085A μ P. Architecture [K1]**
2. Students can learn about 8085 **Instruction Set [K2]**
3. The Student Develops The Skill Of Writing 8085 Microprocessor Programming **[K3]**
4. Ability to design semiconductor memories **[K2]**
5. Students can learn Parallel I/O Interface - 8255 **[K2]**
6. Students can learn Keyboard/Display Interface - 8279 **[K2]**
7. Students can able to understand **The 8086 μ P. Architecture [K1]**
8. Students can learn about 8086 **Instruction Set [K1]**
9. The Student Develops The Skill Of Writing 80865 Microprocessor Programming **[K3]**

SYLLABUS**UNIT-I**

Internal Architecture, functional/signal description of 8085 microprocessor, Instruction set, Addressing modes and programming in 8085.

UNIT-II

Timing diagram, Counters and Time delays, Stacks and Subroutines and Interrupts in 8085

UNIT-III

Classification and interfacing semiconductor memories with 8085 MPU, Interfacing peripherals to INTEL 8085 using Parallel IO interface-8255, Keyboard/DisplayInterface-8279

UNIT-IV

The 8086 Microprocessor architecture & functional /signal description, Segmented memory, Maximum & Minimum modes of 8086.

UNIT-V

Addressing modes, Instruction set and assembly language programming techniques with 8086.

Text Books:

1. Microprocessor Architecture and Applications with the 8085 , Ramesh S. gaonkar, 4th Edition, Penram International, 1999
2. Advanced Microprocessors and Peripherals, A K RAY & K M Bhurchandi , 2nd Edition, The McGraw-Hill companies.

Reference Books:

1. The 80X86 Family , Design, Programming and Interfacing, John E. Uffenbeck, 3rd Edition, Pearson Education Inc., 2002.
2. Walter A . tribal and Avatar Singh. The 8088 and 8086 Microprocessors, Programming interfacing, software, hardware and Applications, 4th Edition Pearson education Inc., 2003
3. Microprocessors and Interfacing. Programming and hardware, 2nd Edition, Douglass V. Hall. MH Edition , 1999.

DATA COMMUNICATIONS

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To study basics of data communication systems.
2. To study the various types of transmission media.
3. To study the various hardware concepts related to data communications.
4. To discuss about multiplexing techniques.

Course Outcomes:

1. Students will have the ability to *use* Data Communications and Networking Protocols and protocol architectures
2. Students will have the ability to *develop* communication models for providing data transmission facility
3. Students will have the ability to *outline* Data Communication terminology
4. Students will have the ability to *classify* various transmission media
5. Students will have the ability to *discriminate* various types of signals for data transmission and ability to *describe* data encoding techniques
6. Students will have the ability to *describe* data communications interface
7. Students will have the ability to *apply* various flow control , error control techniques of data link control protocols
8. Students will have the ability to *use* various data communication terminals and processing hardware
9. Students will have the ability to *demonstrate* multiplexing techniques

SYLLABUS**UNIT-I****Introduction to Data Communications:**

A Communications Model, Data Communications and Data Communications Networking, Protocols and Protocol Architecture, Characteristics of Data Transmission: Concepts and Terminology, Analog and Digital Data Transmission, Transmission Impairments.

UNIT-II**Transmission Media:**

Guided Transmission Media, Wireless Transmission. **Data Encoding:** Digital Data-Digital Signals, Digital Data-Analog Signals, Analog Data-Digital Signals, Analog Data-Analog Signals.

UNIT-III

Data Communication Interface:

Asynchronous and Synchronous Transmission, Line Configurations, Interfacing. Data Link Control Flow Control, Error Detection, Error Control, High-Level Data Link Control (HDLC)

UNIT-IV

Data Communications Hardware & Processing Hardware:

Terminals: Introduction, Basic Terminal Components, Enhanced Terminal Components, General-Purpose Terminals, Remote Job Entry Terminals, Transaction Terminals, Clustering of Terminal Devices.

Communications Processing Hardware:

Introduction, Switching Processors, Multidrop Lines, Multiplexers, Concentrators, Front-End Processors

UNIT-V

Multiplexing:

Frequency-Division Multiplexing, Synchronous Time-Division Multiplexing: Characteristics, TDM Link Control, Digital Carrier Systems, Statistical Time-Division Multiplexing: Characteristics.

Text Books:

1. William Stallings, Data and Computer Communications, 7th Edition, PH/Pearson Edu.Inc.,
2. Mary E.S. Loomis, Data Communications, PHI-N.J.,1983(Chapter 3, Chapter 5)

Reference Books:

1. Behrouz A. Forouzan, Data Communications and Networking, 3rd Edition TMH, 2004
2. William A. Shay, Understanding Data Communications & Networks, 2nd Edition Thomson-Brooks/Cole –Vikas Publishing House, 1999.
3. Michale A. Miller, Data & Network Communications, Thomson/Delmar –Vikas Pub. House, 2000

ADVANCED DATA STRUCTURES

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To study the concepts related to trees such as binary trees, BST , AVL trees etc.
2. To discuss various hashing technique.
3. To study the various external sorting algorithms.
4. To discuss the concepts related to disjoint set ADT.
5. To study several graph algorithms and their time complexities.

Course outcomes:

Student will be able to write programs to implement various trees.

1. Ability to understand various hashing techniques.
2. Ability to write programs to implement sorting techniques.
3. Ability to understand concepts related to graph theory.

SYLLABUS**UNIT-I**

Trees: Definition , operations and applications of Binary search trees, AVL trees, Red-Black Trees, Splay trees, Tries and B-Trees, B+ Trees

UNIT-II

Priority Queues: Heap model and implementations, Binary Heap, Applications of Priority Queues, d-Heaps, Leftist Heaps, Skew Heaps, Binomial Queues structure, operations and Implementation

UNIT-III

Hashing & External sorting: Hash Table Structure, Hash Function, Collision handling, Separate Chaining, Open Addressing, Rehashing, Extendible hashing, Difference between internal and external sorting, Model and simple algorithm for External sorting, Multi-way Merge, Poly-phase Merge, Replacement selection.

UNIT-IV

Graph algorithms: Representation of graphs, Topological sort, Network flow problems, Applications of Depth first search for finding Bi-connectivity, Euler circuits, strong components, Introduction of NP-Completeness

UNIT-V

Disjoint Set ADT & Amortized analysis: Equivalence relations, Dynamic equivalence problem, Basic data structure, smart union algorithms, path compression, Analysis of union/find algorithm, applications of ADT Disjoint set, Introduction to amortized analysis, Basic approaches, binary queues, skew heaps, Aggregate analysis, The accounting method, The potential method and Dynamic tables.

Text Books:

1. Data Structures and Algorithm Analysis in C – Mark Allen Weiss, Pearson Edu Publishers.
2. Data Structures and Algorithms: Concepts, Techniques and Applications – G.A.V.Pai, TataMc Graw Hill Publishers

Reference Books:

1. Advanced Data Structures – Peter Brass, Cambridge University Press, 2008

OPERATIONS RESEARCH

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To discuss about basic Operation Research concepts , Formulation of LPP and its solution using graphical method.
2. To discuss about standard form of LPP. solving LPP using various methods.
3. To study the various solutions of transportation problems and assignment problems.
4. To discuss about PERT and CPM charts .
5. To discuss about replacement problems, inventory problems and game theory.

Course Outcomes:

1. Ability to solve LPP problems using various methods.
2. Ability to solve transportation and assignment problems using several methods.
3. Analyze the PERT and CPM charts.
4. Ability to solve replacement problems and game theory problems.

SYLLABUS**UNIT - I:**

Overview of Operations Research, Types of OR Models, Phases of Operations Research– OR Techniques, Introduction to Linear Programming, Formulation of Linear Programming Problem, Graphical Solution; Standard Form of LPP, Basic Feasible Solutions, Unrestricted Variables, Simplex Algorithm , Artificial Variables, Big M Method, Two Phase Simplex Method, Degeneracy, Alternative Optimal, Unbounded Solutions, Infeasible Solutions, Primal And Dual Problems and Their Relations, Dual Simplex Method.

UNIT - II:

Transportation Problem as LPP, Initial Solutions, North West Corner Rule, Lowest Cost Method, Vogel's Approximation Method, Optimum Solutions of TPP, Degeneracy in Transportation, Transportation Algorithms. Assignment Problem, Assignment Problem as LPP, Hungarian Method, Travelling Salesman Problem, Solutions of TSP.

UNIT - III:

Sequencing Problems, N-Jobs Two Machine Problems, N-Jobs K Machines Problems, Two-Jobs M- Machine Problems, Replacement Problems-Individual and Group Replacement Policy, Reliability & System Failure Problems,

UNIT - IV:

Network Representation of A Project, CPM and PERT, Critical Path Calculations, Time – Cost Optimizations, PERT Analysis and Probability Considerations, Resource Analysis in Network Scheduling. Inventory-Factors Effecting Inventory-EOQ, Inventory Problems with and without Shortages, Inventory Problems with Price Breakups.

UNIT – V:

Game Theory: Two Person Zero Sum Games, Mixed Strategy Games and Their Algorithms.

Text Books:

1. Operations Research, Kanti Swaroop, P.K. Gupta, Man Mohan, Sulthan Chand&Sons Education.
2. Publishers Operations Research – An Introduction, Handy A Taha – Pearson Education.

Reference Books:

1. Operations Research Panneer Selvan Prentice Hall of India.
2. Operations Research by S.D Sharma.
3. Introduction to Operations Research, F.S. Hiller, G.J. Liberman, TMH.
4. Operations Research by V. K. Kapoor.

OPERATING SYSTEMS AND UNIX PROGRAMMING LAB

Lab : 3 Periods

Exam : 3 Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

1. To learn about UNIX/LINUX operating system, its intervals.
2. To learn system programming for UNIX/LINUX Operating System.
3. To understand UNIX/LINUX shell programming.
4. To understand resource management policies, mechanisms and their performance evaluation.

Course Outcomes:

1. The student practices UNIX commands, Vi editor, shell commands.
2. The student develops skill in writing C programs using system calls for process management, inter process communication and memory management aspects.
3. The student learns shell programming and develops skill for writing scripts for batch level tasks.

SYLLABUS

Module I

OS lab familiarization, Home Assignment on Unix commands, Vi editor

Simple C programs using command line arguments, system calls, library function calls, make utility C programs using fork system call to create process and study parent, child process mechanism

C programs to create process chaining, spawning

C programs to handle errors using errno, perror() function

C programs to use pipe system call for inter process communication

Module II

Familiarization of Unix shell programming

Simple shell programming exercises

Shell programming using decision making constructs

Shell programming using loop constructs

Shell programming for file and directory manipulation

Module III

C programs to study process scheduling (FCFS, Shortest Job First, and Round Robin) C

programs to study page replacement (FIFO, Optimal, and LRU page replacement) C programs to

study deadlock avoidance and detection

C Programs to simulate free space management.

Reference Books:

1. Unix concepts and applications by Sumitabha Das, TMH Publications.
2. Unix programming by Stevens, Pearson Education.
3. Shell programming by Yashwanth Kanetkar.
4. Operating System Concepts by Silberschatz, and Peter Galvin.

DIGITAL ELECTRONICS AND MICRO PROCESSORS LAB

Lab : 3 Periods

Exam : 3 Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

1. To learn about logic gates, half adders, full adders and flip -flops.
2. To learn about the microprocessor programming.
3. To learn about the microprocessor interfacing with stepper motor, R-2R ladder.

Course Outcomes:

1. The student understands the logic gates, half adders, full adders and flip-flops to design a circuit.
2. The student develops the skill of writing microprocessor programming.
3. The student understands the interfacing of microprocessor with stepper motor, R-2R ladder.

SYLLABUS

1 DIGITAL EXPERIMENT

Verification of Truth tables of OR, AND, NOT, NAND, NOR, EX-OR gates(by using 7400-series)

Construction of gates using NAND, NOR gates.

Construction of Half and Full adders and verifying their truth tables.

Operation and verifying truth tables of flip-flops-RS, D and JK using IC's

Up/Down counters using JK flip-flops.

4-bit shift right and left registers using JK flip-flops.

2 MICROPROCESSORS: 8085

Binary Addition of 'N' 8-bit numbers.

Binary to BCD conversion

Arranging –Ascending/descending order

To find the largest /smallest numbers in the array.

ASCII to HEXA & HEXA to ASCII conversion.

3 MICROPROCESSORS: 8086

Liner Search

Factorial of a given number

To copy string from S1 to S2

To find GCD and LCD

4 MICROPROCESSOR INTERFACING WITH 8085

Matrix display

Light Rollet

Traffic Light.

Simple calculator

Stepper Motor controller.

Reference Books:

1. Microprocessor Architecture and Applications with the 8085 , Ramesh S. gaonkar, 4th Edition, Penram International, 1999.
2. Advanced Microprocessors and Peripherals, A K RAY & K M Bhurchandi , 2nd Edition, The Mcgraw-Hill companies.

COMPETITIVE PROGRAMMING LAB

Lab : 2 Periods

Exam : 3 Hrs.

Sessionals : 50

Credits : 1

Course Objectives:

1. To Learn basic concepts of Python Programming Language
2. To Learn various Object Oriented Programming Concepts
3. To Learn various advanced Data structures

Course Outcomes:

At the end of the course the student can able to

1. Write programs using python programming
2. Write algorithms
3. Implement various data Structures
4. To apply object oriented mechanisms
5. To Implement various Advance data Structures like AVL trees, B-Trees, Splay trees etc

SYLLABUS

Introduction to Python

- The basic elements of python
- Branching Programs
- Control Structures
- Strings and Input
- Iteration

Functions, Scoping and Abstraction

- Functions and scoping
- Recursion
- Files

Classes and Object-Oriented Programming

- Abstract Data Types and Classes
- Inheritance
- Encapsulation and Information Hiding

Algorithms and Data structures

- Sequences
- Lists
- Item Ordering
- Two-Dimensional Sequences
- The Minmax

Sets and Maps

- Playing Sudoku
- Sets
- Hashing
- The HashSet Class
- Solving Sudoku

Maps

- Memorization
- Correlating Two Sources of Information

Membership Structures

- Bloom Filters
- The Trie Data type

Balanced Binary Search Trees

- Binary Search Trees
- AVL Trees
- Splay Trees
- Iterative Splaying
- Recursive Splaying

B-Trees

- B-Tree Implementation
- B-Tree Insert
- B-Tree Delete

Project

Reference Books :

1. Kent D. Lee, Steve Hubbard, "Data Structures and Algorithms with Python", Springer Publications.

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PROFESSIONAL ETHICS & HUMAN VALUES

(Common to CSE, ECE & IT)

Lecture	: 2 Periods	Int. Marks	: --
Exam	: ---	Credits	: --

Course Objectives:

1. To inculcate Ethics and Human Values into the young minds.
2. To develop moral responsibility and mould them as best professionals.
3. To create ethical vision and achieve harmony in life.

Course outcomes:

By the end of the course student should be able to understand the importance of ethics and values in life and society.

SYLLABUS**UNIT – I**

Ethics and Human Values: Ethics and Values, Ethical Vision, Ethical Decisions, **Human Values** – Classification of Values, Universality of Values.

UNIT – II

Engineering Ethics: Nature of Engineering Ethics, Profession and Professionalism, Professional Ethics , Code of Ethics, Sample Codes – IEEE, ASCE, ASME and CSI.

UNIT – III**Engineering as Social Experimentation:**

Engineering as social experimentation, Engineering Professionals – life skills, Engineers as Managers, Consultants and Leaders Role of engineers in promoting ethical climate, balanced outlook on law.

UNIT – IV**Safety Social Responsibility and Rights:**

Safety and Risk, moral responsibility of engineers for safety, case studies – Bhopal gas tragedy, Chernobyl disaster, Fukushima Nuclear disaster, Professional rights, Gender discrimination, Sexual harassment at work place.

UNIT – V**Global Issues:**

Globalization and MNCs, Environmental Ethics, Computer Ethics, Cyber Crimes, Ethical living, concept of Harmony in life.

Text Books:

1. Govindharajan, M., Natarajan, S. and Senthil Kumar, V.S., Engineering Ethics, Prentice Hall of India, (PHI) Delhi, 2004.
2. Subramainam, R., Professional Ethics, Oxford University Press, New Delhi, 2013.

Reference Books:

1. Charles D, Fleddermann, "Engineering Ethics", Pearson / PHI, New Jersey 2004 (Indian Reprint)

ENGLISH PROFICIENCY-II
(Common to All Branches)

Lecture	: 1Period	Int.Marks	: --
Tutorial	: 1 Periods	Credits	: --
Exam	: ---		

AIM:

To equip the students with the components of the language required and help them gain adequate knowledge so as to become employable and competent.

Course Objectives:

Students will be able to

1. Enhance their Interpretative skills
2. Understand how to prepare a text.
3. Comprehend various types of writing discourses and respond.
4. Produce effectively different write-ups related to various business contexts.
5. Strengthen their emotional make-up.
6. Perceive various writing discourses.

Course Outcomes:

The students will

1. Develop the skills of taking and making notes
2. Interpret the pictures appropriately and effectively.
3. Read, comprehend and infer a given piece of writing effectively.
4. Learn and practice the skills of Research writing.
5. Communicate well through various forms of writing.
6. Be confident in giving presentations and dealing with people.

SYLLABUS**UNIT-1:SPEAKING**

Analyzing proverbs

Enactment of One-act play

UNIT-2:READING

Reading Comprehension

Summarizing Newspaper Article

UNIT-3:WRITING

Note Taking &Note Making

Precis Writing

Essay Writing

Letter Writing

Picture Description

Literary Appreciation– Learning the Language of Literature

UNIT-4: VOCABULARY

Indian-origin English Words

Phrasal Verbs for Day-to-Day Communication

Commonly used Idiomatic Expressions

UNIT-5: PROJECT

Research Writing

Reference Books:

1. English for Colleges by Brendan J. Carroll, Macmillan Publications
2. Effective Technical Communication by M.AshrafRizwi. Tata Mcgraw Hill
3. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
4. The Oxford Guide to Writing and Speaking by John Seely OUP
5. Six Weeks to Words of Power by Wilfred Funk. W.R.Goyal Publishers
6. English for Engineers and Scientists by Sangeeta Sharma and Binod Mishra, PHI Learning



ESTD: 1980

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

III/IV B.TECH

(With effect from **2017-2018** Admitted Batch onwards)

Under Choice Based Credit System

COMPUTER SCIENCE AND ENGINEERING I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Internal Marks	Exam Marks	Total Marks
B17 CS 3101	Computer Networks	3	3	1	--	4	30	70	100
B17 CS 3102	Web Technologies	3	3	1	--	4	30	70	100
B17 CS 3103	Formal Languages & Automata Theory	3	3	1	--	4	30	70	100
B17 CS 3104	Database Management Systems	3	3	1	--	4	30	70	100
B17 CS 3105	Application Development Using Java	3	3	1	--	4	30	70	100
#OE	Open Elective	3	3	1	--	4	30	70	100
B17 CS 3108	Database Management Systems Lab.	2	--	--	3	3	50	50	100
B17 CS 3109	Application Development Lab.	2	--	--	3	3	50	50	100
B17 BS 3101	Problem Solving & Linguistic Competence	1	--	3	--	3	30	70	100
B17 BS 3103	Advanced Coding	1	--	--	3	3	50	50	100
Total		24	18	9	9	36	360	640	1000

#OE	B17CS3106	Embedded Systems
	B17CS3107	Cyber Security
	B17EC3109	Digital Signal Processing
	B17ME3110	Industrial Robotics

COMPUTER NETWORKS

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. Be familiar with basic Computer Network Terminology
2. Able to distinguish between Switching Techniques
3. To Know various types of Networks and Protocols
4. To Know the causes and effects of Congestion
5. Familiar with Internetwork operation.
6. In depth knowledge of various Transport Protocols.

Course Outcomes:

At the end of the course students can able to

1. Distinguish between Circuit Switching and Packet Switching approaches
2. Apply various concepts of ATM networks
3. Distinguish between various types of Networks
4. Apply various Congestion Control Techniques
5. Know Internetwork Operation.
6. Know various Connection Oriented Transport Control Mechanisms.

SYLLABUS**UNIT-I**

Switched Networks: Circuit-Switching Networks, Circuit Switching Concepts, Soft switch Architecture, Packet Switching Principles, X.25, Frame Relay.

Asynchronous Transfer Mode: Protocol Architecture, ATM Logical Connections, ATM Cells, ATM Service Categories, Routing in Switched Networks.

UNIT-II

Congestion Control in Switched Data Networks: Effects of Congestion, Congestion Control, Traffic management, Congestion Control in Packet Switched networks, Principles of Cellular Networks

Local Area Network Overview: Background, Topologies and transmission media, LAN Protocol Architecture, Bridges, Layer 2 and Layer 3 Switches,

UNIT-III

High Speed LANs: The Emergence of High Speed LANs, Ethernet.

Wireless LANs: Overview, Wireless LAN Technology, IEEE802.11 Architecture and Services.

UNIT-IV

Internet Protocols: Basic protocol Functions, Principles of Internetworking, Connectionless Internetworking, Internet Protocol.

Internet Operation: Multicasting, Routing Protocols: Autonomous Systems & Approaches to Routing.

UNIT-V

Transport protocols: Connection oriented Transport Protocol Mechanisms: Reliable Sequencing Network Service, TCP: TCP Services, TCP Header Format, TCP Mechanisms, UDP.

Application Layer protocols: Overview of DNS, Electronic Mail, FTP, TFTP, BOOTP, HTTP Protocols, World Wide Web.

Text Book:

1. Data and Computer Communications, William Stallings 7th Edition, Pearson Education, 2004

Reference Books:

1. Data Communications and Networking, Behrouz A. Forouzan, 3rd Edition, TMH, 2004
2. Computer Networking: A Top-Down Approach Featuring the Internet, James F. Kurose and Keith W. Ross, 2nd Edition, Pearson Education, 2002
3. Computer Networks, Andrew S. Tanenbaum, 4th Edition, Pearson Education, 2003
4. An Engineering Approach to Computer Networking, S. Keshav, Pearson Education, 1997
5. Computer Networks and Internets with Internet Applications, Douglas E. Comer, 4th Edition, Pearson Education, 2003.

WEB TECHNOLOGIES

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

On completing this course student will be able to

1. Understand the principles of Web based application development.
2. Design dynamic content in Web Pages using JavaScript.
3. Understanding the concepts of java Servlets, java Server Pages and design applications using them.
4. Understand the concepts of Component development and design applications by establishing connections to Databases

Course Outcomes:

1. Students will be able to construct web based applications and Identify where data structures are appearing in them.
2. Students will be able to connect java programs to different databases.
3. Students will be able to develop EJB programs.

SYLLABUS**UNIT-I**

Introduction to HTML, Core Elements, Links and Addressing, Images, Text, Colors and Background, Lists, Tables and Layouts, Frames, Forms, Cascading Style Sheets.

Introduction to Java Scripts, Elements of Objects in Java Script, Dynamic HTML with Java Script.

UNIT-II

Document type definition, XML Syntax, XML Schemas, Document Object model, Presenting XML, Using XML Processors.

UNIT-III

Introduction to Servlet, Servlet Life Cycles, Servlet Basics, Tomcat Web Server, Configuring Apache Tomcat, Handling Client Request and Response, Handling Cookies, Session Tracking.

UNIT-IV

Introduction to PHP, Language Basics, Functions, Strings, Arrays.

MYSQL Installation, Accessing MySQL Using PHP, Form Handling, Cookies, Sessions, and Authentication, Tables, Inserting Data into Tables, Selecting Data from a Table, Updating Table, Deleting data from Table, Webpage creation.

UNIT-V

XML AJAX, AJAX introduction, AJAX XMLHttpRequest, AJAX Request, AJAX Response, AJAX XML file, AJAX PHP, AJAX database, AJAX Examples and AJAX applications.

Text Books:

1. Web Programming, building internet applications, 2nd Ed., Chris Bates, Wiley Dreamtech
2. The complete Reference HTML and DHTML, Thomas A. Powey
3. Learning Php, Mysql, Robin Nixon
4. Programming Php, Kevin Tatroe, Peter MacIntyre & Rasmus Lerdorf foreword by Michael Bourque.

Reference Books:

1. Internet, World Wide Web , How to program, Dietel , Nieto, PHI/PEA Web Technologies, Godbole, kahate, 2nd Ed., TMH

FORMAL LANGUAGES AND AUTOMATA THEORY

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. Introduce the student to the concepts of Theory of computation in computer science
2. The students should acquire insights into the relationship among formal languages, formal Grammars and automata.

Course Outcomes:

1. Ability to classify machines by their power to recognize languages,
2. Ability to explain finite state machines to solve problems in computing,
3. Ability to explain deterministic and non-deterministic machines,
4. Ability to explain the concepts of Turing Machines, Undecidability, church thesis.

SYLLABUS**UNIT – I Finite Automata**

Why Study Automata Theory? The Central Concepts of Automata Theory, Finite Automation, Acceptance of a String by a Finite Automation, DFA, Design of DFAs, NFA, Design of NFA, Equivalence of DFA and NFA, Conversion of NFA into DFA, Finite Automata with λ -Transitions, Eliminating λ -Transitions, Minimization of Finite Automata, Mealy and Moore Machines, Equivalence of Mealy and Moore Machines, Applications and Limitation of Finite Automata.

UNIT – II: Regular Expressions

Regular Expressions, Regular Sets, Identity Rules, Equivalence of two Regular Expressions, Finite Automata and Regular Expressions, Conversion of Regular Expression to NFA with λ -Transitions, Conversion of DFA to Regular Expression, Arden's Theorem, Pumping Lemma for Regular Languages, Applications of pumping lemma, Closure Properties of Regular Languages, Regular Grammar, Conversion of Finite Automata to Regular Grammars, Applications of Regular Expressions.

UNIT – III: Context Free Grammars

Grammars, Classification of Grammars, Chomsky Hierarchy, Context Free Grammar, Leftmost and Rightmost Derivations, Parse Trees, Ambiguous Grammars, Simplification of Context Free Grammars-Elimination of Useless Symbols, λ -Productions and Unit Productions, Normal Forms for Context Free Grammars-Chomsky Normal Form and Greibach Normal Form, Pumping Lemma for CFL, Closure Properties, Applications of Context Free Grammars.

UNIT – IV: Pushdown Automata

Pushdown Automata, Definition, Graphical Notation, Instantaneous Description, Language Acceptance of pushdown Automata-Acceptance by empty stack and final state, Design of Pushdown Automata-Deterministic and Non-Deterministic PDA, Conversion of Pushdown Automata to Context Free Grammars, Conversion of Context Free Grammars to Pushdown Automata, Application of Pushdown Automata.

UNIT – V: Turning Machine

Turing Machine, Definition, Representation of Turing Machines-Instantaneous Descriptions, Transition Tables and Transition Diagrams, Language of a Turing Machine, Design of Turing Machines, Techniques for Turing Machine Construction, Types of Turing Machines, Church's Thesis, Universal Turing Machine, Introduction to Decidable and Undecidable Problems, Halting Problem of Turing Machines, Post's Correspondence Problem, Modified Post's Correspondence Problem, Introduction to Classes of P and NP, NP-Hard and NP-Complete Problems.

Text Books:

1. Introduction to Automata Theory, Languages and Computation, J.E.Hopcroft, R.Motwani and J.D.Ullman, 3rd Edition, Pearson, 2008.
2. Theory of Computer Science-Automata, Languages and Computation, K.L.P.Mishra and N.Chandrasekharan, 3rd Edition, PHI, 2007.

Reference Books:

1. Formal Language and Automata Theory, K.V.N.Sunitha and N.Kalyani, Pearson, 2015.
2. Introduction to Automata Theory, Formal Languages and Computation, Shyamalendu Kandar, Pearson, 2013.
3. Theory of Computation, V.Kulkarni, Oxford University Press, 2013.
4. Theory of Automata, Languages and Computation, Rajendra Kumar, McGraw Hill.

DATABASE MANAGEMENT SYSTEMS

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To provide an introduction to DBMS and RDBMS their structure and uses.
2. To illustrate SQL commands.
3. To demonstrate database design using data models.
4. To apply normalization.
5. To explore different techniques of Concurrency control
6. To learn Recovery techniques of DBMS.

Course Outcomes:

After successful completion of the course a student will be able to

1. Generalize the basic concepts of DBMS and RDBMS.
2. Prepare SQL commands for defining, constructing and manipulating databases.
3. Apply conceptual and logical database design using data models.
4. Apply normalization to tables.
5. Manage concurrent transactions.
6. Apply databases Recovery Techniques.

SYLLABUS

UNIT-I: Introduction: Introduction to File Processing System and DBMS, Disadvantages of FPS, Advantages of DBMS, Introduction to RDBMS, Levels of abstraction, Data Independence, types of users, Duties of DBA, Structure of a DBMS.

UNIT-II: Relational Model and SQL: Relational model- relation, relation schema and relation instance, Integrity constraints and relation algebra, SQL Preliminaries, Basic form of SQL Query, Nested Queries, Joins, Set Operations, Aggregate Operators, Data Definition Language Commands-create table, alter table and drop table commands, Integrity Constraints in SQL, Null values, Data Manipulation Language Commands-insert, update and delete commands.

UNIT-III: Database Design: Basic Steps in Database Design, ER Diagrams, Entities, Attributes and Entity Sets, Relationships & Relationship Sets, Features of the ER Model, Conceptual Database Design, Logical Database Design, Triggers, Views, Sequences, JDBC.

UNIT-IV: Normalization: Anomalies of redundancy, Functional Dependencies, Normal Forms- 1NF, 2NF, 3NF and BCNF, Properties of Decomposition, Normalization-Loss-less join decomposition into BCNF, Multi-valued dependencies and 4NF.

UNIT-V: Transaction Management : The ACID Properties, Transactions & Schedules, Concurrent Execution of Transactions, Concurrency Control- Serializability and Recoverability, Two Phase Locking protocol, Dealing with Deadlocks and Time Stamp Ordering Protocol, Crash Recovery- Log-based Recovery, The Log, Other Recovery-Related Structures, The Write-Ahead Log Protocol, Check pointing, ARIES.

Text Book:

1. Database Management Systems, Raghu Ramakrishnann, Johannes Gehrke, 3rd Edition, McGraw hill, 2014.

Reference Books:

1. Database System Concepts, A. Silberschatz, H.F. Korth, S. Sudarshan, McGraw hill Education, VI edition, 2010.
2. Fundamentals of Database Systems 5th edition. Ramez Elmasri, Shamkant B. Navathe, Pearson Education, 2008.
3. Database Management System Oracle SQL and PL/SQL, P.K. Das Gupta, PHI.
4. Database System Concepts, Peter Rob & Carlos Coronel, Cengage Learning, 2008.
5. Database Systems, A Practical approach to Design Implementation and Management Fourth edition, Thomas Connolly, Carolyn Begg, Pearson education.

APPLICATION DEVELOPMENT USING JAVA

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. Study of object oriented programming through JAVA.
2. Learn about web based applications such as AWT components.
3. Study of multitasking by using multithreading concept.
2. Learn about network programming and applications development.
3. Students will have the ability to develop Web Applications using JSP.
4. Students will have knowledge about types of JDBC and they able to access database using JSP
5. Getting the student to be well trained in Advanced Java Programming skills for an easy entry in the IT Industry.

Course Outcomes:

1. Able to do projects for web based and internet applications.
2. Understand multitasking and multiprogramming development
3. Able to do network programming.
4. Able to Construct Web application using Java Server Pages

SYLLABUS**UNIT-I**

Introduction: Classes, Objects and Methods, Constructors, Arrays, Strings and vectors, Inheritance, Interface, Abstract Class, Packages.

Multi Threaded Programming and Exceptions: Introduction on Thread, Life cycle of a Thread, Thread Priorities, Synchronization, Implementing the Runnable interface. **Exceptions:** Types of Errors, Types of Exceptions, throwing our own Exception

UNIT-II

Applet Programming and Basic AWT Components: Introduction to Applet, Applet Life Cycle, Passing parameters to Applet, Displaying Numerical Values. **AWT:** container, Canvas, Panel, Frame, Basic AWT user interface controls: (Button, Checkbox, CheckboxGroup, Scrollbars, TextField, TextArea, Radiobutton and Listbox).

Event-handling and Layout Managers: Handling events with classes, handling events by implementing interfaces, Organizing Windows with Layout Managers (Flow Layout, Border Layout, Card Layout, Grid Layout, GridBagLayout).

UNIT-III

Java Swings: Introduction to JSwings, Components and Containers, the Swing Packages, Exploring Swing (JFrame, JButtons, JTable, JPasswordField, JTabbedPane, JScrollbar, JTrees)

JSP: Introduction to JSP, JSP Elements(JSP Declaration, JSP Scripting, JSP Expression, JSP Comments), JSP Directives, implicit Objects, JSP Program for Database Access.

UNIT-IV

Database Access: Structure of JDBC, JDBC Drivers, JDBC Architecture, JDBC API (java.sql.package), Connecting to the Database, JAVA Database connection program for MS Access, Oracle, MySQL and NOSQL.

UNIT-V

Input /Output, NIO: The I/O Classes and Interfaces, I/O Exceptions, the Stream Classes, the Byte Streams, the Character Streams, NIO Fundamentals

Network Programming: The Networking Classes and Interfaces, InetAddress, TCP/IP Client Sockets, URLConnection, HttpURLConnection, TCP/IP Server Sockets, Datagram, Datagram Socket, DatagramPacket

Text Books:

1. Programming with JAVA a Primer, Third Edition, E. Balagurusamy, Tata McGraw-Hill.
2. The Complete Reference, Java Eighth Edition, Herbert Schildt, Oracle Press.(Chapters: 11, 19, 20, 22, 23, 24, 29, 30).
3. Java Server Pages, Hans Bergstan, Oreilly, 2nd Edition, August 2002. (Chapters: 1-10).
4. Database Programming with JDBC and Java, George Reese, Second Edition, Published by O'Reilly & Associates. (Chapters: 3, 4)

Reference Books:

1. Internet and World wide web- How to program, Dietel and Nieto, Pearson.
2. Java server pages, Pekowsky, Pearson.
3. Beginning Web Programming, Jon Duckett ,Wrox, Wiley.

EMBEDDED SYSTEMS (Open Elective)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To provide in-depth knowledge about Embedded Processor, its hardware and software.
2. To explain the programming concepts, Embedded programming in C & Assembly language.
3. To explain various architectures of Embedded Systems.
4. To explain Real Time Operating Systems.
5. To explain about Inter task communication.
6. To explain about Embedded software development tools.

Course Outcomes:

On successful completion of the course a student will able

1. To describe the differences between general computing system and Embedded System.
2. To recognize the classification of Embedded System.
3. To understand various architectures of Embedded System.
4. To design Real Time Embedded System using the concepts of RTOS.
5. To load embedded software on Host machine.
6. To test Host machine.

SYLLABUS

UNIT-I

Examples of Embedded Systems – Typical Hardware – Memory – Microprocessors – Busses – Direct Memory Access – Introduction to 8051 Microcontroller – Architecture-Instruction set – Programming- Microprocessor Architecture – Interrupt Basics – The Shared-Data problem – Interrupt Latency.

UNIT-II

Round-Robin Architecture - Round-Robin with Interrupts Architecture - Function-Queue-Scheduling Architecture – Real-Time Operating Systems Architecture – Selection of Architecture.

UNIT-III

Tasks and Task States – Tasks and Data – Semaphores and Shared Data – Semaphore Problems – Semaphore variants, Message Queues – Mailboxes – Pipes – Timer Functions – Events – Memory Management – Interrupt Routines in RTOS Environment.

UNIT-IV

RTOS design – Principles – Encapsulation Semaphores and Queues – Hard Real-Time Scheduling Considerations – Saving Memory Space – Saving Power.

UNIT-V

Host and Target Machines – Linker/Locator for Embedded Software- Getting Embedded Software into the Target System, Testing on your Host Machine – Instruction Set Simulators – Laboratory Tools used for Debugging.

Text Books:

1. David A. Simon, An Embedded Software Primer, Pearson Education, Inc., 1999
2. Sriram V Iyer and Pankaj Gupta, Embedded Real Time Systems programming, TMH, 2004

Reference Books:

1. Frank Vahid/ Tony Givargis, Embedded Systems Design – A Unified Hardware/Software Introduction, John Wiley & Sons, 2006
2. Raj Kamal, Embedded Systems, Architecture, Programming and Design, Tata McGraw-Hill Education, 2011

CYBER SECURITY
(Open Elective)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The Cyber security Course will provide the students with foundational Cyber Security principles, Security architecture, risk management, attacks, incidents, and emerging IT and IS technologies.
2. Students will gain insight into the importance of Cyber Security and the integral role of Cyber Security professionals.

Course Outcomes:

1. Cyber Security architecture principles
2. Identifying System and application security threats and vulnerabilities
3. Identifying different classes of attacks
4. Cyber Security incidents to apply appropriate response
5. Describing risk management processes and practices
6. Evaluation of decision making outcomes of Cyber Security scenarios

SYLLABUS

UNIT- I**Introduction to Cybercrime:**

Introduction, Cybercrime: Definition and Origins of the Word, Cybercrime and Information Security, Who are Cybercriminals? , Classifications of Cybercrimes, Cybercrime: The Legal Perspectives, Cybercrimes: An Indian Perspective, Cybercrime and the Indian ITA 2000, A Global Perspective on Cybercrimes, Cybercrime Era: Survival Mantra for the Netizens.

Cyber offenses: How Criminals Plan Them –Introduction, How Criminals Plan the Attacks, Social Engineering, Cyber stalking, Cyber cafe and Cybercrimes, Botnets: The Fuel for Cybercrime, Attack Vector Cloud Computing.

UNIT –II**Cybercrime Mobile and Wireless Devices:**

Introduction, Proliferation of Mobile and Wireless Devices, Trends in Mobility, Credit Card Frauds in Mobile and Wireless Computing Era, Security Challenges Posed by Mobile Devices, Registry Settings for Mobile Devices, Authentication Service Security, Attacks on Mobile/Cell Phones, Mobile Devices: Security Implications for Organizations, Organizational Measures for Handling Mobile, Organizational Security Policies and Measures in Mobile Computing Era, Laptops.

UNIT –III

Tools and Methods Used in Cybercrime:

Introduction, Proxy Servers and Anonymizers, Phishing, Password Cracking, Key loggers and Spywares, Virus and Worms, Trojan Horses and Backdoors, Steganography, DoS and DDoS Attacks, SQL Injection, Buffer Overflow, Attacks on Wireless Networks, Phishing and Identity Theft: Introduction, Phishing, Identity Theft (ID Theft)

UNIT –IV

Cybercrimes and Cyber security:

Why Do We Need Cyber laws: The Indian Context, The Indian IT Act, Challenges to Indian Law and Cybercrime Scenario in India, Consequences of Not Addressing the Weakness in Information Technology Act, Digital Signatures and the Indian IT Act, Information Security Planning and Governance, Information Security Policy Standards, Practices, The information Security Blueprint, Security education, Training and awareness program, Continuing Strategies.

UNIT –V

Understanding Computer Forensics:

Introduction, Historical Background of Cyber forensics, Digital Forensics Science, The Need for Computer Forensics, Cyber forensics and Digital Evidence, Forensics Analysis of E-Mail, Digital Forensics Life Cycle, Chain of Custody Concept, Network Forensics, Approaching a Computer Forensics Investigation, Computer Forensics and Steganography, Relevance of the OSI 7 Layer Model to Computer Forensics, Forensics and Social Networking Sites: The Security/Privacy Threats, Computer Forensics from Compliance Perspective, Challenges in Computer Forensics, Special Tools and Techniques, Forensics Auditing, Antiforensics

Text Books:

1. Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives, Nina Godbole, Sunit Belapure, Wiley.
2. Principles of Information Security, Micheal E. Whitman and Herbert J. Mattord, Cengage Learning.

Reference Book:

1. Information Security, Mark Rhodes, Ousley, MGH.

DIGITAL SIGNAL PROCESSING (Open Elective)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

This course introduces students to the fundamental principles of Digital Signal Processing and develops essential analysis and design tools required for signal processing systems & implementations. Also this subject is an introduction to the graduate-level courses in a broad range of disciplines spanning communications, speech processing & image processing.

The topics include SS basics, sampling theorem, Z-transform, analysis of Discrete-time Linear Time-Invariant Systems, Realization structures, Frequency domain representation of signals and systems, DTFT, DFS, Discrete Fourier Transform (DFT), linear/circular convolutions, Fast Fourier Transform (FFT) algorithms, FIR & IIR digital filter design, Multi-rate DSP and a few DSP applications.

Course Outcomes: At the end of this course, the students will be able to;

1. Describe the DSP fundamental theory and components, Develop an understanding of DSP advantages, limitations and fundamental tradeoffs. Carry-out LTI system analysis using convolution & Z-transform
2. Carryout data analysis & spectrum analysis using FFT
3. Design of IIR digital filters to meet specifications
4. Design of FIR digital filters to meet specifications
5. Knows multi-rate signal processing aspects & DSP applications

SYLLABUS

UNIT-I: Discrete-Time Signals and Systems: (Oppenheim & Proakis)

Introduction to Digital Signal Processing, Basic elements of a DSP system, Advantages of Digital SP over Analogy SP, Discrete-time signals and systems, DT-LTI systems described by Linear constant-coefficient difference equations, Properties & Analysis of DT-LTI systems, Discrete linear convolution, Frequency domain representation of DT Signals and Systems, DTFT, Review of the Z-transform, Properties, Inverse Z-transform, Analysis of DT-LTI systems in Z-Domain, System function, One-sided Z-transform, Solution of difference equations, Structures and Realization of Digital Filters, Direct-I, II, series and parallel forms.

UNIT-II: Discrete Fourier Transform (DFT) and Fast Fourier Transform Algorithms (FFT): (Oppenheim & Proakis)

Frequency analysis of discrete time signals, DFS, Properties of DFS, Sampling of DTFT, DFT, Properties of DFT, Circular and linear convolution of sequences using DFT, Efficient computation of DFT, Radix-2 Decimation-in-Time(DIT) & Decimation-in-Frequency(DIF) FFT Algorithms, Inverse FFT.

UNIT-III: Design of IIR Digital Filters: (Oppenheim & Proakis)

General considerations in Filter design, Analog filter approximations– Butterworth and Chebyshev, Frequency response specifications; Design of IIR digital filters from analog filters, Bilinear Transformation Method, Impulse Invariance Technique, and Low-pass filter Design examples.

UNIT-IV: Design of FIR Digital Filters: (Oppenheim & Proakis)

Characteristics of FIR Digital Filters, Design of Linear Phase FIR digital Filters using Windows, Effect of Window selection & filter length on filter frequency response, Design examples, Comparison of IIR and FIR Filters.

UNIT-V: DSP Applications and Fundamentals of Multirate Digital Signal Processing: (SK Mitra)

Overview of DSP applications, Spectral analysis of sinusoidal signals using FFT, Subband coding of speech signals, Signal compression, Finite precision arithmetic effects.

Introduction to Multirate DSP, Basic sampling rate alteration devices: upsampler, downsampler, Time and Frequency domain characterization of up/down samplers, Interpolator and decimator. Interactive programming based examples.

Text Books:

1. Alan V. Oppenheim, Ronald W. Schaffer, “Digital Signal Processing” – PHI Ed., 2006
2. John G. Proakis, D.G. Manolakis, “Digital Signal Processing: Principles, Algorithms and Applications”, 3rd Ed., PHI, 1996.

Reference Books:

1. Sanjit K. Mitra, "Digital Signal Processing: A Computer Based Approach", Tata McGraw Hill.
2. [Lawrence R. Rabiner](#), [Bernard Gold](#), “Theory and application of digital signal processing”, Prentice Hall.

INDUSTRIAL ROBOTICS (Open Elective)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To give students practice in applying their knowledge of mathematics, science, and Engineering and to expand this knowledge into the vast area of robotics.
2. The students will be exposed to the concepts of robot kinematics and dynamics
3. Mathematical approach to explain how the robotic arm motion can be described.
4. The students will understand the functioning of sensors and actuators.

Course Outcomes:

Students will be able to

1. Distinguish between fixed automation and programmable automation.
2. Identify various components of robot.
3. Select appropriate type of actuator for a joint.
4. Illustrate robot applications in manufacturing.
5. Analyze kinematics of a robot.

SYLLABUS

UNIT - I

INTRODUCTION: Automation and Robotics, Robot anatomy, robot configuration, motions joint notation scheme, work volume, robot drive systems, control systems and dynamic performance, precision of movement.

CONTROL SYSTEM AND COMPONENTS: basic concepts and motion controllers, control system analysis, robot actuation and feedback components, Positions sensors, velocity sensors, actuators, power transmission systems, robot joint control design.

UNIT - II

MOTION ANALYSIS AND CONTROL: Manipulator kinematics, position representation, forward and inverse transformations, homogeneous transformations, manipulator path control, robot arm dynamics, configuration of a robot controller.

UNIT - III

END EFFECTORS: Grippers-types, operation, mechanism, force analysis, tools as end effectors consideration in gripper selection and design. **SENSORS:** Desirable features, tactile, proximity and range sensors, uses sensors in robotics.

MACHINE VISION: Functions, Sensing and Digitizing-imaging devices, Lighting techniques, Analog to digital single conversion, image storage: Image processing and Analysis-image data reduction, Segmentation, feature extraction, Object recognition. Training the vision system, Robotic application.

UNIT - IV

ROBOT PROGRAMMING: Lead through programming, Robot program as a path in space, Motion interpolation, WAIT, SIGNAL AND DELAY commands, Branching, capabilities and Limitations of lead through methods.

ROBOT LANGUAGES: Textual robot Languages, Generations of robot programming languages, Robot language structures, Elements and function.

UNIT - V

ROBOT CELL DESIGN AND CONTROL: Robot cell layouts-Robot centered cell, In-line robot cell, Considerations in work design, Work and control, Inter locks, Error detection, Work cell controller.

ROBOT APPLICATION: Material transfer, Machine loading/unloading, Processing operation, Assembly and Inspection, Future Application.

Text Books:

1. Industrial Robotics /Groover M P /Pearson Edu.
2. Introduction to Robotic Mechanics and Control by JJ Craig, Pearson, 3rd edition.

Reference Books:

1. Robotics / Fu K S/ McGraw Hill.
2. Robotic Engineering / Richard D. Klafter, Prentice Hall.
3. Robot Analysis and Intelligence / Asada and Slotine / Wiley Inter-Science.
4. Robot Dynamics & Control – Mark W. Spong and M. Vidyasagar / John Wiley
5. Introduction to Robotics by SK Saha, The McGraw Hill Company, 6th, 2012.
6. Robotics and Control / Mittal R K & Nagrath I J / TMH.

DATABASE MANAGEMENT SYSTEMS LAB**Lab : 3 Periods****Exam : 3 Hrs.****Int. Marks : 50****Ext. Marks : 50****Credits : 2****Course Objectives:**

1. To introduce to a commercial DBMS such as ORACLE.
2. To learn and practice SQL commands for schema creation, data manipulation.
3. To learn conceptual database design based on a case study.
4. To learn Logical database design based on a case study.
5. To apply database design stages by a case study.

Course Outcomes:

1. The student is exposed to a commercial RDBMS environment such as ORACLE.
2. The student will learn SQL commands for data definition and manipulation.
3. The student applies conceptual design.
4. The student applies Logical data base design.
5. The student takes up a case study and applies the design steps.

SYLLABUS

Features of two commercial RDBMS package such as ORACLE/DB2, MS Access, MYSQL & Structured Query Language (SQL) used with the RDBMS.

I. Laboratory Exercises Should Include

- a. Defining Schemas for Applications
- b. Creation of Database
- c. Writing SQL Queries
- d. Retrieve Information from Database,
- e. Creating Views
- f. Creating Triggers
- g. Normalization up to Third Normal Form
- h. Use of Forms
- i. Report Writing

II. Some sample applications are given below:

1. Accounting Package for Shops
2. Database Manager for Magazine Agency or Newspaper Agency
3. Ticket Booking for Performances
4. Preparing Greeting Cards & Birthday Cards
5. Personal Accounts - Insurance, Loans, Mortgage Payments, Etc.
6. Doctor's Diary & Billing System
7. Personal Bank Account
8. Class Marks Management
9. Hostel Accounting
10. Video Tape Library
11. History of Cricket Scores
12. Cable TV Transmission Program Manager
13. Personal Library
14. Sailors Database
15. Suppliers and Parts Database

Reference Books:

1. Oracle Database 11g The Complete Reference, McGraw Hill Professional, 2008.
2. Database Systems Using Oracle, Nilesh Shah, Pearson Education India, 2016.
3. Introduction to SQL, Rick F Vander Lans, Fourth Edition, Pearson Education, 2007

APPLICATION DEVELOPMENT LAB**Lab : 3 Periods****Exam : 3 Hrs.****Int. Marks : 50****Ext. Marks : 50****Credits : 2****Course Objectives:**

1. To develop a static web page using html.
2. To develop a dynamic web page using html, css, java script.
3. To understand mysql database connectivity.
4. To implement hosting a website.

Course Outcomes:

1. Compare and Contrast HTML, DHTML, CSS, JavaScript and other Web technologies.
2. Implement JavaScript Language to perform functionalities at client side validations.
3. Assess and evaluate the role of “WEBSERVERS” for the management and delivery of electronic information.
4. Develop Web based applications by PHP to have an interactive applications such as Client Server Architecture.

SYLLABUS**CYCLE-1:**

1. Write a java program to implement Interface.
2. Write a java program to implement Packages.
3. Write a java program to manipulate the Strings.
4. Write a java program that implements thread class methods.
5. Write a java program that handles all mouse events and shows the event name at the center of the window when a mouse event is fired. Use adapter classes.
6. Write a java program to demonstrate the key event handlers.
7. Write a java program to display the table using labels in grid layout and Flow layout.
8. Write an applet program to Pass a parameters to Applet
9. Write a program to implement an application using AWT Components
10. Write a java program to implement JTree.
11. Write a java program to implement JTabbedPane
12. Write a java program to implement JScrollPane
13. Write JSP Program to validate user name and password on server side?
14. Write an online book purchase application using JSP. Consider a login validation page and one billing page for bill payment process. Assume any information if required?
15. Write JSP Program for Database Access.
16. Write a java program by using JDBC to execute a SQL query for a database and display the results.
17. Write a java program to display the Header information of the given URL
18. Write a java program to split the given URL.
19. Implementing one-one chat Application without threads
20. Write a java program for Datagram server and Client interaction as per given below.
 - i). A program to create Datagram server to send a message to client.
 - ii). A program to create Datagram client to receive the message sent by the server

CYCLE-2:

Each batch (only two members) should develop one project out of this list.
Project has to develop by using HTML, CSS, JS, PHP and MYSQL.

1. Design Airlines Ticket Reservation System
2. Online Shopping
3. Design Library Information system.
4. Design Gram Panchayat Information system for House tax, water tax, wealth tax, Library tax collection, phone bill, Electricity bill collection.
5. Design student information system portal which maintain attendance, marks etc.
6. Design online examination system.
7. Event management System.
8. Car Rental System.
9. Cinema Booking System.
10. Hotel Management System.
11. Complaint management System.
12. Online voting system.
13. Student Result System.
14. Car Comparison System Project.
15. Selling your old stuff.
16. Aquaculture Updates.
17. Timesheet using PHP
18. Online Help Desk using PHP
19. Online marriage bureau system
20. EAMCET web counseling

Final copy of Documentation has to submit by the batch as following:

- | | | |
|--------------------------|------------|--|
| 1. Abstract | 2. Modules | 3. Software Requirement Specifications |
| 5. Database Connectivity | 6. Output | 7. Code |

Reference Books:

1. P. J. Deitel, H. M. Deitel, —Java for Programmers, Pearson Education, PHI, 4th Edition, 2007.
2. P. Radha Krishna, —Object Oriented Programming through Java, Universities Press, 2nd Edition, 2007
3. Bruce Eckel, —Thinking in Java, Pearson Education, 4th Edition, 2006.
4. Sachin Malhotra, Saurabh Chaudhary, —Programming in Java, Oxford University Press, 5th Edition, 2010.

PROBLEM SOLVING & LINGUISTIC COMPETENCE
(Common to all Branches)

Tutorial	: 3 Periods (VA-2+QA-1)	Int.Marks	: 30
Exam	: 3 Hrs.	Ext.Marks	: 70
		Credits	: 1

Part-A: Verbal and Soft Skills-I

Course Objectives:

1. To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.
2. To familiarize the learner with high frequency words as they would be used in their professional career.
3. To inculcate logical thinking in order to frame and use data as per the requirement.
4. To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.
5. To familiarize students with soft skills and how it influences their professional growth.

Course Outcomes:

The student will be able to

1. Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.
2. Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.
3. Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.
4. Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.
5. Apply soft skills in the work place and build better personal and professional relationships making informed decisions.

SYLLABUS

Grammar: (VA)

Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns-number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.

Vocabulary: (VA)

Synonyms and synonym variants(with emphasis on high frequency words), antonyms and antonym variants(with emphasis on high frequency words), contextual meanings with regard to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.

Reasoning: (VA)

Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms).

Usage: (VA)

Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.

Soft Skills:

Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.

Text Books:

1. Oxford Learners' Grammar – Finder by John Eastwood, Oxford Publication.
2. R S Agarwal's books on objective English and verbal reasoning
3. English Vocabulary in Use- Advanced , Cambridge University Press.
4. Collocations In Use, Cambridge University Press.
5. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. English Grammar in Use by Raymond Murphy, CUP
2. Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from _IMS, Career Launcher and Time' institutes for competitive exams.
4. The Art of Public Speaking by Dale Carnegie
5. The Leader in You by Dale Carnegie
6. Emotional Intelligence by Daniel Golman
7. Stay Hungry Stay Foolish by Rashmi Bansal
8. I have a Dream by Rashmi Bansal

Part-B: Quantitative Aptitude -I

Course objectives:

The objective of introducing quantitative aptitude-1 is:

1. To familiarize students with basic problems on numbers and ratio's problems.
2. To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.
3. To enable the students to work efficiently on percentage values related to shares, profit and loss problems.
4. To inculcate logical thinking by exposing the students to reasoning related questions.
5. To expose them to the practice of syllogisms and help them make right conclusions.

Course Outcomes:

1. The students will be able to perform well in calculating on number problems and various units of ratio concepts.
2. Accurate solving problems on time and distance and units related solutions.
3. The students will become adept in solving problems related to profit and loss, in specific, quantitative ability.
4. The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry.
5. The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies.

SYLLABUS

Numbers, LCM and HCF, Chain Rule, Ratio and Proportion Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion.

Time and work, Time and Distance Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.

Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends

Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.

Introduction, number series, number analogy, classification, Letter series, ranking, directions Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.

Data sufficiency, Syllogisms Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi

References:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from _IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

Websites:

1. www.m4maths.com
2. www.Indiabix.com
3. www.800score.com
4. Official GRE site
5. Official GMAT site

ADVANCED CODING

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 1

Course Objectives

1. To understand the basics of modular programming
2. To learn about ADT, Linked Lists and Templates
3. To investigate different methods to find time complexities
4. To learn about Java collections and Libraries

Course Outcomes

At the end of the course, a student should be able to:

1. Acquire coding knowledge on essential of modular programming
2. Acquire Programming knowledge on linked lists
3. Acquire coding knowledge on ADT
4. Acquire knowledge on time complexities of different methods
5. Acquire Programming skill on Java libraries and Collections

SYLLABUS**UNIT I** Review Coding essentials and modular programming

Introduction to Linear Data, Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding.

Introduction to modular programming: Formation of methods, Methods: Signature and definition, Inter-method communication, Data casting & storage classes, Recursions

UNIT II Linear Linked Data

Introduction to structure pointer, Creating Links Basic problems on Linked lists, Classical problems on linked lists. Circular Linked lists, Operations on CLL, Multiple links, Operations on Doubly linked lists

UNIT III Abstract Data-structures

Stack data-structure, Operations on stack, Infix/Prefix/Post fix expression evaluations, Implementation of stack using array, Implementation of stack using linked lists.

Queue data-structure: Operations on Queues, Formation of a circular queue, Implementation of queue using stack, Implementation of stack using array, Implementation of stack using linked lists

UNIT IV Running time analysis of code and organization of linear list data

Code evaluation w.r.t running time, Loop Complexities, Recursion complexities, Searching techniques: sequential Vs. binary searching.

Organizing the list data, Significance of sorting algorithms, Basic Sorting Techniques: Bubble sort, selection sort, Classical sorting techniques: Insertion sort, Quick sort, Merge sort.

UNIT V Standard Library templates and Java collections

Introduction to C++ language features, Working on STLs, Introduction to Java as Object Oriented language, Essential Java Packages, Coding logics.

Note: This course should focus on Problems

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific .
4. Object Oriented Programming in C++: N. Barkakati, PHI.
5. Object Oriented Programming through C++ by Robat Laphore.
6. <https://www.geeksforgeeks.org/>.
7. <https://www.tutorialspoint.com/>

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R17)

III/IV B.TECH
(With effect from **2017-2018** Admitted Batch onwards)

COMPUTER SCIENCE AND ENGINEERING

II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 CS 3201	Data Warehousing & Data Mining	3	3	1	--	4	30	70	100
B17 CS 3202	Object Oriented Software Engineering	3	3	1	--	4	30	70	100
B17 CS 3203	Design and Analysis of Algorithms	3	3	1	--	4	30	70	100
B17 CS 3204	Artificial Intelligence	3	3	1	--	4	30	70	100
B17 CS 3205	Compiler Design	3	3	1	--	4	30	70	100
ELE-I	Elective-I	3	3	1	--	4	30	70	100
B17 CS 3210	Software Engineering Mini Project Lab	2	--	--	3	3	50	50	100
B17 CS 3211	Network Programming Lab	2	--	--	3	3	50	50	100
B17 BS 3201	Employability Skills	1	--	3	--	3	30	70	100
B17 BS 3204	Competitive Coding	1	--	--	3	3	50	50	100
B17 BS 3206	IPR & PATENTS	--	--	2	--	2	--	--	--
Total		24	18	11	9	38	360	640	1000

Elective-I	B17CS3206	Cloud Computing
	B17CS3207	Mobile Computing
	B17CS3208	Distributed Systems
	B17CS3209	Information Retrieval Systems

DATA WAREHOUSING & DATA MINING

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To understand the evolution of data warehousing and data mining techniques.
2. To understand sourcing , cleaning and transformation of data into a warehouse.
3. To learn the principles of statistics, information theory, databases, machine learning and other areas for design and implementation of data mining techniques.
4. To understand pattern mining classification and clustering methods.

Course Outcomes:

1. The student understands the differences between OLTP and OLAP.
2. The student learns how data cube technology supports structuring and querying high dimensional data.
3. The student is introduced to similarity , distance, information gain and other performance and error metrics used for data mining.
4. The student is introduced to association rule mining , supervised and unsupervised learning and the corresponding classification and clustering approaches involving decision trees, Bayesian approaches, model based and agglomerative approaches.

SYLLABUS**UNIT-I**

Introduction to Data Mining: Evolution of I T into DBMS, Motivation and importance of Data Warehousing and Data Mining, Kinds of Patterns, Technologies, Basic Data Analytics: Data Objects and Attributes Types, Statistical Descriptions of Data, Data Visualization, Estimating Data Similarity and Dissimilarity, Major Issues in Data Mining, Data Mining Applications.

UNIT-II

Data Processing and OLAP Technology: Data Cleaning, Data Integration, Data Reduction, Data Transformation, Discretization and Concept Hierarchy Generation. Basic Concepts of Data warehouse, Data Modelling using Cubes and OLAP, DWH Design and usage, Implementation using Data Cubes and OLAPs.

UNIT-III

Mining Frequent Patterns Based on Associations and Correlations: Basic Concepts, Frequent Item set Mining Methods: Apriori Algorithm, Association Rule Generation, Improvements to A Priori, FP- Growth Approach, Mining Frequent Patterns using Vertical Data Formats, Mining Closed and Max Patterns, Pattern Evaluation Methods, mining in multilevel, multi-dimensional space.

UNIT-IV

Classification & Prediction: Basic Concepts, Decision Tree Induction, Bayes Classification, Rule-Based Classification, Model Evaluation and Selection, Techniques to Improve Classification Accuracy Advanced Methods: Classification by Back Propagation.

UNIT-V

Cluster Analysis: Basic Concepts and issues in clustering, Types of Data in Cluster Analysis, Partitioning Methods-K-Means, K-Medoids, Hierarchical Methods-Agglomerative versus Divisive-Distance Measures in Algorithmic Methods, Birch, Density Based Methods- DBSCAN, OPTICS, Grid Based Methods-STING.

Text Book:

1. Data Mining- Concepts and Techniques by Jiawei Han, Micheline Kamber and Jian Pei – Morgan Kaufmann publishers ---3rd edition

Reference Books:

1. Introduction to Data Mining, Adriaan, Addison Wesley Publication.
2. Data Mining Techniques, A.K.Pujari, University Press Data mining concepts by Tan, Steinbech, and Vipin Kumar - Pearson Edu publishers.
3. Data Mining –Introductory and Advanced by Margaret Dunham -- Pearson Education publishers.
4. Data Warehousing for Real –world by Sam Annahory-- Pearson Education publishers.

OBJECT ORIENTED SOFTWARE ENGINEERING

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To explain the importance of OOSE in Software development.
2. To explain the students the importance of Requirements Engineering.
3. To explain the role of UML and Testing in Software Development.
4. To explain the entire Software Development Process with aid of case studies.

Course Outcomes:

1. Ability to define a problem and perform Requirements Engineering.
2. Ability to draw UML diagrams for the requirements gathered.
3. Ability to design various aspects of the system
4. Ability to implement the designed problem in Object Oriented Programming Language and test whether all the requirements specified have been achieved or not.
5. Able to apply various testing approaches to test the system
6. Able to use various Process management activities.

SYLLABUS**UNIT-I**

Introduction to Object Oriented Software Engineering: Nature of the Software, Types of Software , Software Engineering Projects, Software Engineering Activities, Software Quality, Introduction to Object Orientation

Software Process Models: Waterfall Model, Opportunistic Model , Phased Released Model, Spiral Model, Evolutionary Model, Concurrent Engineering Model.

UNIT-II

Requirements Engineering: Domain Analysis, Problem Definition and Scope, Requirements Definition, Types of Requirements, Techniques for Gathering and Analyzing Requirements, Requirement Documents, Reviewing, Managing Change in Requirements.

Unified Modeling Language: Introduction to UML, Modeling Concepts, Types of UML Diagrams with Examples;

UNIT-III

User-Centred Design: Characteristics of Users, Developing Use - Case Models of Systems, Use-Case Diagram, Use- Case Descriptions, Basics of User Interface Design, Usability Principles, User Interfaces.

Class Design and Class Diagrams: Essentials of UML Class Diagrams, Associations and Multiplicity, Other Relationships, Generalization, Instance Diagrams, Advanced Features of Class Diagrams, Interaction and Behavioural Diagrams: Interaction Diagrams, State Diagrams, Activity Diagrams, Component and Deployment Diagrams.

UNIT-IV

Software Design and Architecture: Process of Design, Principles Leading to Good Design, Techniques for Making Good Design Decisions, Good Design Document; Pattern Introduction, **Design Patterns:** Abstraction-Occurrence Pattern, General Hierarchical Pattern, Play-Role Pattern, Singleton Pattern, Observer Pattern, Delegation Pattern, Adaptor Pattern, Façade Pattern, Immutable Pattern, Read-Only Interface Pattern and The Proxy Pattern; Software Architecture Contents of Architecture Model, Architectural Patterns: Multilayer, Client-Server, Broker, Transaction Processing, Pipe & Filter and MVC Architectural Patterns

UNIT-V

Software Testing: Overview of Testing, Testing Concepts, Testing Activities, Testing Strategies, Unit Testing, Integration Testing, Function Testing, Structural Testing, Class Based Testing Strategies, Use Case/Scenario Based Testing, Regression Testing, Performance Testing, System Testing, Acceptance Testing, Installation Testing, OO Test Design Issues, Test Case Design, Quality Assurance, Root Cause Analysis, Post-Mortem Analysis.

Software Process Management: Introduction to Software Project Management, Software Engineering Teams, Project Scheduling, Tracking and Monitoring.

Text Books:

1. Object-Oriented Software Engineering Practical software development using UML and Java by Timothy C. Lethbridge & Robert, Langanieri McGraw-Hill
2. Object-Oriented Software Engineering: Using UML, Patterns and Java, Bernd Bruegge and Allen H. Dutoit, 2nd Edition, Pearson Education Asia.

Reference Books:

1. Software Engineering: A Practitioner's Approach, Roger S Pressman.
2. A Practical Guide to Testing Object-Oriented Software, John D. McGregor; David A Sykes, Addison-Wesley Professional.
3. Software Engineering, K.K. Agarwal, New Age Publications 2008

DESIGN AND ANALYSIS OF ALGORITHMS

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. Analyze the asymptotic performance of algorithms.
2. Write rigorous correctness proofs for algorithms.
3. Demonstrate a familiarity with major algorithms and data structures.
4. Synthesize efficient algorithms in common engineering design situations

Course Outcomes:

1. Students will be able to Argue the correctness of algorithms using inductive proofs and invariants and Analyze worst-case running times of algorithms using asymptotic analysis.
2. Describe the various paradigms of design when an algorithmic design situation calls for it. Recite algorithms that employ this paradigm and synthesize them
3. Students will be able to Compare between different data structures. Pick an appropriate data structure for a design situation.

SYLLABUS**UNIT-I**

INTRODUCTION: Algorithm specification, Recursive Algorithms, Performance analysis, Space Complexity, Time Complexity, Asymptotic Notation, Practical Complexities Performance Measurement, Priority queues, Heaps, Heap Sort, Sets and Disjoint set union, Union and Find Operations

DIVIDE AND CONQUER :General method, Binary search, Finding maximum and minimum, Mergesort, Quicksort, Performance Measurement , Selection Problem, A Worst-Case Optimal Algorithm , Implementation of Select2, Strassen's matrix multiplication, Convex hull Problem, QuickHull Algorithm

UNIT-II

THE GREEDY METHOD : General method, Knapsack problem, Tree vertex splitting, Job sequencing with deadlines, Minimum cost spanning trees, Prim's algorithm, Kruskal's algorithm, Optimal storage on tapes, Optimal merge patterns , Huffman coding, Single source shortest paths

UNIT-III

DYNAMIC PROGRAMMING: General method, Multistage graphs, All pairs shortest paths, Single source shortest paths with general weights, Optimal binary search trees, String editing, 0/1Knapsack, Reliability design, The travelling salesperson problem

UNIT-IV

BACKTRACKING: General method, 8-Queens problem, Sum of subsets, Graph coloring, Hamiltonian cycles, Knapsack problem

BRANCH AND BOUND: The method, Least Cost(LC) Search, The 15-puzzle problem, Control abstractions for LC-Search, FIFO Branch-and-Bound, LC Branch-and-Bound, 0/1 Knapsack problem, Travelling sales person problem

UNIT-V

ALGEBRAIC PROBLEMS and LOWER BOUND THEORY: The method, Evaluation and interpolation, Fast Fourier Transform, Modular arithmetic, Comparison trees, Ordered Searching, Sorting, Selection, Oracles and adversary arguments, Merging, Largest and Second largest

NP-HARD AND NP-COMPLETE PROBLEMS: Basic concepts, Nondeterministic Algorithms, The Classes NP-hard and NP-complete

Text Book:

1. Fundamentals of Computer Algorithms 2nd edition by Ellis Horowitz, Sartaj Sahni, S. Rajasekharan

Reference Book:

1. Introduction to Algorithms 3rd edition by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein

ARTIFICIAL INTELLIGENCE

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To learn about AI problem, Production Systems and their characteristics.
2. To understand the importance of search and the corresponding search strategies for solving AI problem.
3. To introduce to Planning, Natural Language Processing and Expert Systems.

Course Outcomes:

1. The Student understands AI problem characteristics, state space approach for solving AI problem, Production System framework.
2. The student learns several optimal search strategies and the use of heuristics.
3. The student learns relational, inferential, inheritable and procedural knowledge and the corresponding knowledge representation approaches.
4. The student is introduced to applying AI problem solving approaches to natural language processing, planning and expert systems

SYLLABUS

UNIT-I

Introduction to Artificial Intelligence: Artificial Intelligence, AI Problems, AI Techniques, Defining the Problem as a State Space Search, Problem Characteristics, Production Systems.

Search Techniques: Issues in The Design of Search Programs, Un-Informed Search, BFS, DFS; Heuristic Search Techniques: Generate-And- Test, Hill Climbing, Best-First Search, A* Algorithm, Problem Reduction, AO*Algorithm, Constraint Satisfaction, Means-Ends Analysis.

UNIT-II

Knowledge Representation using Rules: Procedural Vs Declarative Knowledge, Logic programming, Forward Vs Backward Reasoning, Matching Techniques, Partial Matching, RETE Matching Algorithm ,Overview of LISP and PROLOG.

Structured Representations of Knowledge: Semantic Nets, Partitioned Semantic Nets, Frames, Conceptual Dependency and Scripts.

UNIT-III

Symbolic Logic: Propositional Logic, First Order Predicate Logic: Representing Instance and is-a Relationships, Computable Functions and Predicates, Unification & Resolution, Natural Deduction.

Reasoning under Uncertainty: Introduction to Non-Monotonic Reasoning, Truth Maintenance Systems, Logics for Non-Monotonic Reasoning

UNIT-IV

Statistical Reasoning: Bayes Theorem, Certainty Factors and Rule-Based Systems, Bayesian Probabilistic Inference, Bayesian Networks, Dempster- Shafer Theory, Fuzzy Logic: Crisp Sets, Fuzzy Sets, Fuzzy Logic Control, Fuzzy Inferences & Fuzzy Systems.

Natural Language Processing: Steps in the Natural Language Processing, Syntactic Processing and Augmented Transition Nets, Semantic Analysis, NLP Understanding Systems;

UNIT-V

Planning Components of a Planning System, Goal Stack Planning, Non-linear Planning using Constraint Posting, Hierarchical Planning.

Experts Systems: Overview of an Expert System, Architecture of an Expert Systems, Different Types of Expert Systems- Rule Based, Frame Based, Decision Tree based, Case Based, Neural Network based, Black Board Architectures.

Text Books:

1. Artificial Intelligence, Elaine Rich and Kevin Knight, Tata Mcgraw-Hill Publications
2. Introduction To Artificial Intelligence & Expert Systems, Patterson, PHI publications

Reference Books:

1. Artificial Intelligence, George F Luger, Pearson Education Publications
2. Artificial Intelligence: A modern Approach, Russell and Norvig, Printice Hall
3. Artificial Intelligence, Robert Schalkoff, Mcgraw-Hill Publications
4. Artificial Intelligence and Machine Learning, Vinod Chandra S.S., Anand Hareendran S.

COMPILER DESIGN

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To explain the basic understanding of grammars and language definition and Introducing various phases of designing a compiler which will be helpful to construct new tools like LEX, YACC, etc.
2. To make the student to understand the concepts underlying the design and implementation of language processors and its mechanisms.
3. To extend the knowledge of parser by parsing LL parser and LR parser.
4. To enrich the knowledge in various phases of compiler and its use, code optimization techniques, loop optimization techniques, machine code generation, use of symbol table, and Runtime-Environments.

Course Outcomes:

1. Acquire knowledge in different phases and passes of Compiler, and specifying different types of tokens by lexical analyzer, and also able to understand the Compiler tools like LEX, YACC, etc.
2. Ability to describe the different types of parsers. i.e. Top-down, Bottom-up parsers, Construction of SLR, CLR and LALR parse table.
3. Ability to explain Syntax directed translation, synthesized and inherited attributes.
4. Ability to explain code optimization techniques and code generation techniques to improve the performance of a program in terms of speed & space.

SYLLABUS**UNIT-I**

Introduction Language Processing, Structure of a Compiler, Compiler-Construction Tools, The Evaluation of Programming language, The Science of Building a Compiler

Lexical Analysis: The Role of Lexical Analysis, Input Buffering, Specification of Tokens, Recognitions of Tokens, The Lexical Analyzer Generator Lex.

UNIT-II

Syntax Analysis: The Role of a Parser, CFG(Definition of CFG, Derivations and Parse Trees, Ambiguity), Writing Grammar(Eliminating Ambiguity, Elimination of Left Recursion and Left Factoring in CFG), Top-down Parsing (Recursive-Descent Parsing and Predictive Parsing), Bottom-up Parsing(Shift Reduce Parsing).

UNIT-III

Introduction to LR Parser(Simple LR Parsing), More Powerful LR Parser(Canonical LR and LALR Parsing), Using Ambiguous Grammars, Error Recovery in LR parser, The Parser Generator Yacc.

Syntax-Directed Translation: Syntax Directed Definitions(Inherited and Synthesized Attributes, Evaluating an SDD at Nodes of Parse Tree), Evolution Order of SDTS(Dependency Graphs, Ordering the Evaluation of Attributes, S-Attributed Definitions and L-Attributed Definitions), Application of SDTS(Construction of Syntax Trees), Syntax Directed Translation Schemes(Postfix Translation Schemes, Parser Stack Implementation of Postfix SDT's).

UNIT-IV

Intermediate Code Generation: Variants of Syntax Trees(DAG for Expressions, The Value-Number Method for Constructing DAG's), Three-Address Code(Address and Instructions, Quadruples, Triples), Type Checking(Rules for Type Checking and Type Conversion).

Code Optimization: Basic Blocks and Flow Graphs, Optimization of Basic Blocks, The Principal Sources of Optimization, Introduction to Data-Flow Analysis

UNIT-V

Code Generation: Issues in the Design of a Code Generator, The Target Language, A Simple Code Generator, Code Generation from DAG, Peephole Optimization, Register Allocation and Assignment.

Runtime Environments: Storage Organization, Stack Allocation of Space, Heap Management, Symbol Tables(Symbol Table Per Scope, Use of Symbol Tables)

Text Books:

1. Compilers, Principles Techniques and Tools.Alfred V Aho, Monical S. Lam, Ravi Sethi Jeffery D. Ullman,2nd edition,pearson,2007
2. Compiler Design K.Muneeswaran, OXFORD
3. Principles of compiler design,2nd edition,Nandhini Prasad,Elsevier.

Reference Books:

1. Compiler Construction, Principles and practice, Kenneth C Loudon, CENGAGE
2. Implementations of Compiler, A New approach to Compilers including the algebraic methods, Yunlinsu ,SPRINGER

CLOUD COMPUTING (Elective-I)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. This course is designed to examine the system models for cloud computing.
2. Understand the concepts of virtualization, hardware and storage
3. Identify cloud platform architecture and programming.
4. Develop cloud applications

Course Outcomes:

1. Define basic networking concepts for distributed and cloud computing. (K1).
2. Understand the importance of Virtualization concept in cloud computing. (K2)
3. Explain the architecture of Cloud platform. (K2)
4. Make use of some important cloud computing driven commercial systems such as Google Apps, Microsoft Azure and Amazon Web Services and other cloud software environments. (K3)
5. Utilize infrastructure, storage and tools to access the cloud to develop cloud application (K3)

SYLLABUS

UNIT I

Systems modeling, Clustering and virtualization: Scalable Computing over the Internet, Technologies for Network based systems, System models for Distributed and Cloud Computing, Software environments for distributed systems and clouds, Performance, Security And Energy Efficiency.

UNIT II

Virtual Machines and Virtualization of Clusters and Data Centers: Implementation Levels of Virtualization, Virtualization Structures/ Tools and mechanisms, Virtualization of CPU, Memory and I/O Devices, Virtual Clusters and Resource Management, Virtualization for Data Center Automation.

UNIT III

Cloud Platform Architecture: Cloud Computing and service Models, Architectural Design of Compute and Storage Clouds, Public Cloud Platforms, Inter Cloud Resource Management, Cloud Security and Trust Management. Service Oriented Architecture, Message Oriented Middleware.

UNIT IV

Cloud Programming and Software Environments: Features of Cloud and Grid Platforms, Parallel & Distributed Programming Paradigms, Programming Support of Google App Engine, Programming on Amazon AWS and Microsoft Azure, Emerging Cloud Software Environments.

UNIT V

Hardware and Infrastructure: Clients, Security, Network, Services. Accessing the Cloud: Platforms, Web Applications, Web APIs, Web Browsers. Cloud Storage: Overview, Cloud Storage Providers.

Developing Applications: Google, Microsoft, Intuit Quick Base, Cast Iron Cloud, Bungee Connect, Development, Troubleshooting, Application Management.

Text Books:

1. Distributed and Cloud Computing, Kai Hwang, Geoffrey C. Fox, Jack J. Dongarra MKElsevier.
2. Cloud Computing, A Practical Approach, Anthony T Velte, Toby J Velte, RobertElsenpeter, TMH.

Reference Books:

1. Mastering Cloud Computing, Foundations and Application Programming, Raj KumarBuyya, Christen vecctiola, S Tammaraiselvi, TMH.
2. Cloud Computing, A Hands on approach, ArshadeepBahga, Vijay Madiseti, University Press.

**MOBILE COMPUTING
(Elective-I)**

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. Introduction of an advanced element of learning in the field of wireless communication.
2. Introduce the students to the concepts of wireless devices and mobile computing.
3. To introduce wireless communication and networking principles, that support connectivity to cellular networks, wireless internet and sensor devices.
4. To introduce various platforms for mobile computing.

Course Outcomes:

1. A working understanding of the characteristics and limitations of mobile hardware devices including their user-interface modalities
2. The ability to develop applications that are mobile-device specific and demonstrate current practice in mobile computing contexts.
3. A comprehension and appreciation of the design and development of context-aware solutions for mobile devices.
4. A student will be able to understand various protocols for mobile computing
5. A student will be able to understand various platforms for mobile computing
6. A student will be able to understand various routing algorithm.

SYLLABUS**UNIT- I**

Introduction: Mobile Communications, Mobile Computing – Paradigm, Promises/Novel Applications and Impediments and Architecture; Mobile and Handheld Devices, Limitations of Mobile and Handheld Devices. GSM – Services, System Architecture, Radio Interfaces, Protocols, Localization, Calling, Handover, Security, New Data Services, GPRS.

UNIT –II

(Wireless) Medium Access Control (MAC): Motivation for a specialized MAC (Hidden and exposed terminals, Near and far terminals), SDMA, FDMA, TDMA, CDMA, Wireless LAN/(IEEE 802.11)

UNIT –III

Mobile Network Layer: IP and Mobile IP Network Layers, Packet Delivery and Handover Management, Location Management, Registration, Tunneling and Encapsulation, Route Optimization, DHCP.

UNIT –IV

Mobile Transport Layer: Conventional TCP/IP Protocols, Indirect TCP, Snooping TCP, Mobile TCP, Other Transport Layer Protocols for Mobile Networks. Database Issues: Database Hoarding & Caching Techniques, Client-Server Computing & Adaptation, Transactional Models, Query processing, Data Recovery Process & QoS Issues.

UNIT- V

Data Dissemination and Synchronization: Communications Asymmetry, Classification of Data Delivery Mechanisms, Data Dissemination, Broadcast Models, Selective Tuning and Indexing Methods, Data Synchronization – Introduction, Software, and Protocols.

Mobile Ad hoc Networks (MANETs) : Introduction, Applications & Challenges of a MANET, Routing, Classification of Routing Algorithms, Algorithms such as DSR, AODV, DSDV, etc. , Mobile Agents, Service Discovery. Protocols and Platforms for Mobile Computing: WAP, Bluetooth, XML, J2ME, Java Card, PalmOS, Windows CE, SymbianOS, Linux for Mobile Devices, Android.

Text Books:

1. Jochen Schiller, “Mobile Communications”, Addison-Wesley, Second Edition, 2009
2. Raj Kamal, “Mobile Computing”, Oxford University Press, 2007, ISBN:0195686772

Reference Books:

1. Asoke K Talukder, Hasan Ahmed, Roopa R Yavagal, “Mobile Computing, Technology Applications and Service Creation” Second Edition, Mc GrawHill.
2. UWE Hansmann, Lothar Merk, Martin S. Nocklous, Thomas Stober, “Principles of Mobile Computing,” Second Edition, Springer.

DISTRIBUTED SYSTEMS (Elective-I)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objective:

This course provides an introduction to the fundamentals of distributed computer systems, assuming the availability of facilities for data transmission.

Course Outcomes:

1. Scale as the number of entities in the system increase
2. Can sustain failures and recover from them
3. Work with distributed, fault tolerant file systems
4. Can handle and process large data volumes
5. Are secure and handle certain classes of distributed denial of service attacks
6. Are Loosely coupled, transactional and eventually stable

SYLLABUS

UNIT-I

Introduction to Distributed Systems, What is a Distributed System, Hardware concepts, Software concepts, Design issues. Communication in Distributed Systems, Layered Protocols, ATM networks, The Client – server model, Remote Procedure call, Group communication.

UNIT-II

Synchronization in Distributed System, Clock Synchronization, Mutual Exclusion, Election algorithms, Atomic transactions, Deadlocks in Distributed Systems.

UNIT-III

Process and processors in Distributed System threads, System Models, Processors allocation, Scheduling in Distributed System, Fault tolerance, Real time Distributed System.

UNIT-IV

Distributed File Systems, Distributed File System Design, Distributed File System implementation, Trends in Distributed File System.

UNIT-V

Distributed Shared Memory, Introduction, What is Shared memory?, Consistency models, Page based Distributed Shared memory, Shared – variable Distributed Shared memory, Object based Distributed Shared Memory.

Text Book:

1. Distributed Operating Systems, Andrew S. Tanenbaum

Reference Book:

1. Advanced Concepts in Operating Systems, Makes Singhal and Niranjana G. Shivaratna

INFORMATION RETRIEVAL SYSTEMS (Elective-I)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To provide the foundation knowledge in information retrieval.
2. To equip students with sound skills to solve computational search problems.
3. To appreciate how to evaluate search engines.
4. To appreciate the different applications of information retrieval techniques in the Internet or Web environment.
5. To provide hands-on experience in building search engines and/or hands-on experience in evaluating search engines.

Course Outcomes:

1. Identify basic theories in information retrieval systems
2. Identify the analysis tools as they apply to information retrieval systems
3. Understands the problems solved in current IR systems
4. Describes the advantages of current IR systems
5. Understand the difficulty of representing and retrieving documents.
6. Understand the latest technologies for linking, describing and searching the web.

SYLLABUS

UNIT - I

Introduction to Information Storage and Retrieval System: Introduction, Domain Analysis of IR systems and other types of Information Systems, IR System Evaluation.

Introduction to Data Structures and Algorithms related to Information Retrieval: Basic Concepts, Data structures, Algorithms

UNIT- II

Inverted files: Introduction, Structures used in Inverted Files, Building Inverted file using a sorted array, Modifications to Basic Techniques.

Signature Files: Introduction, Concepts of Signature Files, Compression, Vertical Partitioning, Horizontal Partitioning.

UNIT- III

New Indices for Text: PAT Trees and PAT Arrays: Introduction, PAT Tree structure, algorithms on the PAT Trees, Building PAT trees as PATRICA Trees, PAT representation as arrays.

UNIT- IV

Stemming Algorithms: Introduction, Types of Stemming Algorithms, Experimental Evaluations of Stemming to Compress Inverted Files

UNIT- V

Thesaurus Construction: Introduction, Features of Thesauri, Thesaurus Construction, Thesaurus construction from Texts, Merging existing Thesauri.

Text Books:

1. Frakes, W.B., Ricardo Baeza-Yates: Information Retrieval Data Structures and Algorithms, Prentice Hall, 1992.
2. Modern Information Retrieval by Yates Pearson Education.
3. Information Storage & Retrieval by Robert Korfhage – John Wiley & Sons.

Reference Books:

1. Kowalski, Gerald, Mark T Maybury: Information Retrieval Systems: Theory and Implementation, Kluwer Academic Press, 1997.
2. Information retrieval Algorithms and Heuristics, 2ed, Springer

SOFTWARE ENGINEERING MINI PROJECT LAB

Lab : 3 Periods	Int. Marks : 50
Exam : 3 Hrs.	Ext. Marks : 50
	Credits : 2

The purpose of the Software Engineering Lab course is to familiarize the students with modern software engineering methods and tools, **Rational Products**. The course is realized as a project-like assignment that can, in principle, by a team of three/four students working full time. Typically the assignments have been completed during the semester by each project team.

The goal of the Software Engineering Project is to have a walk through from the requirements, design to implementing and testing. An emphasis is put on proper documentation. Term projects are projects that a group student might take through from initial specification to implementation by giving equal importance to both design and implementation.

Cycle I: Practicing UML diagrams using IBM Rational Rose. 6*3 periods= 18periods

Before developing a mini-project, in this cycle, the student is acquainted with different UML diagrams using Rational Rose. The experiments should include drawing UML diagrams listed below for two demo/example applications assigned by the lab Instructor. The input for the following experiments is problem statement for any two demo projects supplied by the instructor.

1. Introduction to Rational Rose and Practicing the following diagrams
 - A. Activity diagrams for the overall business process of the projects
 - B. Use-case diagram for the demo projects along with Use-case descriptions and sub-diagrams for Use-cases.
2. Class diagram- Class diagrams including the features like classes, relationships, attributes and methods along with their visibilities.
3. Interaction diagrams- Sequence diagrams and Collaboration diagrams for different scenarios of the systems with all features like actors, objects and interactions.
4. Activity diagrams, State chart and other diagrams - Activity diagrams including the features like fork join and swim lanes. State diagrams including composite states and transitions. Component diagrams, Package diagrams and Deployment diagrams.
5. Forward and Reverser Engineering- Forward Engineering Class diagrams to classes in C++ and java and persistent classes to a database. Reverse Engineering C++ code, java code and a database.
6. Documentation using Rational Rose clear quest.

Cycle II: Mini-Project

8*3 periods= 24periods

The project deliverables include

- Problem statement
- Requirements Analysis
- Design
 - A Software Design Description and a System Design.
 - A test specification.
- Implementation
 - Implement the assigned project with one of the following web technologies
 - **Front end:** Java technologies/PHP/MS.NET Technologies
 - **Backend:** Oracle/My-SQL/SQL-Server
- Testing

Reference Books :

1. Project-based software engineering: An Object-oriented approach, Evelyn Stiller, Cathie LeBlanc, Pearson Education
2. Visual Modeling with Rational Rose 2002 and UML, Terry Quatrini, Pearson Education

Code: B17 CS 3211
NETWORK PROGRAMMING LAB

Lab : 3 Periods	Int. Marks : 50
Exam : 3 Hrs.	Ext. Marks : 50
	Credits : 2

Course Objectives

1. To understand the use of client/server architecture, inter process communication and to explain the basic communication protocols.
2. To understand elementary socket system calls, advanced socket system calls and Java Socket API and to explain the basic concepts relating to TCP and UDP based sockets
3. To Understand e-mail Programming like SMTP and POP
4. To understand File transfer protocol, remote login using pseudo terminal and RPC.

Course Outcomes

1. Students will be able to write Socket based Network application programs
2. Students will be able to design and develop Client Server applications using Java
3. Students will be able to write network applications like One-One chat ,Broadcasting and Multicasting
4. Students will be able to understand e-mail programming (SMTP, POP).

SYLLABUS

1. Write a program to identify well known ports on a Local/Remote System.

By trying to listen to the various well-known ports by opening client connections.

If the exception does not occur then the remote port is active else the remote port is inactive.

2. Write a program to implement Chat Application.
 - i. One-One Chat: By opening socket connection and displaying what is written by one System to other.
 - ii. Many-Many Chat (Broad Casting): Each Client opens a socket connection to the chat server and writes to the socket. Whatever is written by one system can be seen by all other systems.

3. Write a program to retrieve data from a remote database using Java.

At the remote database a server listens for client connections. This server accepts SQL queries from the client, executes it on the database and sends the response to the client.

4. Write a program to implement Mail Client
 - i. SMTP Client: Gives the server name, send email to the recipient using SMTP Commands
 - ii. POP Client: Gives the server name, user name and password, retrieve the mails and allow manipulation of mailbox using POP commands.

5. Write a program to simulate Telnet Client which allows to connect to well known servers and send and receive information.

Provide a user interface to contact well known ports so that client server interaction can be seen by the user.

6. Write a program to implement IP multicasting.
7. Write a program to implement simple file transfer between two systems (without protocols)

By opening socket connection to our server on one system and sending a file from one system to another

8. Write a program to implement TFTP Client and TFTP Server for file transfer
9. Write a program to implement HTTP-Server and HTTP Client.
The Server has to process the following commands: GET, POST, HEAD, and DELETE.
The server must handle multiple clients.
10. Write a program to implement UDP Echo Server and UDP Echo Client.
11. Write a program to get the attributes and contents of a web page using URL Connection class.
12. Write a program to implement DNS.

Reference Books:

1. "Java Network Programming", Elliott Rusty Harold, Fourth Edition, O'Reilly 2013
2. "An Introduction to Computer Networking", Kenneth C. Mansfield Jr and James L. Antonakos Pearson Education Asia
3. William Stallings, "Data and Computer Communication", Sixth Edition, Pearson Education, 2000.
4. Ferouz.A.Ferouzan, "Computer Communication and Networking", Fifth Edition, Pearson Education, 2001
5. James F. Kuross, Keith W. Ross, "Computer Networking, A Top-Down Approach Featuring the Internet", Third Edition, Addison Wesley, 2004.

EMPLOYABILITY SKILLS
(Common to all Branches)

Tutorial : 3 Periods (VA-2+QA-1)

Int.Marks : 30

Exam : 3 Hrs.

Ext.Marks : 70

Credits : 1

Part-A: Verbal Aptitude and Soft Skills-II

Course objectives:

1. To expose the students to bettering sentence expressions and also forming equivalents.
2. To instill reading and analyzing techniques for better comprehension of written discourses.
3. To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.
4. To inculcate conversational skills, nuances required when interacting in different situations.
5. To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.

Course Outcomes:

The students will be able to

1. Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.
2. Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences
3. Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).
4. Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.
5. Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.

SYLLABUS

UNIT -I (VA)

Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing.

UNIT- II (VA)

Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author's intention of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.

UNIT- III (VA)

Punctuation, discourse markers, general Essay writing, writing Issues and Arguments (with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a 'Statement of purpose', 'Letters of Recommendation', business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.

UNIT-IV (VA)

Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.

UNIT-V (SS)

Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.

Reading/ Listening material:

1. Guide to IELTS, Cambridge University Press
2. Barron's GRE guide.
3. Newspapers like 'The Hindu', 'Times of India', 'Economic Times'.
4. Magazines like Frontline, Outlook and Business India.
5. News channels NDTV, National News, CNN

Text Books:

1. Objective English and Verbal Reasoning by R S Agarwal.
2. Communication Skills by Sanjay Kumar and Pushpa Latha, Second Edition, OUP.
3. Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. Oxford Guide to Effective Writing and Speaking by John Seely.
2. Collins Cobuild English Grammar by Collins
3. The Art of Public Speaking by Dale Carnegie
4. The Leader in You by Dale Carnegie
5. Emotional Intelligence by Daniel Golman
6. Stay Hungry Stay Foolish by Rashmi Bansal

Part-B: Quantitative Aptitude-II

Course objectives:

The objective of introducing quantitative aptitude-II is:

1. To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages.
2. To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.
3. To familiarize the students with finding out accurate date and time related problems.
4. To enable the students solve the puzzles using logical thinking.
5. To expose the students to various problems based on geometry and mensuration.

Course Outcomes:

1. The students will be able to perform well in calculating different types of data interpretation problems.
2. The students will perform efficaciously on analytical and logical problems using various methods.
3. Students will find the angle measurements of clock problems with the knowledge of calendars and clock.
4. The students will skillfully solve the puzzle problems like arrangement of different positions.
5. The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.

SYLLABUS

UNIT I: Averages, mixtures and allegations, Data interpretation Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.

UNIT II: Puzzle test, blood Relations, permutations, Combinations and probability Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.

UNIT III: Periods,Clocks, Calendars, Cubes and cuboids Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.

UNIT IV: Puzzles Selective puzzles from previous year placement papers, sitting arrangement, problems- circular arrangement, linear arrangement, different puzzles.

UNIT V: Geometry and Mensuration Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals-Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of mensuration-Introduction of cylinder, cone, sphere, hemi sphere.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi
4. More puzzles by shakunataladevi
5. Puzzles by George summers.

Reference Books:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from _IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

COMPETITIVE CODING

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 1

Course Objectives:

1. To understand the essentials of competitive coding
2. To learn about competitive programming like time and space complexities
3. To investigate different methods like dynamic Arrays, Set & Map structures and sorting
4. To learn about String, Tree, Graph Theory algorithms

Course Outcomes:

At the end of the course, a student should be able to:

1. Acquire coding knowledge on essential of competitive coding
2. Acquire Programming knowledge on time & space complexities
3. Acquire coding knowledge on dynamic Arrays, Set & Map structures and sorting
4. Acquire knowledge on time complexities of different methods
5. Acquire Programming skill on String, Tree, Graph Theory algorithms

SYLLABUS**UNIT-I Introduction to Competitive Coding**

Introduction to Competitive coding and coding Platforms. Coding solution Vs. Efficient Coding solution. Types of solution approaches. Analyzing problem specific data requirement, Various data representations. Essential Data structures for fast coding. Various Syntactical I/O techniques comparison. Numbers, operations (including exponentiation). Integer properties(positive, negative, even, odd, divisible, prime, etc), Fractions, percentages and ratios. Point, vector, Cartesian coordinates(2D integer grid).

UNIT-II Essentials to Competitive coding

Basic data structures: Arrays, Strings, Stacks, Queues, Linked Lists . Asymptotic Notations – (Big-O), Evaluating the runtime complexity – Space Complexity - Towers-of-Brahma – Standard Template Libraries - Square root functions, primality testing and related techniques. Euclidean algorithms. Recursion techniques. Organizing data in $O(n \log n)$. Binary search techniques. Red-Black Trees. Fenwick tree, Segment Tree.

Basic Techniques

Dynamic Arrays, Set structures, Map structures, Iterators and ranges, Generating Subsets, Generating permutations, Backtracking techniques, Pruning the search. Bit masking. Disjoint set union.

UNIT-III Essential Coding Algorithms

Selection based algorithms: sorting, Coin change problem, Fractional selections, Schedules matching, Activity marking, heap sort, Huffman coding techniques, Spanning Trees, Minimizing sums, Data compression. Finding method count, Subsequence and related problems, paths in grid. DP with Bit mask

UNIT-IV String & Tree Algorithms

TRIE data structure, Naïve string searching, z-algorithm, Manacher's algorithm, Rabin-Karp, KMP Algorithm, Tree Traversals, Diameter, All longest paths, Binary trees, Applying search property to tree structures. Suffix arrays.

UNIT-V Graph Algorithms

Graph Algorithms – DFS, BFS. Depth First and Breadth First Traversals - Shortest paths: Dijkstra's algorithm Bellman-Ford Algorithm – Floyd Warshall Algorithm - Adjacency List Representation – Euler path, tour, cycle – Eulerian Graph - Johnson's Algorithm for All-pairs shortest path – Shortest path in Directed Acyclic Graph. Bridges and articulation points. Topological sorting, Strongly connected components in directed graphs. 2-SAT.

Note: Introduce C++ STL/Java Collections and let students solve problems using C++ STL/Java Collections

References:

1. Data Structures and Algorithm Analysis in C – Mark Allen Weiss, Pearson Edu Publishers.
2. Data Structures and Algorithms: Concepts, Techniques and Applications – G.A.V.Pai, Tata Mc Graw Hill Publishers.
3. Advanced Data Structures – Peter Brass, Cambridge University Press, 2008.
4. Introduction to Algorithms by Thomas H. Cormen, 3rd Edition, PHI.
5. <https://www.geeksforgeeks.org/>.
6. <https://www.tutorialspoint.com/>

IPR & PATENTS
(Common to CSE, ECE & IT)

Tutorial : 2 Periods

Credits : 0

Course Objectives:

1. To introduce the idea of tangible and intangible property and its protection.
2. To familiarize with the frameworks for protection of intellectual property.
3. To layout the procedures to claim rights over intellectual property.

Course Outcomes:

After successful completion of the course, the student shall be able to

1. Identify various types of intangible property that an engineering professional could generate in the course of his career.
2. Distinguish between various types of protection granted to Intellectual Property such as Patents, Copy Rights, Trademarks etc.,
3. List the steps involved in getting protection over various types of intellectual property and maintaining them.
4. Take precautions in writing scientific and technical reports without plagiarism.
5. Help micro, small and medium entrepreneurs in protecting their IP and respecting others IP as part of their business processes.

SYLLABUS

UNIT I

Intellectual Property Law: Basics - Types of Intellectual Property - Innovations and Inventions - Trade related Intellectual Property Rights – Agencies Responsible for Intellectual Property Registration – Infringement - Compliance and Liability Issues

UNIT II

Principles of Copyright – Subject Matters of Copyright – Rights Afforded by Copyright Law – Copyright Ownership– Copyright Formalities and Registration – Limitations – Infringement of Copyright - Plagiarism and difference between Copyright infringement and Plagiarism

UNIT III

Introduction to Trade Mark – Trade Mark Registration Process – Post registration procedures – Trade Mark maintenance – Infringement – Dilution of Ownership of Trade Mark – Likelihood of confusion – Trade Mark claims – Trade Marks Litigation – International Trade Mark Law

UNIT IV

Introduction to Patent Law – Rights and Limitations – Rights under Patent Law – Patent Requirements – Ownership and Transfer – Patent Application Process and Granting of Patent – Patent Infringement and Litigation – International Patent Law – Double Patenting

UNIT V

Introduction to Trade Secrets – Maintaining Trade Secret – Physical Security – Employee Access Limitation – Employee Confidentiality Agreement – Trade Secret Law – Unfair Competition – Trade Secret Litigation – Breach of Contract – Applying State Law.

Text Books:

1. Kompal Bansal & Parikshit Bansal "Fundamentals of Intellectual Property for Engineers", BS Publications
2. Prabhuddha Ganguli: "Intellectual Property Rights" Tata McGraw – Hill, New Delhi
3. R. Radha Krishnan, S. Balasubramanian: "Intellectual Property Rights: Text and Cases", Excel Books, New Delhi.

Reference Books:

1. Deborah E. Bouchoux: "Intellectual Property". Cengage learning , New Delhi
2. Richard Stim: "Intellectual Property", Cengage Learning, New Delhi.